

Session-1-August 17, 2020-Monday

Course:

Microsoft SQL Server Administration

Example:

Amazon.com -- Shopping Website -- A to Z Products.

1. Developers built the website.
2. Testers will test the application performance.
3. Application team will take care of application code.
4. **SQL Server DBA Team will take care of DB Servers.**
5. Windows Team will take care of Windows Servers.
6. Network Team will take care of Internet related.
7. Active Directory team takes cares of Accounts, Groups.
8. Storage Team will take care of Drives.
 - a. Project Manager takes care of entire project.

Roles in IT:

1. Developer
2. Tester
3. Application Developer
4. **DBA (SQL Server\Oracle\MySQL\Sybase/etc)**
5. Windows Server Admin
6. Network Admin
7. Active Directory Admin
8. Storage Admin

IT Infrastructure Team: Windows\Network\Storage\A.D

This is now available in small organizations.

Construct New House:

1. Select a Site or Place
2. Visit an architecture - gives us plans/sketches/layout.
3. Engage a Mansion
4. Purchase the equipment.
5. Engage a Plumber, Electrician, etc.
6. Engage a Painter.

DBA

1. Installs SQL Server Software and Configure as per std.
2. Administer the existing SQL Server Softwares.
3. Taking Caring of Databases.
4. Performing Backups and Restores.
5. Working on DB Automation Jobs/Tasks(Repetitive work).
6. SQL Server Security.
7. SQL Server HA & DR Setup and Troubleshooting.
HA - High Availability -- Inverter
DR -- Disaster Recovery -- Generator.

- a. DB Log Shipping
- b. DB Mirroring -- Generator\Inventory
- c. DB Replication

- d. SQL Server Cluster
- e. SQL Server Alwayson

8. SQL Server & DB Migrations.

Example:

John lives in an old house.

John moved to a new house.

9. SQL Server & DB Upgradations.

Example:

John lives in an old house

John demolished his old house and built a new house.

Windows 7 ---> Windows 10 (Upgrade)

SQL Server 2012 → SQL Server 2014 (Overwrite\Replaced)

10. SQL Server & DB Decommissioning.

Uninstalling/Removing SQL Server Software

DB Decommission --

- 1) Take DB Full backup for safer side
- 2) take db into offline status.
- 3) drop the db after 1 week.

11. SQL Server Performance Tuning.

12. SQL Server & DB Issues Troubleshooting & fixing.

Note: All above tasks should be done based on tickets/Approvals and Change Requests.

Phone -- Nokia, Samsung, AT & T, Apple,

DBMS/RDBMS -- Sybase, SQL Server, Oracle, MySQL, DB2

DBMS - Database Management System

RDBMS - Relational Database Management System

World Bank -- Main Bank which defines Rules & Regulations

Example: Praveen wants to open a bank.

Rules: 1,2,.....100 --- Conditions.

In Late 1980, DBMS Softwares:

Save, Update, Retrieve data. -- Security

1. Sybase

2. SQL Server - Microsoft

Roles:

a. Developer -- Write Queries/Codes

b. Administrator--Simple -- Take care of SQL Servers, DB's.

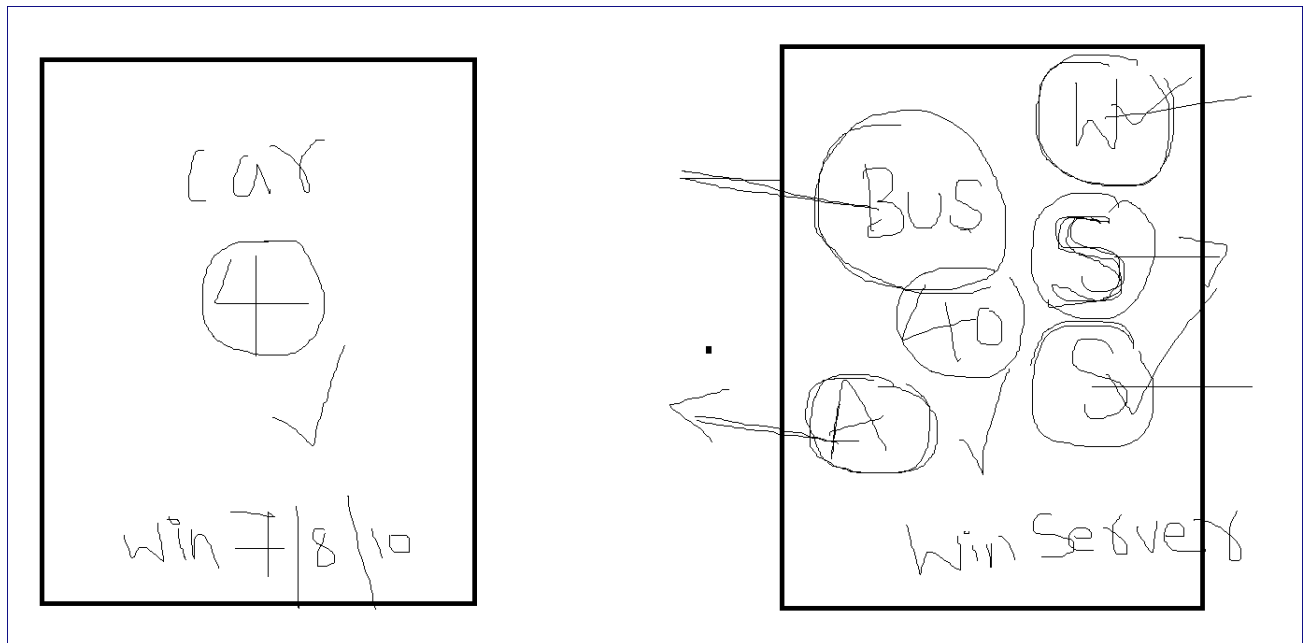
3. Oracle -- Oracle

4. MySQL -- MySQL Team

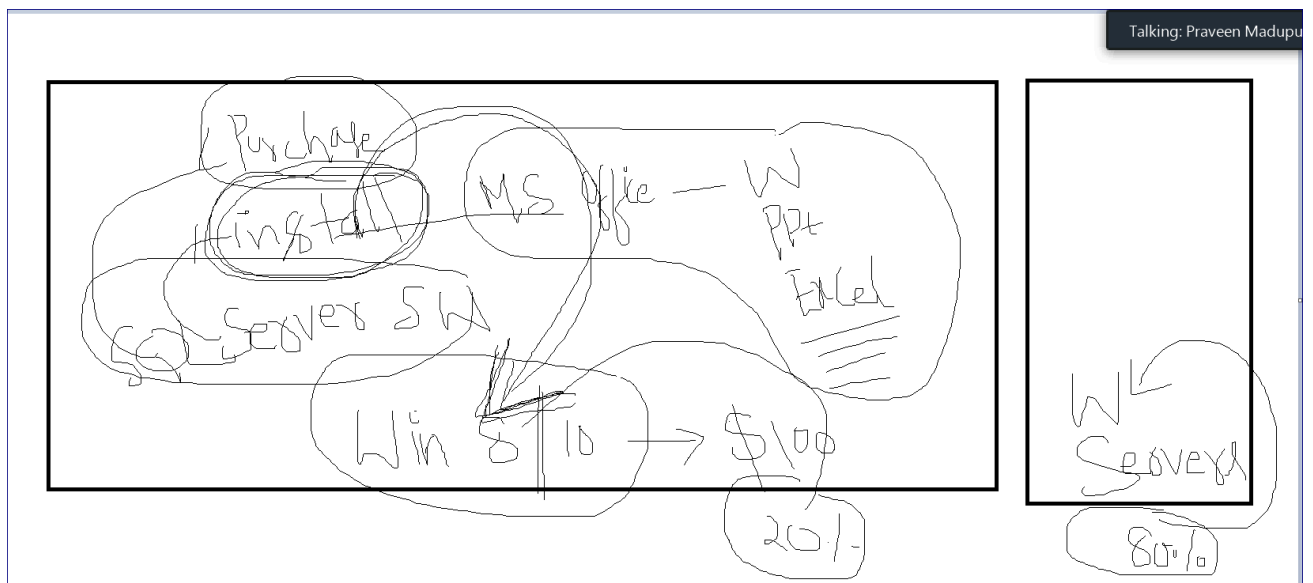
... 350 plus softwares.

Some Knowledge on below.

1. Operating Systems--Windows XP, 7, 8, 10 --- Home.
2. OS -- Business Purpose -- Windows Servers
2003, 2008, 2012, 2016.



Personal User -- Software Cost - \$ 10
Business Purpose -- \$100



Session-2-August 18, 2020-Tuesday

The history of Microsoft SQL Server begins with the first Microsoft SQL Server product—SQL Server 1.0, a 16-bit server for the OS/2 operating system in 1989—and extends to the current day.

Milestones[[edit](#)]

- MS SQL Server for OS/2 began as a project to port [Sybase SQL Server](#) onto OS/2 in 1989, by [Sybase](#), [Ashton-Tate](#), and [Microsoft](#).
- SQL Server 4.2 for NT is released in 1993, marking the entry onto [Windows NT](#).
- SQL Server 6.0 is released in 1995, marking the end of collaboration with [Sybase](#); Sybase would continue developing their own variant of *SQL Server*, Sybase [Adaptive Server Enterprise](#), independently of Microsoft.
- SQL Server 7.0(2000) is released in 1998, marking the conversion of the source code from C to C++.

SQL Server DB Replication

SQL Server DB Logshipping

SQL Server Clustering

- SQL Server 2005, released in 2005, finishes the complete revision of the old Sybase code into Microsoft code.

DB Mirroring

DB Snapshots

- SQL Server 2017, released in 2017, adds Linux support for these Linux platforms: [Red Hat Enterprise Linux](#), [SUSE Linux Enterprise Server](#), [Ubuntu](#) & [Docker Engine](#).^[4]

Currently^[edit]

As of May 2020, the following versions are supported by Microsoft:

- SQL Server 2012^[5]
- [SQL Server 2014](#)
- [SQL Server 2016](#)
- [SQL Server 2017](#)
- [SQL Server 2019](#)
- [SQL Server 2022](#)

From SQL Server 2016 onward, the product is supported on x64 processors only.^[6]

The current version is Microsoft SQL Server 2019, released November 4, 2019. The RTM version is 15.0.2000.5.^[7]

Microsoft & Sybase divided and released their own DBMS Softwares.

1. [Sybase software](#)
2. [Microsoft SQL Server - 1.0 Version](#)

[Windows O/S -- NT, XP, 7, 8, 10 -- Diff versions](#)

[SQL Server 1.0](#)

The **history of Microsoft SQL Server** begins with the first Microsoft SQL Server product - SQL Server 1.0, a 16-bit server for th

Version ↕	Year ↕	Release ↕	Code name ↕	Internal database version ↕
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4.2B (OS/2)	1993	SQL Server 4.2B (16-bit)	–	–
4.21a (WinNT)	1993	SQL Server 4.21a	SQLNT	–
6.0	1995	SQL Server 6.0	SQL95	406
6.5	1996	SQL Server 6.5	Hydra	408
7.0	1998	SQL Server 7.0	Sphinx	515
-	1999	SQL Server 7.0 OLAP Tools	Plato	–
8.0	2000	SQL Server 2000	Shiloh	539
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9.0	2005	SQL Server 2005	Yukon	611/612
10.0	2008	SQL Server 2008	Katmai	655
10.25	2010	Azure SQL database (initial release)	Cloud database or CloudDB	–
10.50	2010	SQL Server 2008 R2	Kilimanjaro (aka KJ)	661
11.0	2012	SQL Server 2012	Denali	706
12.0	2014	Azure SQL database	–	–
12.0	2014	SQL Server 2014	SQL14	782
13.0	2016	SQL Server 2016	SQL16	852
14.0	2017	SQL Server 2017	Helsinki	869
15.0	2019	SQL Server 2019	Seattle	895
Old version Older version, still maintained Latest version				

Release Version

SQL Server 1.0 --- 1.0 -- This is outdated

SQL Server 4.2 ----4.2 -- This is outdated

SQL Server 6.0 --- 6.0-- This is outdated

SQL Server 6.5 --- 6.5 -- This is outdated

SQL Server 7.0 --- 7.0 -- This is outdated

=====

SQL Server 2000 --- 8.0 -- This is outdated but still used

SQL Server 2005 --- 9.0 -- This is outdated but still used

SQL Server 2008 --- 10.0 -- This is outdated but still used

SQL Server 2008 R2 -10.5 -- This is outdated but still used

=====

Latest Versions

SQL Server 2012 -- 11.0

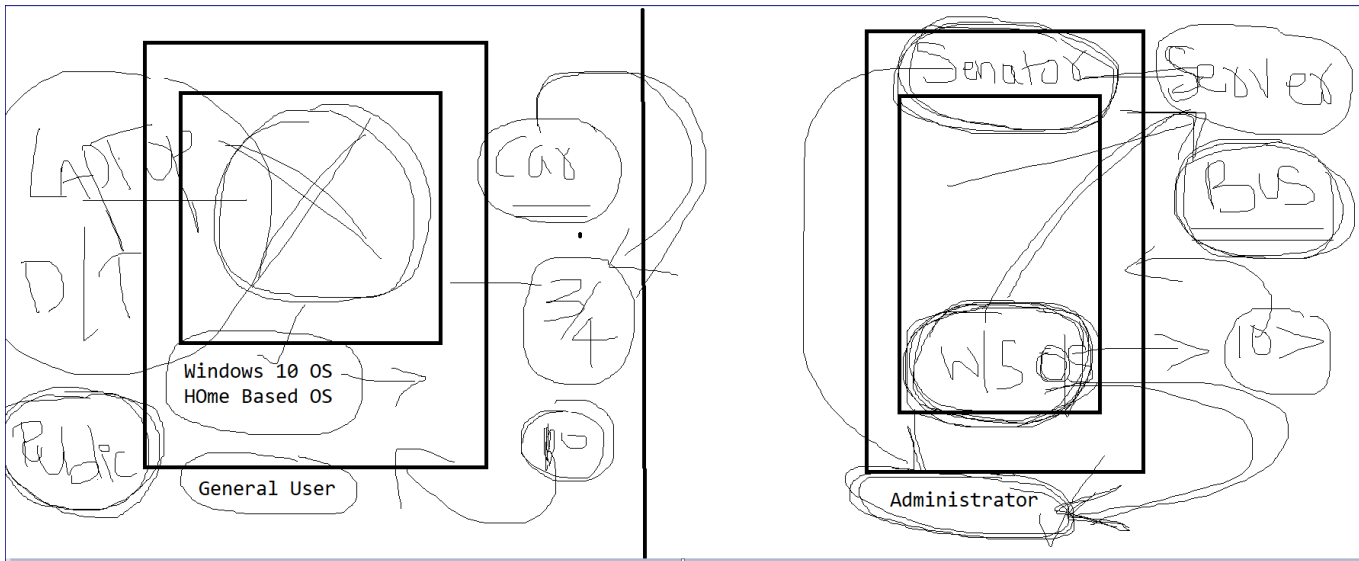
SQL Server 2014 -- 12.0

SQL Server 2016 -- 13.0

SQL Server 2017 -- 14.0

SQL Server 2019 - 15.0

Store Incharge -- 1980 -- old Software - \$ 500 Huge



Student Name: Praveen

Institute: Amazon Training Institute

Course-1-Meeting: 1111--Server name---XYZ1--IP Address: 192.168.1.111

Course-2-meeting: 2222--Server name---XYZ2--IP Address: 192.168.1.112

Course-3-meeting: 3333--Server name---XYZ3---IP Address: 192.168.1.113

Session-3-August 20, 2020-Thursday

- Environment Details
- Ticketing Tools & Process to handle tickets

Ticket:

General Local Terms used in Daily Life

Technical Knowledge/Terms used in Office work.

Agenda: Travel from New York to Washington.

1. Source & Destination.
2. When to travel ? on which date
3. What is the schedule time ? Time ?
4. Mode of travel from Source to Destination
5. Ticket purchase -- Ticket Number
6. Ticket Category -- Economy,premium Economy, Business, First class.

Technical Work:

1. Ticket No
2. Ticket Status
3. Priority
4. Schedule
5. SQL Server Name ?
6. DB Name ?
7. What to be done? Action to be taken by DBA?
8. Change Request -- Business Approvals\Application App

Tickets:

1. Incident ---- Unwanted event
2. Service Request -- Requesting for only details
3. Change Request -- Modifying any existing values, changing configuration.

Incident Types:

- a. Low -- P3
- b. Medium -- P2
- c. High -- P1

SLA - Service Level Agreement

- d. Low -- P3 Ack with in 15 mins -- 8 hours/24 hours
- e. Medium -- P2 Ack with in 15 mins -- 4 hours / 8 hours
- f. High -- P1 Ack with in 15 mins -- 1 to 4 hours.

Job Place: Employee works for 8 hours per day

8 hours/day

40 hours/week

40 hours * 4 weeks = 160 hours/month

Incidents Models:

- a. Automatic Incidents generated from 3rd Party Monitoring Tools.
- b. User Created Incidents.

Note: When a user creates a ticket, it goes to HelpDesk Team.

Helpdesk\Command Center\Operations Team:

Tickets -- DBA Team, Windows, Storage, Network.

Ticketing Tools:

1. Service Now
2. ITSM
3. HPSM
4. Jira
5. Samanage

Ticket Status:

1. New
2. Assigned
3. In-progress
4. Waiting on 3rd Party
5. Waiting for customer
6. Resolved \ Closed

Bridge call will be opened by Operations Team when issue is going beyond 1 hour or issue is critical:

1. Windows Team -- oncall Person
2. DBA Team -- oncall Person

3. Network Team -- oncall Person
4. Application Team -- oncall Person
5. Development -- oncall Person
6. Storage Team -- oncall Person

System Requirements:

Laptop

Windows/Mac:

1. VMWare Workstation Software -- In real time, SQL DBA never install
2. Windows Server OS 2012 -- In real time, SQL DBA never install
3. SQL Server 2012 Software --- SQL DBA Installs always as per req.

8 GB RAM

Free space 300 / 500 GB Hard Disk.

Session-4-August 20, 2020-Friday -- 7:00 PM EST.

SLA

Example: Customer is purchasing LAPTOP -- Dell

Free Shipping -- 3 days to 1 week

Prime Shipping - 1 day to 2 days

Expedite Shipping -- 1 day or overnight.

Incident -- Types: Low, Medium, High

-- States: New, Assigned, Inprogress, W on 3rd, WoC, R/Closed.

Human Being -- States: New Born, child, teen, old age, deceased.

Service Request -- Type: Low

Change Request:

Types:

1. Normal CR --- well planned before 1 week or 2 weeks.
2. Emergency CR -- This is unplanned CR
3. Urgent CR -- This is unplanned CR

CR States:

1. Analysis Phase
2. Planning phase
3. Approval Phase --- CAB (Change Advisory Board)
4. Implementation Phase -- SQL DBA who works on instructions.

Activity Start Time: 6 PM (6 PM)

Activity End Time: 7 PM (10 PM)

Duration: 1 hour 4 hours

5. Validation phase
6. Closure phase

Schedule Downtime: (week offs)

US - Business Hours -- 8/9 AM to 4/5 PM EST -- Monday to Friday.

Task: DB Files moving from one drive E:\ to another drive F:

Schedule-1: Sat or Sun any time

Schedule-1: Mon to Friday -- off business hours -- post 5 pm & before 8 AM

NOTE: Server will have its dedicated downtime -- 12 AM to 4 AM.(Sat & Sun)

Ticketing Tools: (shopping websites: Amazon, flipkart,

1. HPSM
2. Service now
3. Jira.
4. samanage
-etc.

Create tickets, SR's, Change Requests to Track all our ACTIVITIES

Agenda:

1. **Operating System**
2. **Virtual Machines**
3. **Windows Servers**
4. **SQL Servers.**

Home Based Usage -- Windows XP / 7 / 8/ 10
-- Mac - OS

Business Based Usage -- Windows Servers
-- 2000
-- 2003
-- 2008
-- 2012
-- 2016

VM -- Virtual Machine is created using VMWare Workstation Software.

All Softwares:

1. **Evaluation Copy (Free Trial for certain period -- 30days\90 \ 180 days.)**
2. **Enterprise edition --- Business -- not free--customer has to purchase them.**

Session-5-August 24, 2020-Monday -- 7:00 PM EST.

DBA -- Support a Project -- Small Online Website.

Nile -- (Shopping website)

Multiple Teams supporting this website.

SQL DBA -- Team or individual

Small Org: 5 approximately for a single project

Medium org: 8 to 10 app for a single project

Huge org: 15 to 30 app for a single/multiple project.

Will DBA works 24 x 7 x 365 ?

1) Onshore -- USA or UK -- 8/9 AM to 4/5 PM EST/CST/PST/GST/EU

-- Monday to Friday -- Business days

-- Sat & Sun -- Weekends.

-- Oncall Person (will receive a call from Ops Team)

SQL Server Administrators (DBA)

Alice

Bob

Charlie

Emmanuel

Fisher

A -- 6 AM EST to 2 PM EST

B -- 2 PM EST to 10 PM EST

C -- 10 PM EST to 6 AM EST.

2) Offshore -- India or others --- DBA works in 3 shift (24 x7)

- **1st shift - A Shift - Morning Shift -- 6 AM IST to 2 PM IST**
- **2nd shift -- B Shift- Afternoon Shift - 2 PM IST to 10 PM IST**
- **3rd shift -- C shift - Night shift - 10 PM IST to 6 AM IST.**

Application: 2 faces

1. Front End -- the interface we use to access it

-- .net \ java \ HTML \ etc...

2. Back End -- we use this to save data, update data and Retrieve data.

-- SQL Server \ Oracle \ Sybase \ MySQL

Combination:

Application = Front end + Back end

= .net s/w + SQL Server

= Java s/w + Oracle

Windows O/S Servers: (Windows Admin/Server Admin)

1. Windows Server -- 2000 -- Editions: Enterprise and Data Center
2. Windows Server -- 2003 -- Editions: Enterprise and Data Center
3. Windows Server -- 2008 -- Editions: Enterprise and Data Center
4. Windows Server -- 2012 -- Editions: Enterprise and Data Center
5. Windows Server -- 2016 -- Editions: Enterprise and Data Center

SQL Server Versions & Editions -- SQL DBA

Versions:

SQL Server 2012/SQL Server 2014

SQL Server 2016/SQL Server 2019

SQL Server Editions:

1. Enterprise -- \$10,000 (Commercial)
2. Developer -- free of cost. (Development Purpose)
3. Standard -- \$4000 (Commercial)

Environment:

1. Production Env --- Enterprise/Standard
2. Pre-prod or Stage Env --- Enterprise/Standard
3. QA (Quality Assurance) Env -- Developer Edition
4. Test Env -- Developer Edition
5. Dev Env -- Developer Edition

SQL Server Versions

SQL Server 2012 / 2014 --- 32 bit or 64 bit type softwares.

SQL Server 2016 /2019 --- 64 bits.

Windows Server OS -- 2012 (32 bit): -- Ground Floor

--> SQL Server 2012 only 32 bit can be installed.

→ SQL Server 2012 64 bit can't be installed. (As per design)

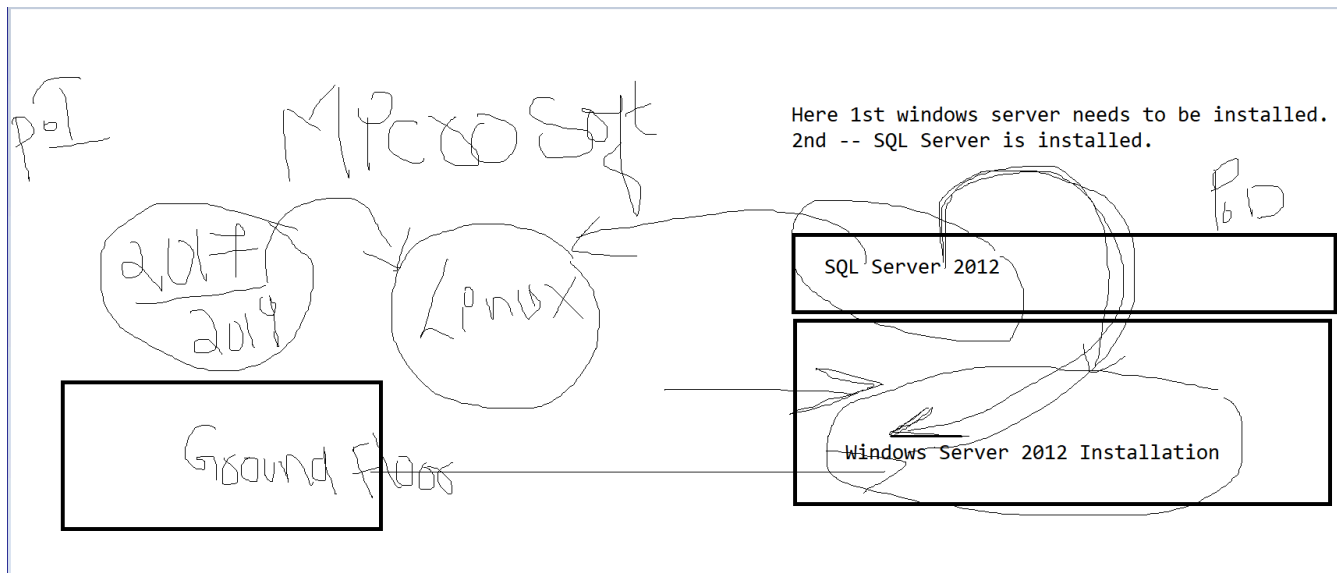
Windows Server OS -- 2012 (64 bit): -- Ground Floor

--> SQL Server 2012 32 bit can be installed.

→ SQL Server 2012 64 bit can be installed.

Platform Dependent -- SQL Server can only be installed on windows Server OS

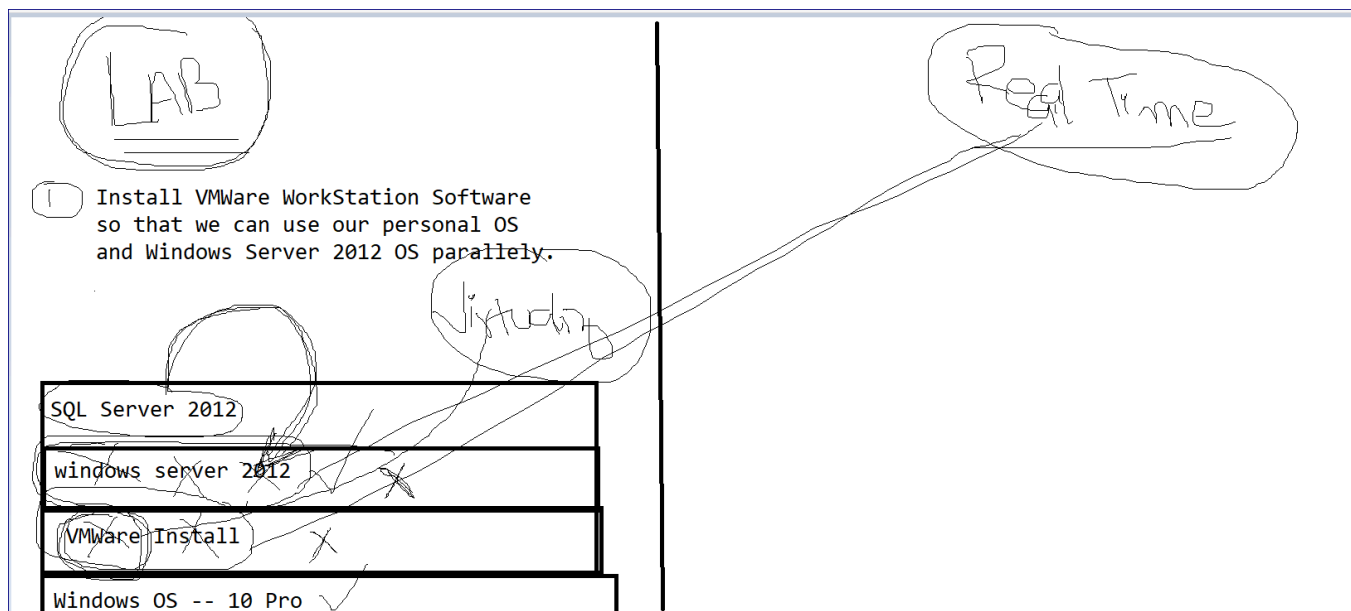
Platform Independent -- SQL Server 2017, 2019 can be installed on other OS like Linux.



Softwares from GoogleDrive:

<https://drive.google.com/drive/folders/1iXwOOwTTg5at3N5shnQnmqTprDQdHU11?usp=sharing>

My Drive > 1-1-Softwares-2020 ▾		
Name ↑	Owner	Last modified
1-Windows Servers	me	May 15, 2020 me
2-SQL_Server_SW's_Enterprise	me	May 16, 2020 me
3-SQL_Server_SW's-Evaluation	me	May 16, 2020 me
4-SQL_ServicePacks	me	May 5, 2020 me
5-SQL-Server-Sample-DB's	me	May 9, 2020 me
6-VMWare-WorkStation-15	me	May 15, 2020 me
7-Starwind-SW-SAN	me	May 5, 2020 me



1. Download VMWare Workstation (Folder 6)
2. Download Windows Server 2012 (Folder 1)
3. Download SQL Server 2012 (Folder 3) -- Developer

Session-6-August 25, 2020-Tuesday -- 7:00 PM EST.

Host -- Windows 10 Pro (Personal Machine)

Guest-- Virtual Machine -- Windows Server 2012 OS

Session-7-August 25, 2020-Thursday -- 7:00 PM EST.

Earlier Market:

Person -- Learning - 1 Technology was fine -- SQL Server Administration.

Current Market:

Person -- Should be well versed with multiple technologies ?

Competition ?

Priority?

Package?

Skillset: 1) Primary 2) Secondary.

Primary : SQL Server Administration.

Secondary: Cloud Technology (AWS, Azure, Google, Alibaba etc)

Original Path:

C:\Users\pmadupu\Documents\Virtual Machines\DBAClass7PMESTVM

Customized:

C:\10-DBAClass\DBAClass7PMESTVM

Segregation:

Project Manager:

Windows Server Admin --- Can only Restart or Shutdown any system as per policy.

-- Do not have permission to Stop or start SQL Services.

**SQL Server DBA --- Will not have permission to restart or shutdown any system.
--- Permission to Start stop SQL Services.**

**VMWare installation -- Completed -- VM Admin/Windows Admin
Windows Server 2012 OS -- Completed -- Windows Admin
Drives/Disk --- Storage Admin / Windows Admin**

Session-8-August 28, 2020-Friday

Agenda:

**Configured Windows Server Name -- Naomi --- Restart Server
Added new drives using VM Settings and Computer Management.**

**SQLData -- E:\
SQLLogs -- F:\
SQLTempDB
SQLBackups
SQLBinaries**

SQL Server Installation

- 1. Pre-checklist**
- 2. Installation Process**
- 3. Post checklist -- Which feature**

Session-9-August 29, 2020-Friday

SQL Server Versions-Editions

SQL Server 2012 -- Enterprise, Standard, Developer, Express (Free)

SQL Server 2014 -- Enterprise, Standard, Developer

SQL Server 2016 -- Enterprise, Standard, Developer

SQL Server 2017 -- Enterprise, Standard, Developer

SQL Server 2019 -- Enterprise, Standard, Developer

SQL Server Installation

1. Pre-checklist
2. Installation Process
3. Post checklist -- Which features

SQL Server 2012 -- Developer Edition.

Pre-requisites --

Softwares needed -- .net framework 2.0 / 3.0 / 3.5 / 4.0 / 4.5 --- in this case: .net f/w 3.5

Hardware needed -- Harddisk, DVD Drive,

Memory *	Minimum: Express Editions: 512 MB All other editions: 1 GB Recommended: Express Editions: 1 GB All other editions: At least 4 GB and should be increased as database size increases to ensure optimal performance.
Processor Speed	Minimum: x64 Processor: 1.4 GHz Recommended: 2.0 GHz or faster
Processor Type	x64 Processor: AMD Opteron, AMD Athlon 64, Intel Xeon with Intel EM64T support, Intel Pentium IV with EM64T support

SQL Server 2014 -- .net framework 4.0

SQL Server 2016/2017/2019 -- .net framework 4.6

<https://www.sqlskills.com/blogs/glenn/operating-system-support-for-sql-server-versions/>

SQL Server Version	WS 2008/2008 R2	WS 2012/2012 R2	WS 2016
SQL Server 2005	Yes	No	No
SQL Server 2008	Yes	Yes	No
SQL Server 2008 R2	Yes	Yes	No
SQL Server 2012	Yes	Yes	No
SQL Server 2014	Yes	Yes	Yes
SQL Server 2016	No	Yes	Yes

<https://docs.microsoft.com/en-us/sql/sql-server/install/hardware-and-software-requirements-for-installing-sql-server?view=sql-server-ver15>

SQL Server Version -- Edition.

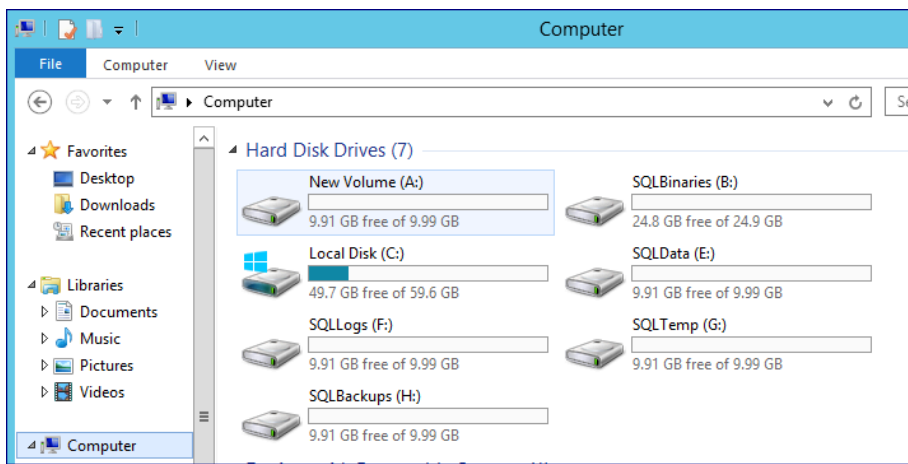
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15.0	2019	SQL Server 2019	Seattle	895

■ Old version
 ■ Older version, still maintained
 ■ Latest version

Multiple Drives as follows:

SQLData -- E:\
 SQLLogs -- F:\
 SQLTempDB -- G:\
 SQLBackups -- H:\
 SQLBinaries -- B:\
 Operating System -- C:\



Microsoft says:

Business Purpose -- SQL Server Enterprise or Standard Edition-- Paid Softwares--\$10,000

Dev/Testing Purpose -- SQL Server Developer Edition -- Paid / Free -- \$3000

Session-10-August 31st, 2020-Monday

Agenda: SQL Server Installation.

1. Pre-Checklist -- Software & Hard Ware Req.
 - Praveen -- Full Access on Windows Server -- Admin
 - Software -- .net framework -- 3.5 version for SQL Server 2012
2. Installation --- Need to choose required components.
 - SQL Server Engine (Service) -- Dev / DBA

Developers -- Mostly use below components/DBA works very less

- SSIS (SQL Server Integration Services)-- Packages
- SSRS (SQL Server Reporting Services)--Reports
- SSAS (SQL Server Analysis Services) -- Cubes/DWH
- Management Tools (SSMS - SQL Server Management Studio)
- Dev / DBA does their activities

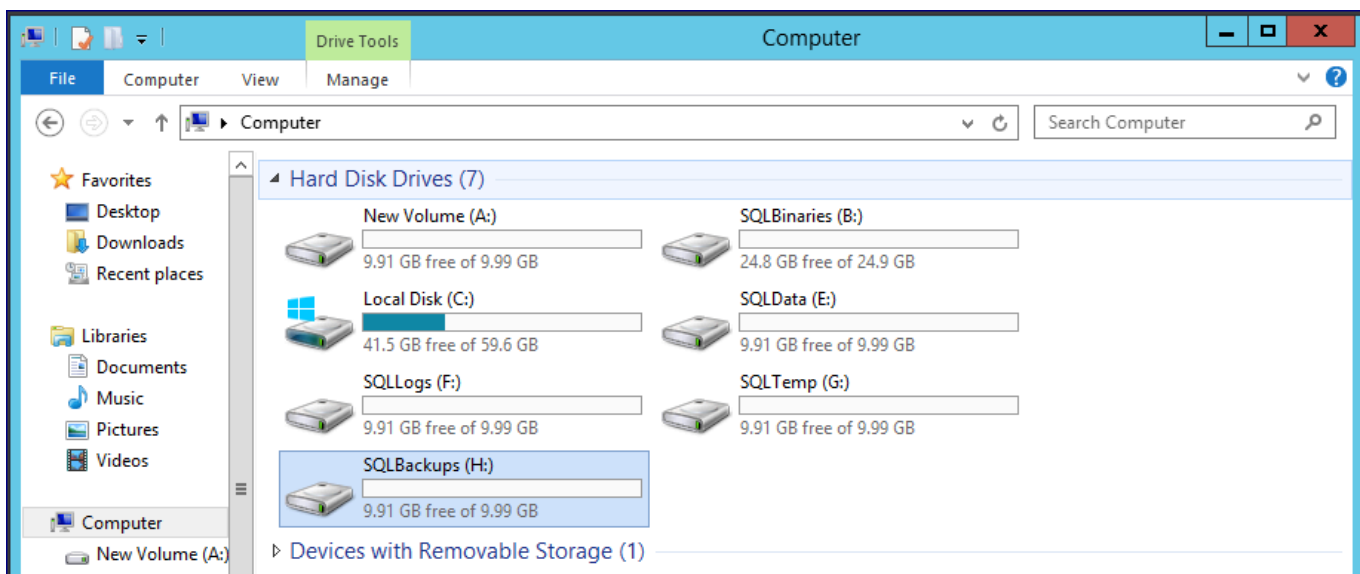
Dev -- Writes code, creates tables, creates Stored Procedures, etc.
DBA -- Creates DB, Takes DB Backups, Restores, Creates Logins,
Creates Jobs, Creates HA & DR Strategies, etc.

HA - High Availability --- Inventor --- Automatic
DR - Disaster Recovery --- Generator --- Manually

MS Office -- MS Word, **MS Excel**, MS Powerpoint, MS Access, MS OneNote.

3. Post Checklist

Few options -- needs to be enabled.
-- needs to be disabled.



Required Softwares are purchased by your organization and kept in some shared path.
Licenses

\\Servernamexyz.com\softwares\

1. Windows Server OS Softwares -- 2012, 2016, 2019
-- Standard, Enterprise, Data Center
2. SQL Server -- 2014, 2016, 2017, 2019
-- Standard, Developer, Enterprise
3. Visual Studio -- is used to create SSIS Packages etc.

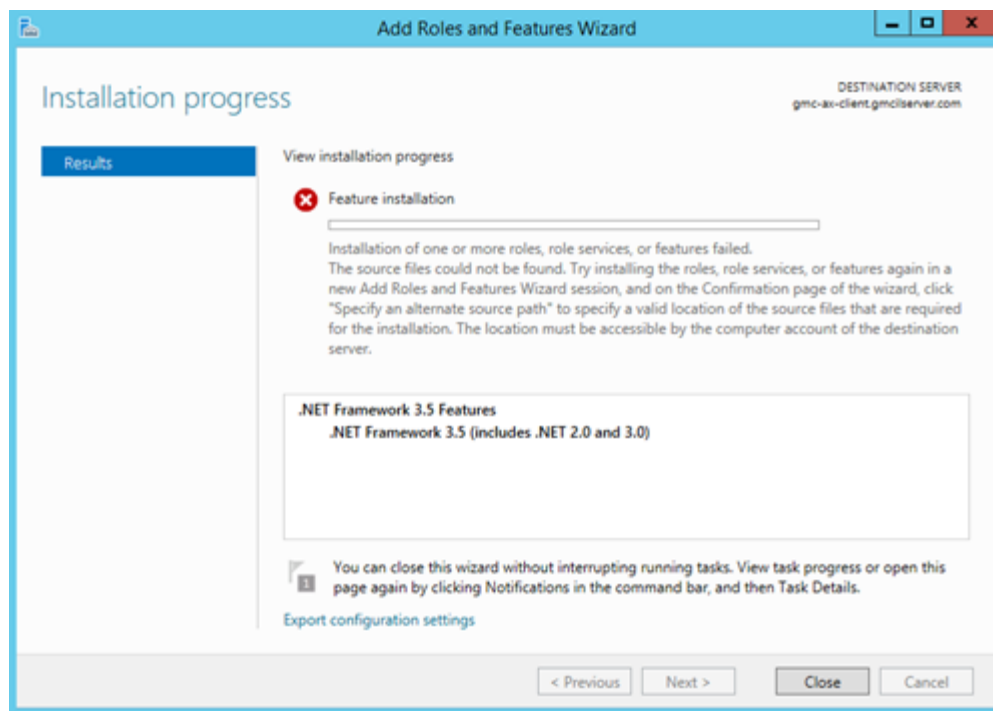
2 Situations here.

1. LAB Practice -- we always use Evaluation Versions or Trial Version -- Free to use
 - 30 days or 90 days or 180 days
 - software becomes unusable
2. Business/Enterprise/Realtime -- use always licensed softwares. (Purchased)
 - only fewer times we use free softwares
 - Example: Dev env -- Dev Edition of SQL Server.
 - Dev edition is available for free and licensed one.

Note: Server Manager -- is a tool used by Windows Admin or System Admin

When we try to install .NET Framework 3.5 Features on Windows Server 2012 R2 we might get below error if we overlooked the warning message given at Confirm installation selections page of the Add Roles and Features Wizard.

Installation of one or more roles, role services, or features failed. The source files could not be found. Try installing the roles, role services, or features again in a new Add Roles and Features Wizard session, and on the Confirmation page of the wizard, click "Specify an alternate source path" to specify a valid location of the source files that are required for the installation. The location must be accessible by the computer account of the destination server.



Install .net framework software using powershell tool.

Note:

1. Go to Windows Server 2012 Setup location "Windows_Server_2012_Datacenter\sources"
2. Copy the folder **sxs** from "Windows_Server_2012_Datacenter\sources" to C:\ drive in your server.
3. Then run the following commands using power shell as "Run As Administrator"

dism.exe /online /enable-feature /featurename:NetFX3Serverfeatures

**Dism.exe /online /enable-feature /featurename:NetFXdism.exe /online /enable-feat3 /Source:C:\sxs
/LimitAccess**

Session-11--September 01, 2020 - Tuesday.

What we need ?

Software ?

1. .net framework - 3.5 -- SQL Server 2012
2. SQL Server 2012 -- 2014 --2016 -- 2017 -- 2019

Hardware -- Hard disk , DVD Drive, Processor, etc.

Session-12--September 03, 2020 - Thursday.

SQL Server/DB Server

SQL Engine/DB Engine

SQL Box/DB Box

SQL instance/DB Instance

SQL Server 2000 -- This software we can install 16 times on a windows server os.

-- Default SQL instance -- 1st time we install

-- here DBA will not provide SQL Server Name

-- Default SQL Instance will take the name of the windows server itself

-- 15 times -- we install named instances

Example: Windows Server OS Machine -- name -- Amazon

1-Default --- MSSQLServer (Internally) -- SSMS --> Connect--> Amazon

---- No Name is provided by DBA

-- Ground Floor

15-Named -- DBA has to provide SQL Instance Name.

-- Pentagon ----> SSMS ---> Connect --> Amazon\Pentagon -- 1st Floor

-- Mercury ----> SSMS --> Connect --> Amazon\Mercury -- 2nd Floor

SQL Server 2005 till SQL Server 2019 -- 50 SQL Instances can be installed.

1- Default SQL Instance

49 -- Named SQL Instance

SQL Server Instance --- 5 comp -- 20 to 30 min
2 comp (DB Engine+Mgmt tools) 10 to 15 min

Components

1. DB Engine
2. SSIS --One time
3. SSAS
4. SSRS
5. Management Tools -- SSMS - SQL Server Management Studio.

SQL Server Softwares comes in 2 ways.

1. Normal SQL Server Setup Folder
2. Disk Image File -- Right click and choose to MOUNT.

Windows Server OS -- 64 Bit

SQL Server 64 bit

Windows Server OS -- 32 bit ----> SQL Server 32 bit only can be installed.

Windows Server OS -- 64 bit ----> SQL Server 32/64 bit both can be installed.

Session-13--September 04, 2020 - Friday.

We just completed new SQL Server Installation .

Named Instance -- PENTAGON

Summary Log File -- This will show us installation report

- Success
- Failure
- what components are installed.
- Installation Start time, Installation End time

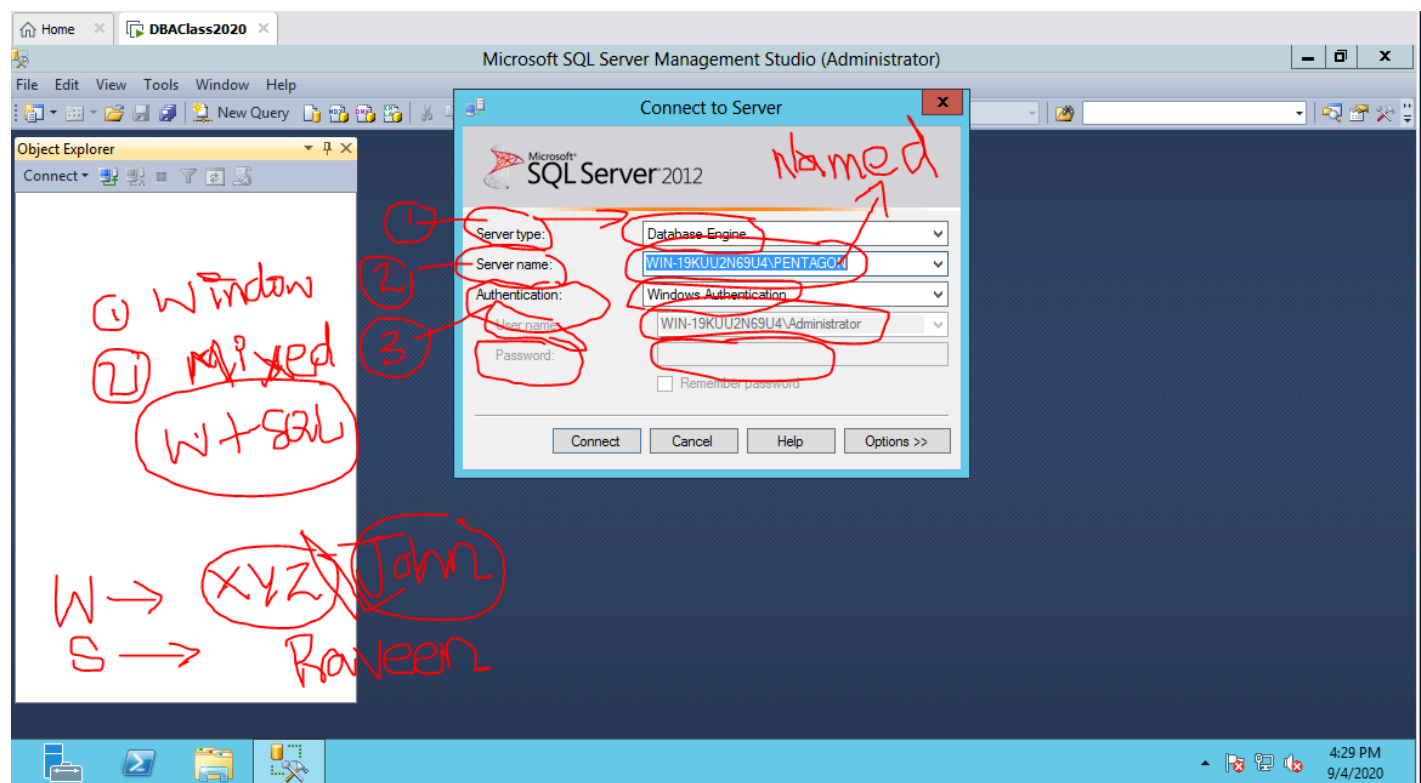
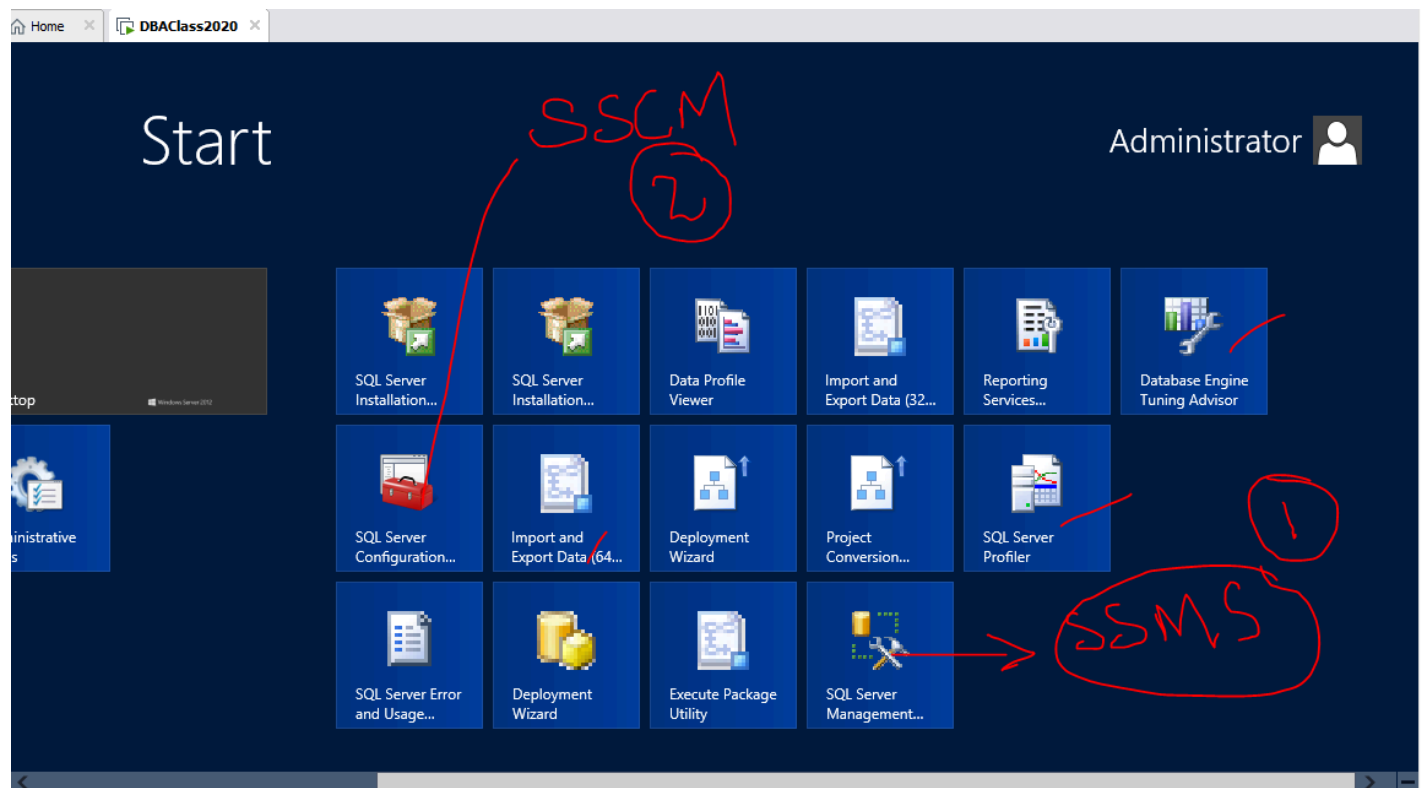
Post SQL Server installation -- we have few things which needed to be configured.

Note: Important Tools DBA uses every day.

1. SSMS -- SQL Server Management Studio. --- 95 %
-- DBA can create a DB, Create a table, Create a Login, Jobs, etc.

Note: To open SSMS,

- 1) Windows/Start button → Search for SSMS
- 2) open command prompt or powershell ----> Type SSMS



2. SSCM -- SQL Server Configuration Manager -- 5 %
 - used to manage SQL Services
 - DB Engine
 - SQL Agent
 - SSRS, SSIS, SSAS
 - SQL Browser.

Home DBAClass2020

Sql Server Configuration Manager

File Action View Help

SQL Server Configuration Manager (Local)

- SQL Server Services
 - SQL Server (PENTAGON)
 - SQL Server (MSSQLSERVER)
 - SQL Server Agent (PENTAGON)
 - SQL Server Browser
 - SQL Server Agent (MSSQLSERVER)
- SQL Server Network Configuration (32bit)
- SQL Native Client 11.0 Configuration (32bit)
- SQL Server Network Configuration
- SQL Native Client 11.0 Configuration

Name	State	Start Mo...	Log On As	Pr...	Service Type
SQL Server (PENTAGON)	Running	Automatic	NT Service\MSSQL\$PENTAGON	1388	SQL Server
SQL Server (MSSQLSERVER)	Running	Automatic	NT Service\MSSQLSERVER	1440	SQL Server
SQL Server Agent (PENTAGON)	Running	Automatic	NT Service\SQLAgent\$PENTAGON	1648	SQL Agent
SQL Server Browser	Running	Automatic	NT AUTHORITY\LOCALSERVICE	1712	
SQL Server Agent (MSSQLSERVER)	Running	Automatic	NT Service\SQLSERVERAGENT	2372	SQL Agent

Home DBAClass2020

Sql Server Configuration Manager

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Name	State	Start Mo...	Log On As	Pr...	Service Type
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SQL Server Browser	Running	Automatic	NT AUTHORITY\LOCALSERVICE	1712	
SQL Server Agent (MSSQLSERVER)	Running	Automatic	NT Service\SQLSERVERAGENT	2372	SQL Agent

Default

Home DBAClass2020

Sql Server Configuration Manager

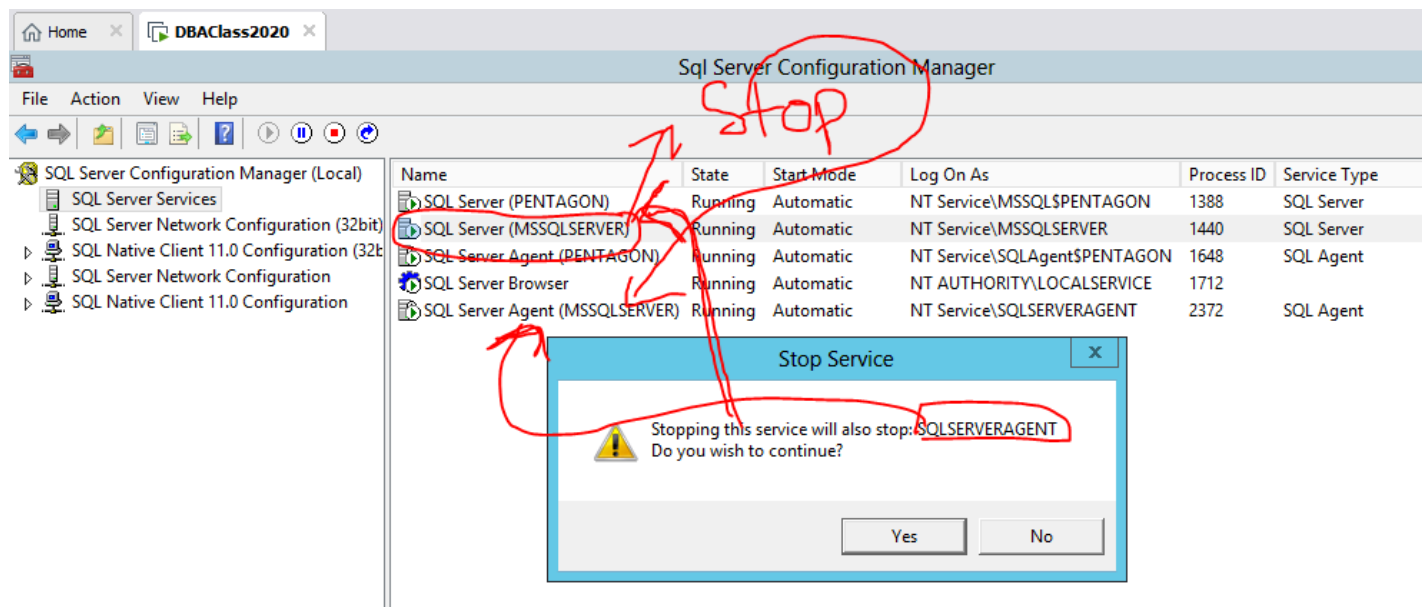
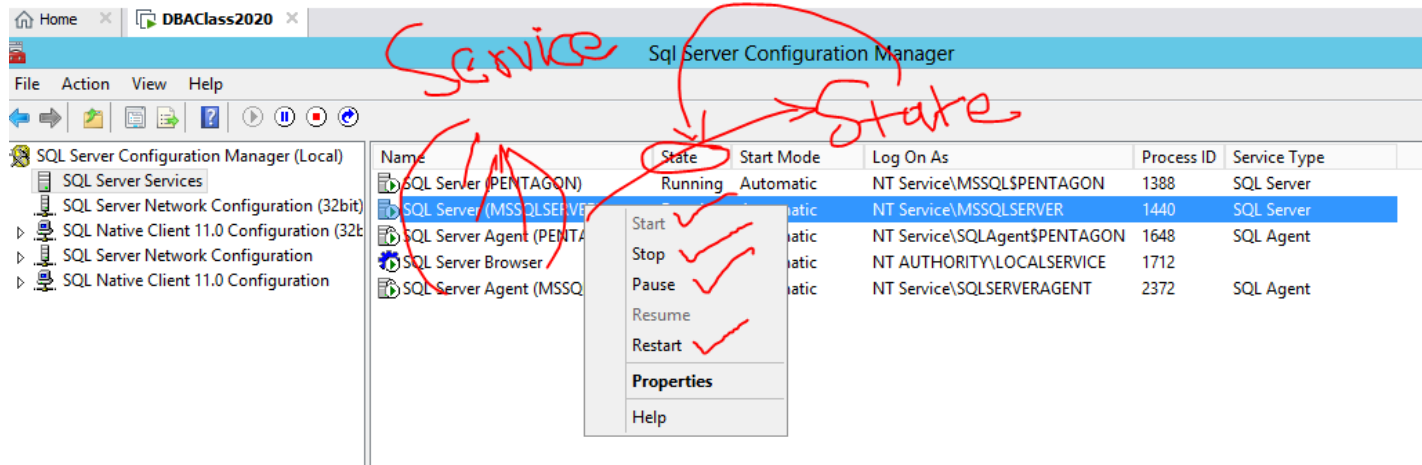
File Action View Help

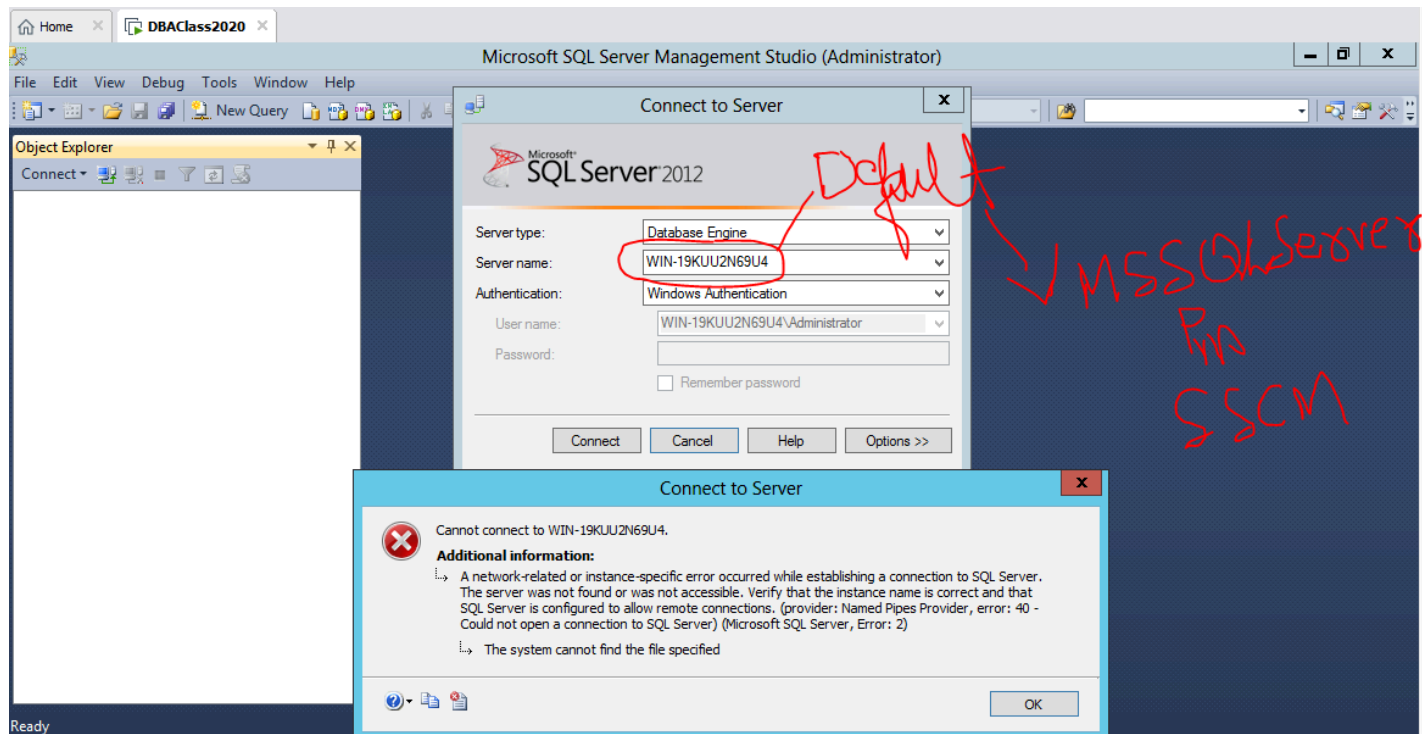
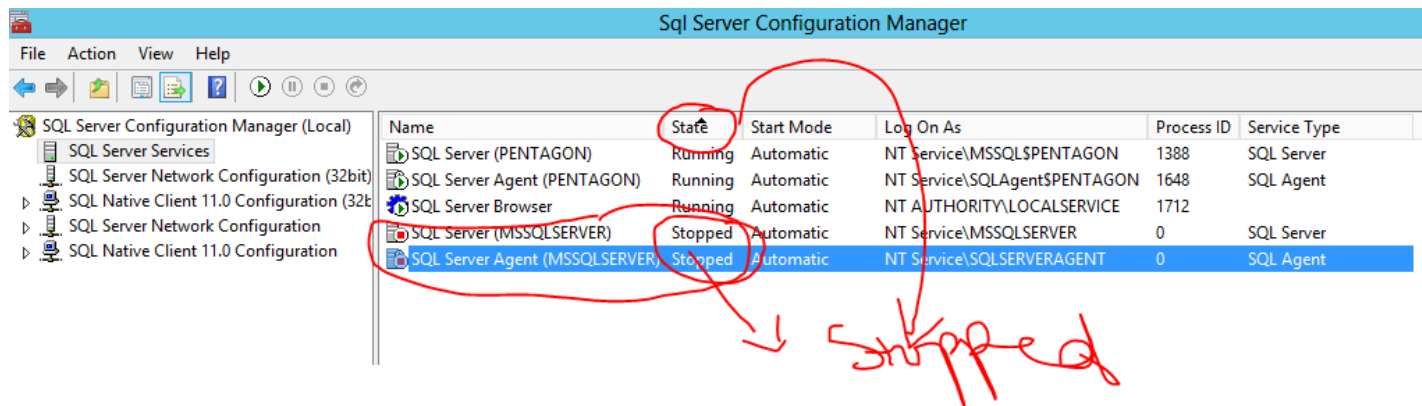
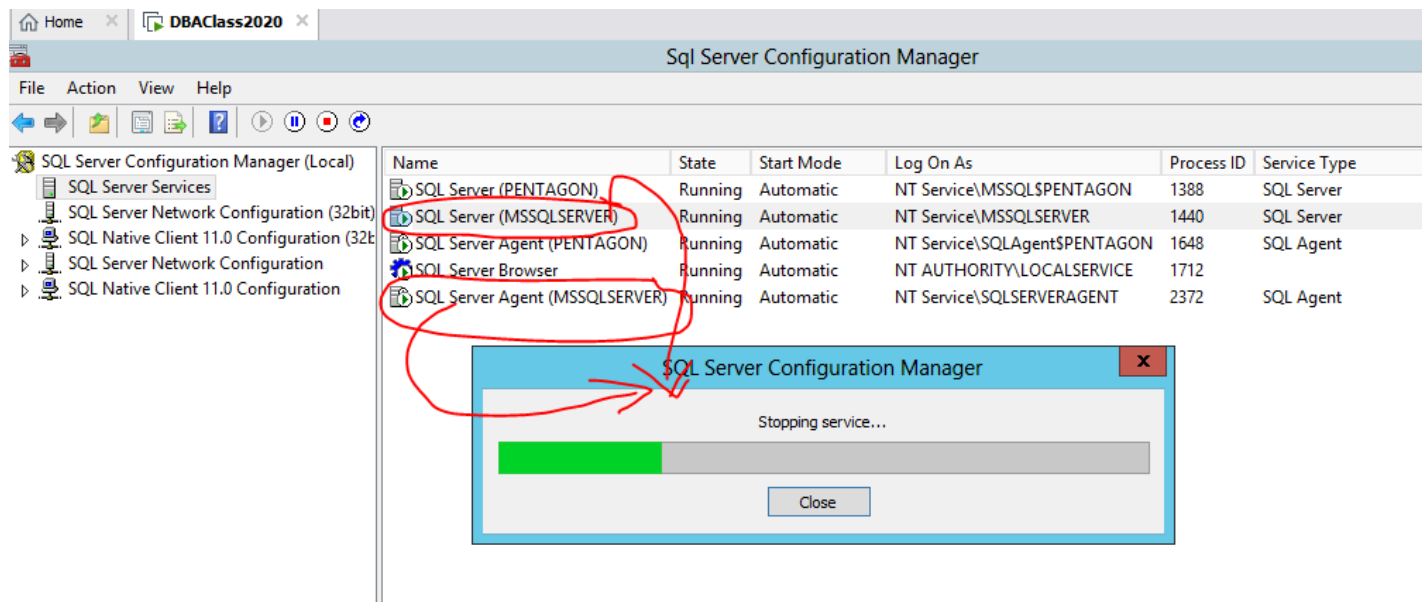
SQL Server Configuration Manager (Local)

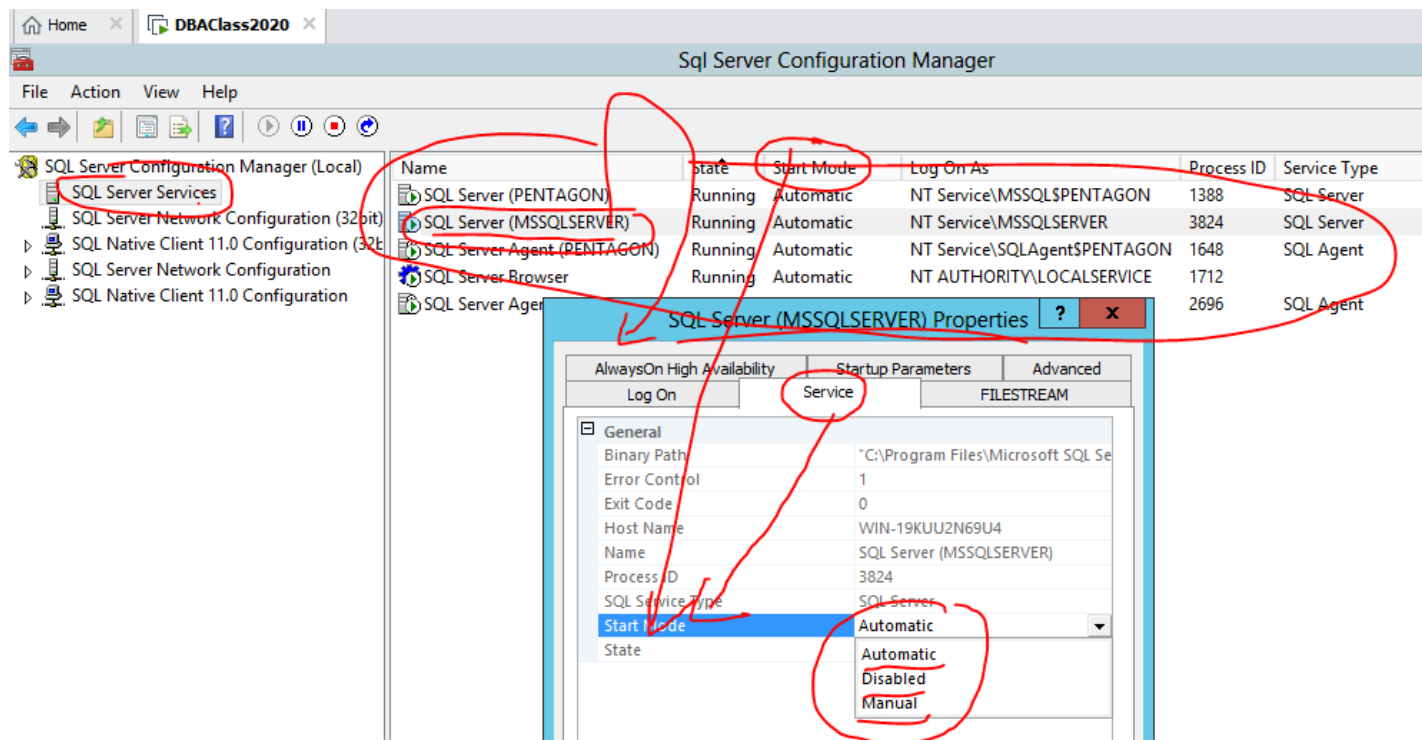
- SQL Server Services
 - SQL Server (PENTAGON)
 - SQL Server (MSSQLSERVER)
 - SQL Server Agent (PENTAGON)
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Name	State	Start Mo...	Log On As	Pr...	Service Type
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SQL Server Browser	Running	Automatic	NT AUTHORITY\LOCALSERVICE	1712	
SQL Server Agent (MSSQLSERVER)	Running	Automatic	NT Service\SQLSERVERAGENT	2372	SQL Agent

Named







Session-14--September 07, 2020 - Monday.

Post SQL Server Configuration Checklist:

E:\SQLData --- Default, PENTAGON, Newyork

F:\SQLLogs --Default, PENTAGON, Newyork

H:\SQLBackups --Default, PENTAGON, Newyork

T:\SQLTempDB--Default, PENTAGON, Newyork

I:\SQLBinaries--Default, PENTAGON, Newyork

SQL Server Supports 2 types of Authentications

- 1. Windows Auth -- only Windows accounts are allowed to Access SQL Server.**

Windows Account --- Example: Wonderla\PraveenM

praveen.m@wonderla.com

- these accounts will be created by Windows Admin\Sys Admin
- SQL DBA will never create windows accounts.

2. Mixed mode auth -- both windows + sql accounts are allowed to access sql server.
 - a) Windows auth -- same as above notes
 - b) SQL Accounts -- these accounts are created by SQL DBA
 - Example: SQL Account: **JohnBiller**

Session-15--September 08, 2020 - Tuesday.

Windows Server OS:

1. Windows Server 2000
2. Windows Server 2003
3. Windows Server 2008 R2
4. **Windows Server 2012**
5. Windows Server 2016
6. Windows Server 2019

SQL Server: 2008, **2012**, 2014, 2016, 2017, 2019

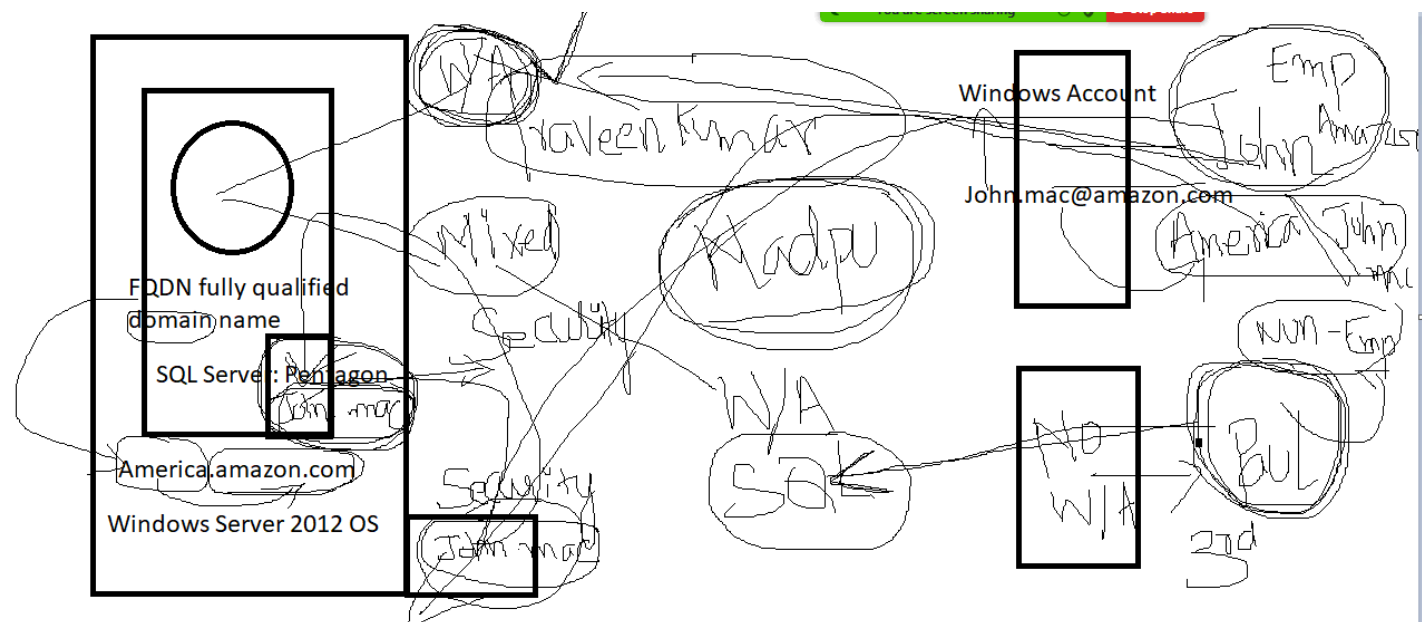
Windows Server

Supported DBMS Version	SQL Server 2005	SQL Server 2008	SQL Server 2008 R2	SQL Server 2012	SQL Server 2014	SQL Server 2016	SQL Server 2017
Supported OS Version							
Win 2003 SP2	Supported	Supported	Supported	N/A	N/A	N/A	N/A
Win 2008	Supported	Supported	Supported	Supported	Supported	N/A	N/A
Win 2008 SP2	Supported	Supported	Supported	Supported	Supported	N/A	N/A
Win 2008 R2	Supported	Supported	Supported	Supported	Supported	N/A	N/A
Win 2008 R2 SP1	Supported	Supported	Supported	Supported	Supported	N/A	N/A
Win 2012	N/A	Supported	Supported	Supported	Supported	Supported in 5.1.4+	Not Supported
Win 2012 R2	N/A	Supported	Supported	Supported	Supported	Supported in 5.1.4+	Not Supported
Win 2016	N/A	N/A	N/A	Supported in 5.1.8+	Supported in 5.1.8+	Supported in 5.1.7+	Not Supported

<https://docs.delphix.com/docs-old/delphix-administration/sql-server-environments-and-data-sources/setting-up-and-configuring-delphix-for-sql-server/tasks-for-the-windows-system-administrator/windows-server-os-and-sql-server-dbms-compatibility-matrices>

1. **SQL Server Enterprise** -- Costly and this has almost many features
2. **SQL Server Developer** -- purchased & free of cost versions.
3. **SQL Server Standard** -- less cost, purchase, less features when compared to enterprise.

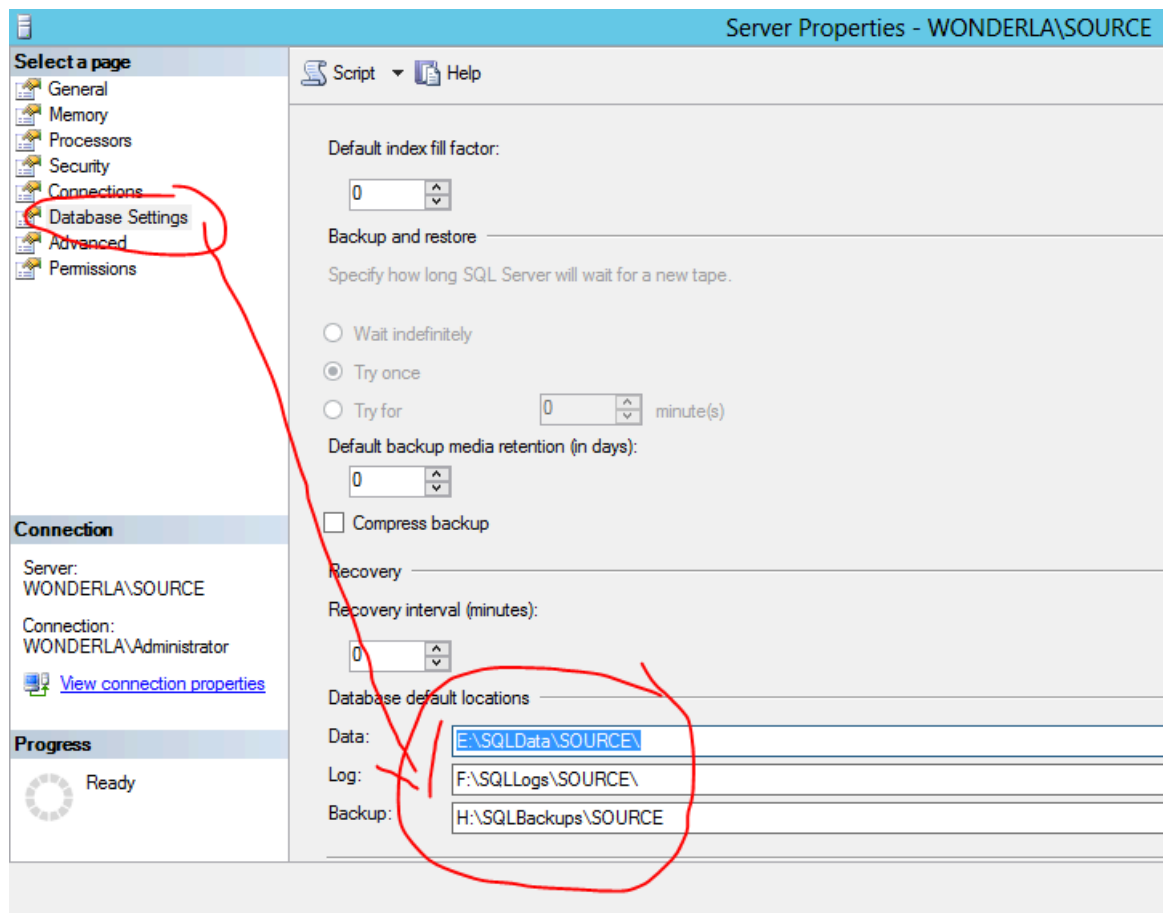
1. **SQL Server Authentication Mode**
 - a) **Windows Auth**
 - b) **Mixed Mode (Windows + SQL Server)**



Note:

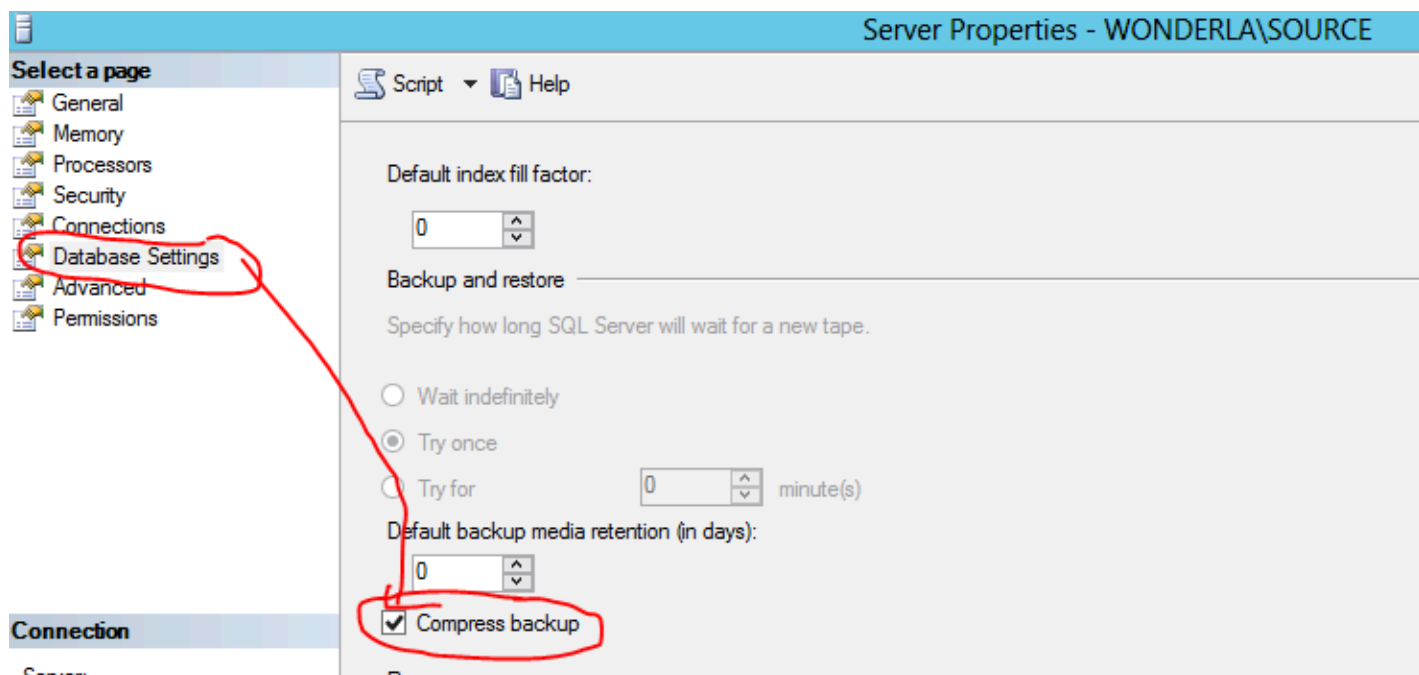
Find the screen displaying

- ## 2. Open powershell -- type logoff



3. Enable DB Backup compression feature if not enabled.

Note: this is available in Enterprise Edition only and not in Standard edition.



Example:

DB Size -- 100 GB Container -- 90 GB Data -- Normal Backup -- 85 GB Backup size.

-- compress backup -- 10 to 20 Gb backup size.

Note: Compression backup feature is available from SQL Server 2008 Version.

4. Configure SQL Server TempDB Files based on no of processors.

Server Properties - WONDERLA\SOURCE

Name	Value
Name	WONDERLA\SOURCE
Product	Microsoft SQL Server Developer (64-bit)
Operating System	Microsoft Windows NT 6.2 (9200)
Platform	NT x64
Version	11.0.5058.0
Language	English (United States)
Memory	2499 (MB)
Processors	8
Root Directory	B:\Program Files\Microsoft SQL Server\MSSQL11.SOURCE\MSSQL
Server Collation	SQL_Latin1_General_CP1_CI_AS
Is Clustered	False
Is HADR Enabled	False

Database Properties - tempdb

Database name: tempdb
Owner: sa
☒ Use full-text indexing

Database files:

Logical Name	File Type	Filegroup	Initial Size (MB)	Autogrowth / Maxsize	Path	File Name
tempdev	Rows Data	PRIMARY	8	By 10 percent, Unlimited	G:\SQLTemp\SOURCE	tempdb.mdf
templog	Log	Not Applica...	1	By 10 percent, Unlimited	G:\SQLTemp\SOURCE	templog.ldf

SQL Server Has 2 Types of Databases:

- A. **System Databases** --- Master, Model, MSDB, Tempdb, Resource DB -- 5 databases
- B. **User Databases** --- these are created as per business requirement.

Note: Resource database is in hidden status.

DB -- Container-- collection of files -- Min -- 2 files
-- Data File -- Used to save actual data
-- Log File -- used to hold transactions details.

TempDB -- 2 Files
-- Data File
-- Log file.

Additional data files for better performance of SQL Server.

Case-1:

How many data files are needed for Tempdb ?
No of tempdb data files = no of processors/2
= 8 core processors/2
= 4 files. (data)

TempDB -- min 2 files.

-- data file (mdf -- master data file) -- .mdf (extension).

-- log file (ldf -- log data file) -- .ldf (extension)

=====

4-1 = 3 (i need additional 3 data files for tempdb

-- data file (ndf -- next data file) -- .ndf (extension).

-- data file (ndf -- next data file) -- .ndf (extension).

-- data file (ndf -- next data file) -- .ndf (extension).

Case-2:

How many data files are needed for Tempdb ?

No of tempdb data files = no of processors/2

= 4 core processors/2

= 2 files. (data)

TempDB -- min 2 files.

-- data file (mdf -- master data file) -- .mdf (extension).

-- log file (ldf -- log data file) -- .ldf (extension)

=====

2-1 = 1 (i need additional 1 data files for tempdb

-- data file (ndf -- next data file) -- .ndf (extension).

Database -- collection of files.

Case-1: DB has only 2 files

Data file - mdf

Log file - ldf

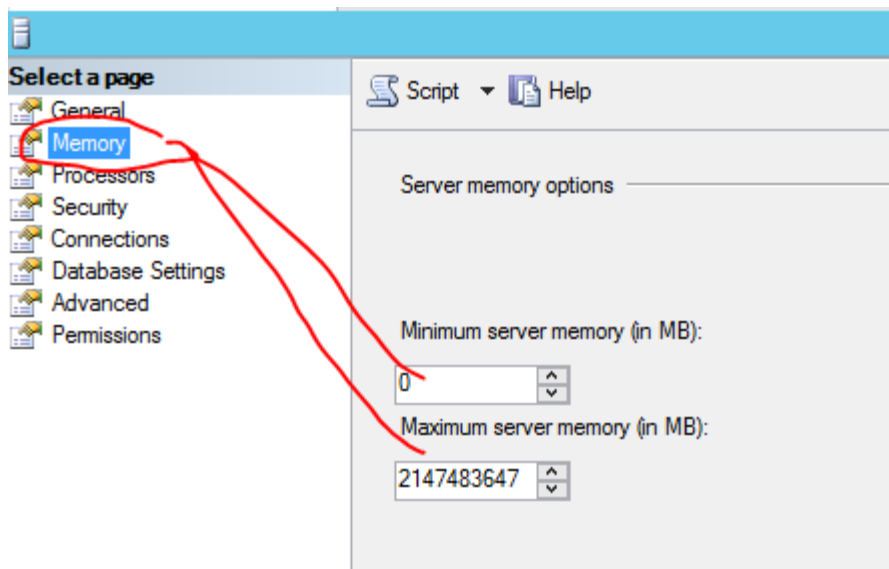
Case-2: DB has more than 2 files.

Data file - mdf

Data file (secondary) - ndf

Log file -- ldf

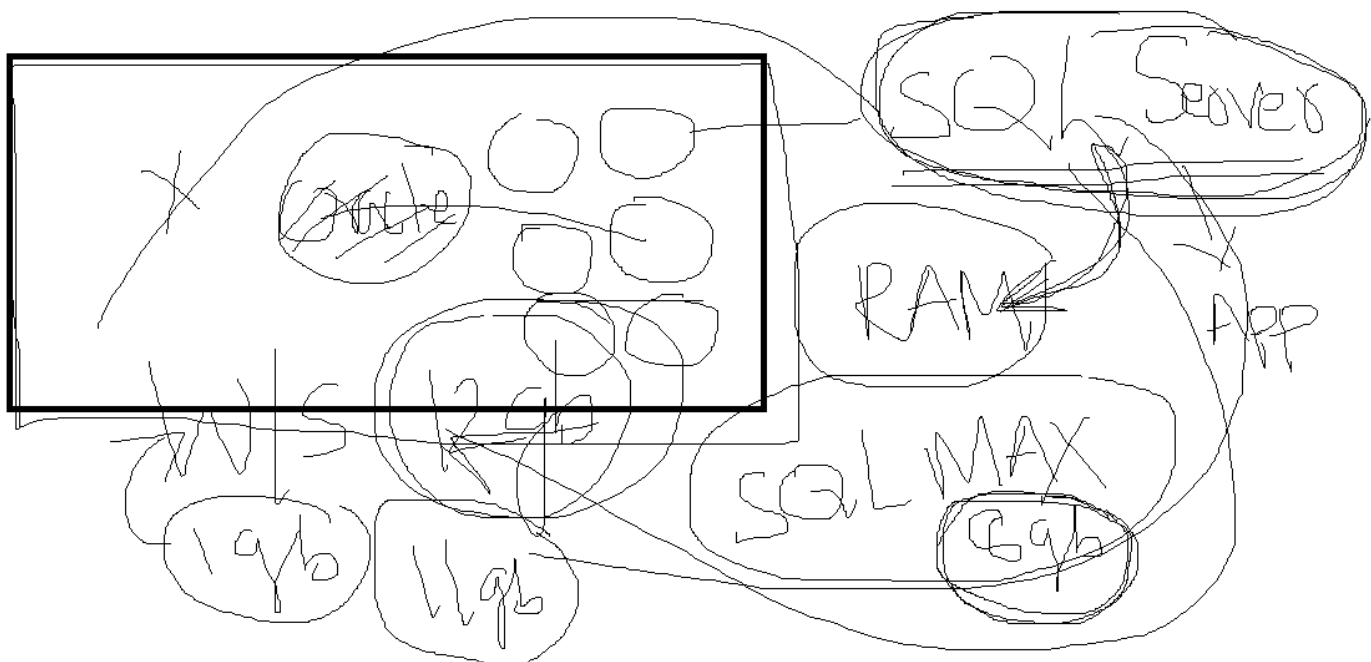
5. Configure SQL Server Max Memory and Min Memory.



Windows Server -- 12 GB

SQL Server Min:

SQL Server Max:



1	2.
3	4 SQL ①

12gb RAM

SQL MAX

$$= \frac{3}{4} \times \text{RAM Size}$$

$$= \frac{3}{4} \times \frac{3}{2}$$

$$= \frac{9}{8} \text{ RAM}$$

1	2. SQL 4.5 ② ③
3	4 4.5 SQL ①

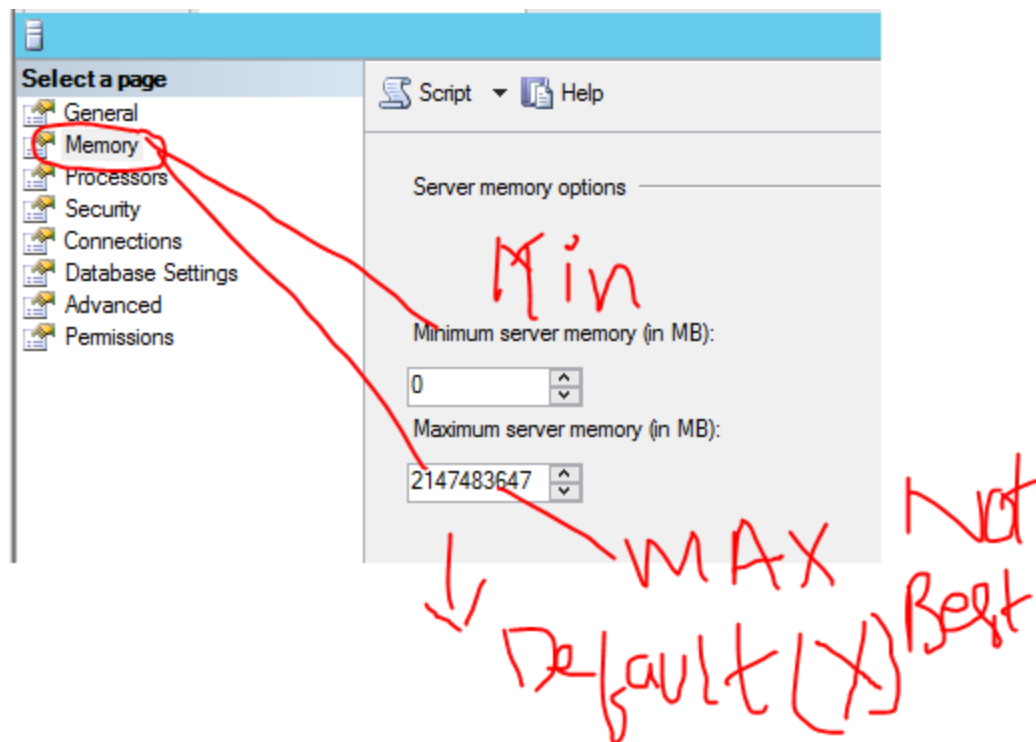
12gb RAM

SQL MAX

$$= \frac{3}{4} \times \text{RAM Size}$$

$$= \frac{3}{4} \times \frac{3}{2}$$

$$= \frac{9}{8} \text{ RAM}$$



Bit

Byte

KB - 1024 bites

MB - 1024 KB

GB - 1024 MB

TB - 1024 GB

Milligram

Gram

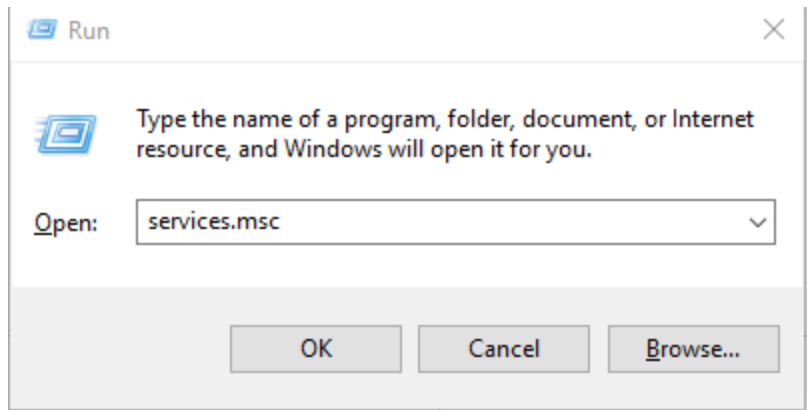
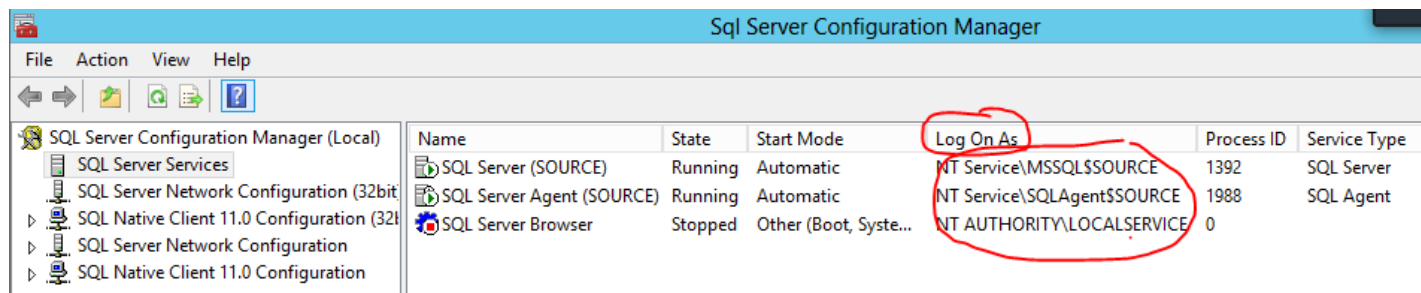
KG

Note: Whenever we configure SQL Max or min memory, There is NO need to go for SQL Service restart.

6. Configure SQL Services to run on a dedicated Service Account

SQL Server Software -- combination of components.

- DB Engine + SQL Agent -- Store Data, Retrieve Data, Process Data, Access to users.
- SSIS -- SQL Server Integration Services.-- create packages-- collection of tasks.
- SSRS -- SQL Server Reporting Services.-- creates Reports, View Reports
- SSAS -- SQL Server Analysis Services. -- Creates cubes related to Dataware House.
- Full Text Services --- Service
- SQL Server Browser -- Service.



Note: By default, all SQL Services when installed will run on Local Accounts.

Action: Change all SQL Service Accounts to a Dedicated Windows SQL Service Account.

Account -- Windows Account

--Example: WindowsMachineName\SQLService
 --wonderla\SQLService
 --Wonderla\SA-SQLService

Praveen --- Windows Account

--- Wonderla\praveen -- This is a dedicated windows account used by praveen.

Note: To view all services running at windows server level

Service Account:

1. Local Account
2. Domain Account

Note:

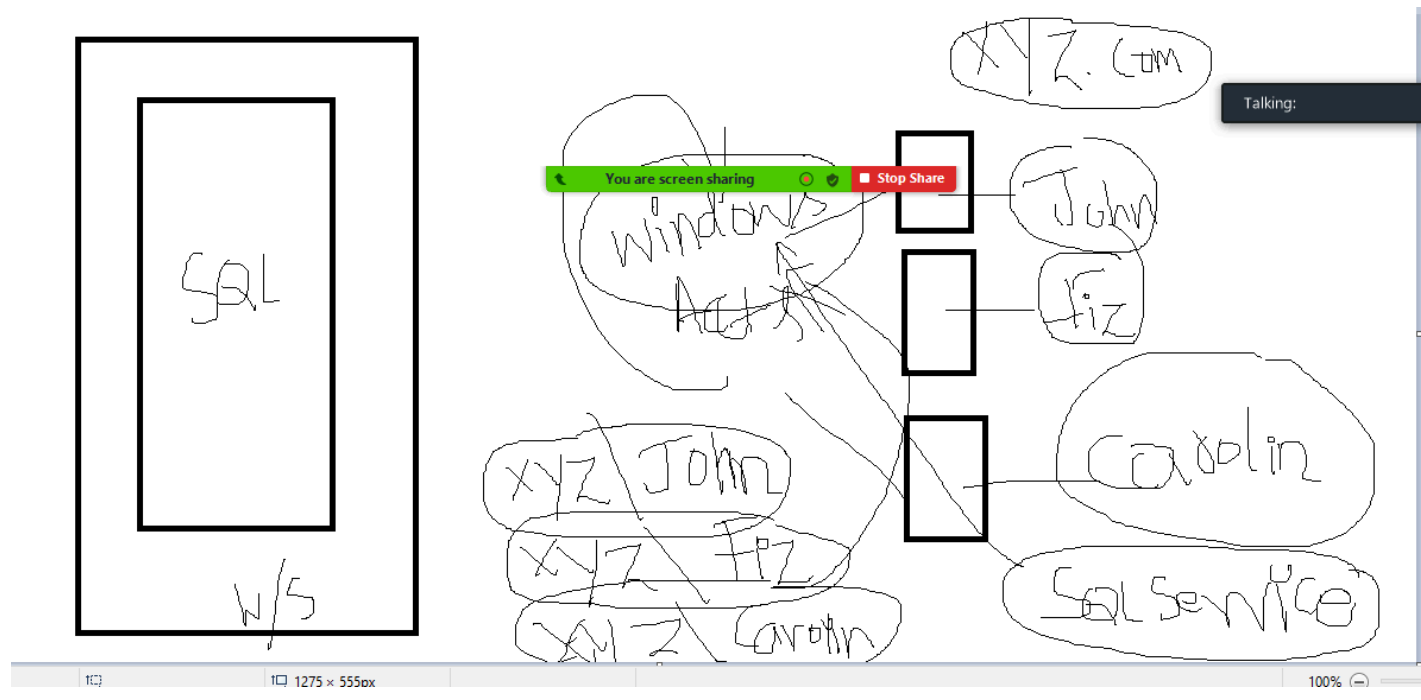
Windows Server -- Wonderla ---> DBA has to create a ticket requesting to create a Dedicated Windows SQL Service Account and assign this ticket to WINDOWS Server Team.

Request to Windows Team from DBA Team.

1. Please create a dedicated windows sql service account for default\named instance
2. Server name: Wonderla
3. Ticket no: 1234

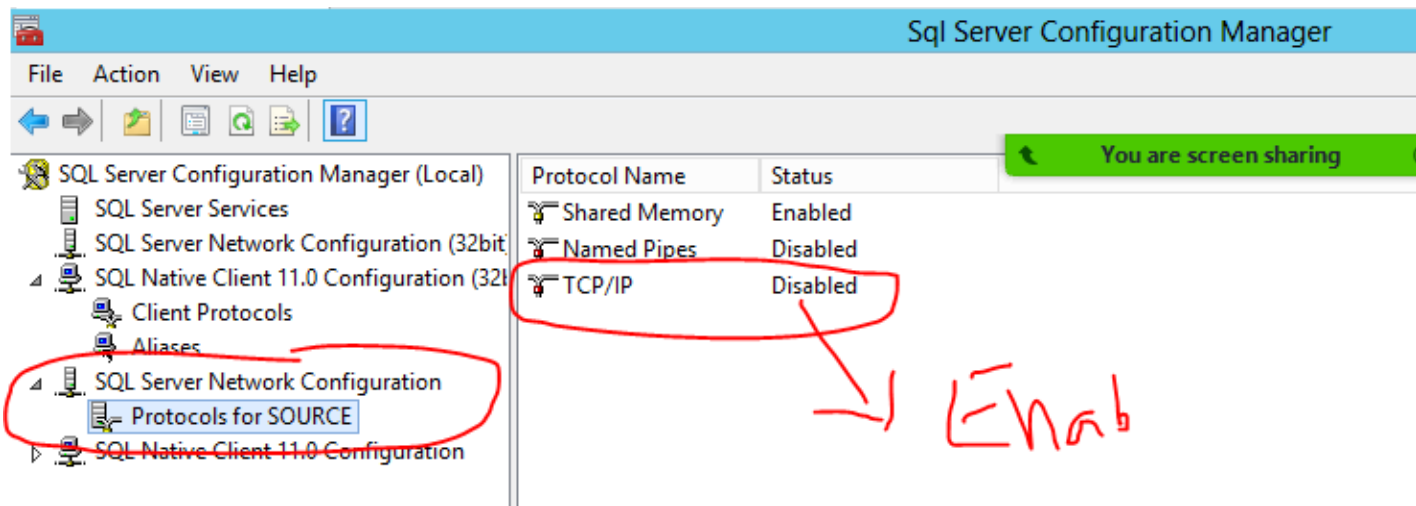
Windows Team:

1. Create Windows SQL Service Account: Wonderla\SQLService
2. This new account will never be given full permission on server.



7. Enable TCP/IP Protocol under SQL Server Configuration Manager if its in disabled status.

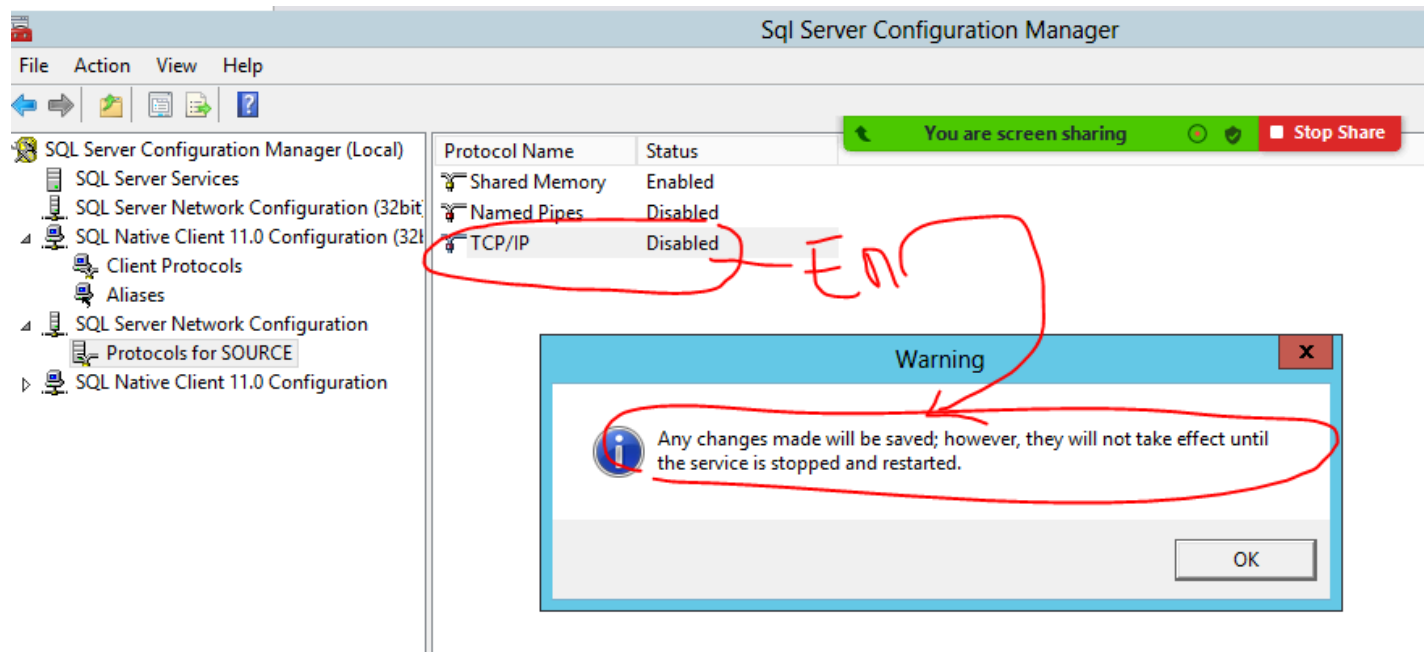
Zoom -- meeting id -- password



Protocols

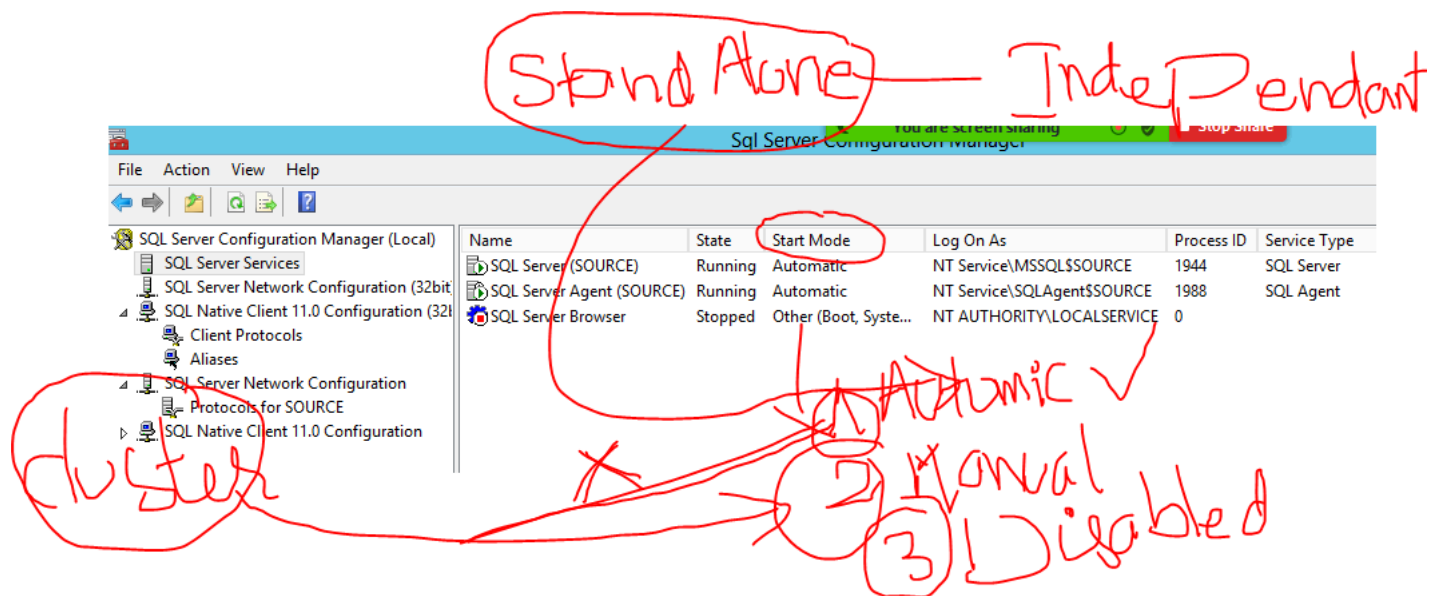
1. Shared Memory
2. Named Pipes
3. TCP/IP -- Transmitted Control Protocol. -- Default SQL Instance -- 1433 TCP Port no
Named instance -- Port no will be different -- 45678, 34567.

Enable it and then for sure Restart/RECYCLE/BOUNCE SQL Services of that particular SQL Engine.



Note: TCP/IP -- Protocol no: Default -- 1433 ---> 45678 given Network Admin

Note: I installed Named Instance: SOURCE → 34513 → Network Admin

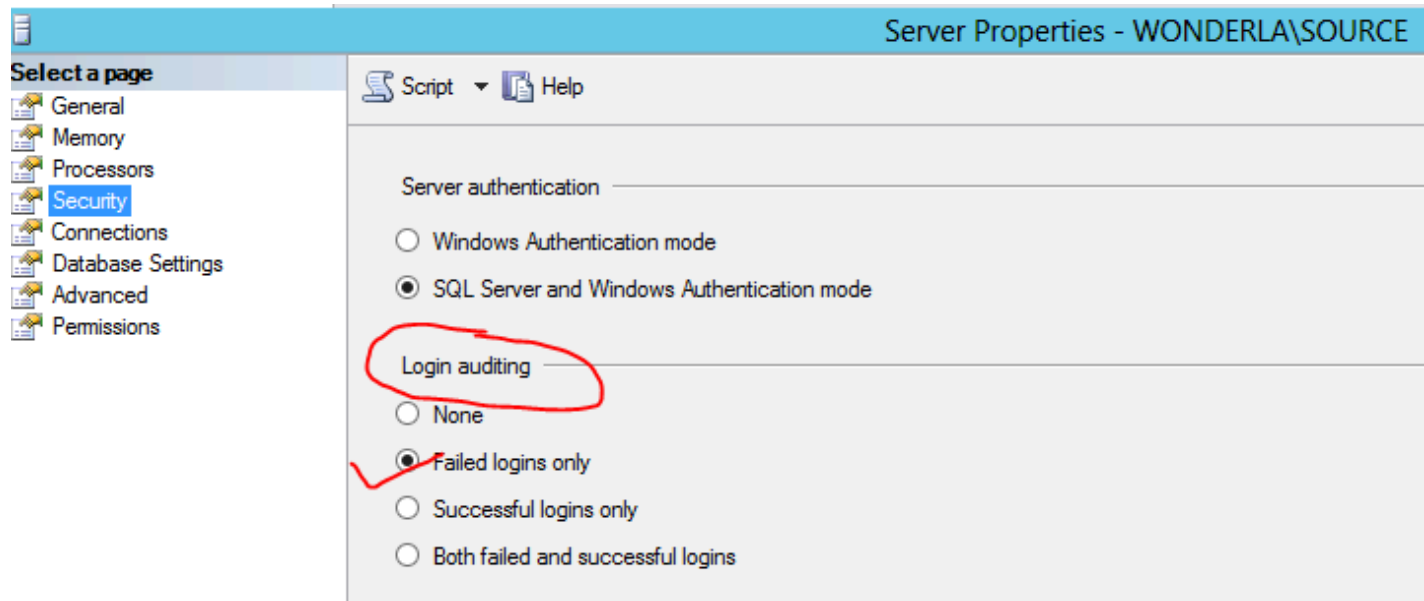


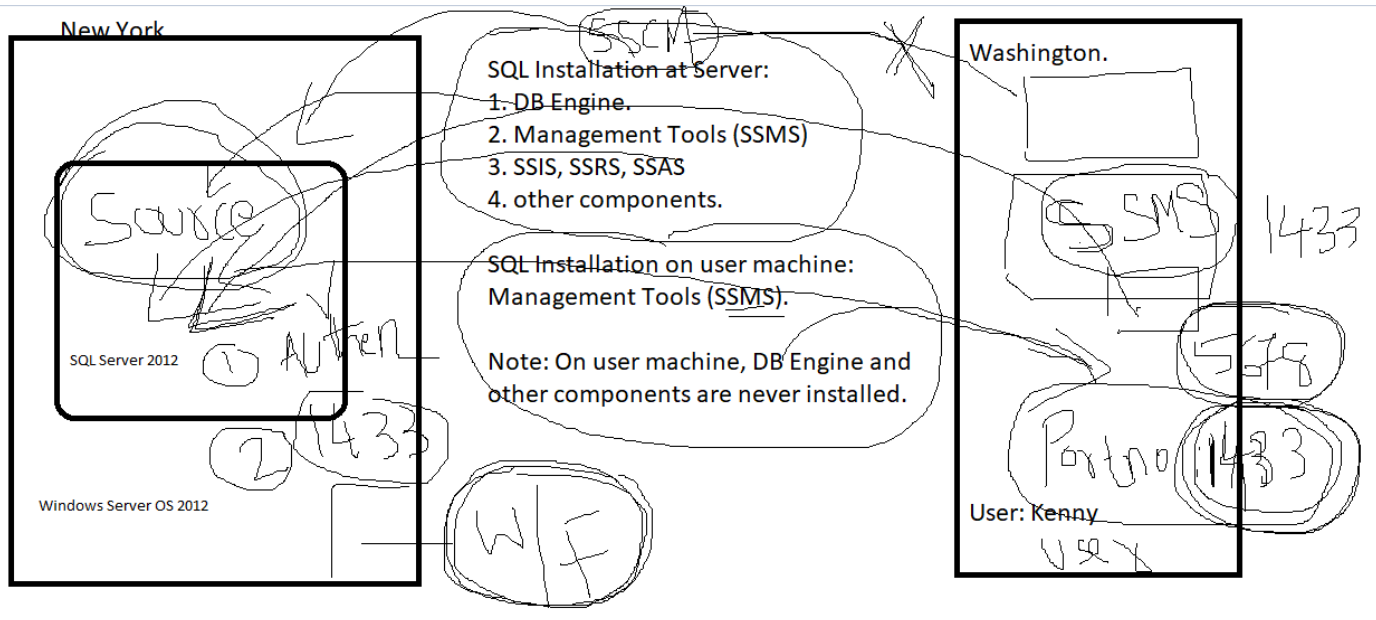
Under SSCM --- SQL Services Start Mode -- Automatic, Manual & Disabled.
 -- State -- Start, stop, Pause, Resume, Restart

9. Disable unwanted SQL Services.

- g. Full Text Services
- h. SQL Server Browser

10. Configure SQL Server Login Auditing





Few Options more to enable....Tomorrow.

Connection to a SQL Server instance

The **server/instance** name syntax used in the **server** option is the same for all SQL Server connection strings.

```
Server=myServerName\myInstanceName; Database=myDataBase; User Id=myUsername; Password=myPassword;
```

SQL Server 2019

SQL Server 2017

SQL Server 2016

SQL Server 2014

SQL Server 2012

SQL Server 2008

SQL Server 2005

SQL Server 2000

SQL Server 7.0

Session-16--September 10, 2020 - Thursday.

SQL Server 2005 till 2014 -- SSMS is inbuilt.

SQL Server 2012 -- Installation

SQL Server 2016 -- SSMS is not inbuilt.

-- SSMS download separately and install.

Note: Few Org:

SQL Server 2000 -- Old Version -- 8.0 -- Enterprise Manager

SQL Server 2005 till 2019 -- SSMS

SQL Server 2005 to 2014 -- SSMS is in built

SQL Server 2016 -- Need to install SSMS separately.

Session-17--September 11, 2020 - Friday.

=====

Lab Practice -- We would be an all rounder
-- Build VM Workstation.
-- Creating Windows OS/ Building/Install
-- Adding required Disks/drives
-- Network Team will assign ip address to server
-- SQL DBA - Installs SQL Server

Real Time -- Once everything is ready then finally it would be handed over to SQL DBA Team.

Servers -- Physical Server or Virtual Machine -- windows team.

Storage -- physical or virtual storage -- Storage Team.

IP Address -- Each Server is assigned a unique ip address by N/w Team.

=====

SQL DBA Team --- we only install SQL Software.

Drives -- SQL DBA Team will request initially some standard drives config.

=====

C:\ -- 100 GB

B:\ -- SQL Binaries -- 50 GB

E:\ -- SQL Data Files -- 200 GB

F:\ -- SQL Log Files -- 100 GB

H:\ -- SQL Backups -- 200 GB

T:\ -- SQL Tempdb -- 100 GB

=====

1 TB = 1024 GB

Virtual Machine:

Max Size for single Drive -- 2 TB.

Purchased: 2 TB of hard disk.

Server build: 1 TB of space for all drives. (1 TB is remaining)

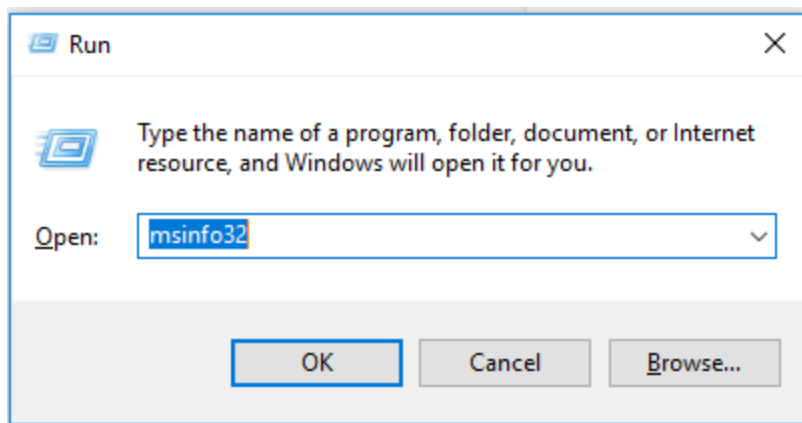
Server: Amazon.xyz.com

C:\ 100 GB

E:\ 500 GB (Container with blank data) - insert data --- 50 GB every day

Note: To Know your server internal details:

Press Windows + R == Open Run prompt window and type: msinfo32

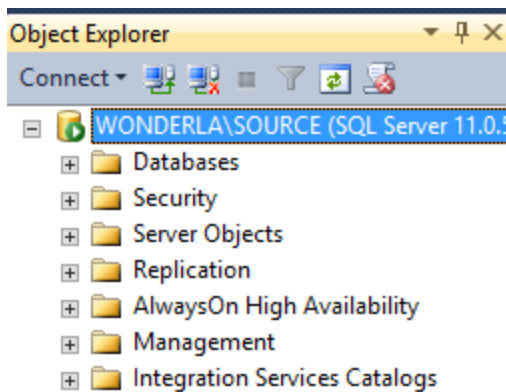


Else open powershell window and type: msinfo32

QA: what is default & named SQL Instances ?

Below Screenshot refers:

Named SQL Instance -- Machinename\SQLInsName



Default SQL Instance:

- + Databases
- + Security
- + Server Objects
- + Replication
- + AlwaysOn High Availability
- + Management
- + Integration Services Catalogs

Named SQL Instance:

- + Databases
- + Security
- + Server Objects
- + Replication
- + AlwaysOn High Availability
- + Management
- + Integration Services Catalogs

Case-1: Consistency -- Speed is compromised

Case-2: More Speed -- Consistency is compromised.

Windows Server 2012 -- SQL Server Software can be installed 50 times Max.

SQL Server 2000 -- only 16 times can be installed on Windows server OS.

-- 1 -- default instance -- DBA will not give SQL Name

-- MSSQLServer (Automatically)

-- 15 -- named instances -- DBA has to give unique names

Machine name: Amazon

Default SQL Instance: Amazon (SSMS)

MSSQLServer (SQL Server Configuration manager)

Named Instances:

Amazon\Source1

Amazon\Source2

.....

.....

Amazon\Source15

=====

SQL Server 2005 onwards

-- only 50 times can be installed on Windows server OS.

-- 1 -- default instance -- DBA will not give SQL Name

-- MSSQLServer (Automatically)

-- 49 -- named instances -- DBA has to give unique names

Machine name: Amazon

Default SQL Instance: Amazon (SSMS)

MSSQLServer (SQL Server Configuration manager)

Named Instances:

Amazon\Source1

Amazon\Source2

.....

.....

Amazon\Source49

Windows Server -- 4 GB RAM -- 500 GB Hard Disk.

2 SQL Servers

Windows Server -- 256 GB RAM -- 8 TB GB Hard Disk.

10 SQL Servers

Windows Server -- 64 GB RAM -- 2 TB GB Hard Disk.

10 SQL Servers

Note:

Default -- Here there is no need to specify Any application name.

Named -- if you are specific that your Names SQL Instance should reflect business names then choose proper names.

Example: MachineName\SQLMobileApp

Servername\Washington

Windows Server -- Amazon.xyz.com

Amazon\SQLDev

Amazon\SQLTest

Session-18--September 15, 2020 - Tuesday.

=====

TempDB files configuration

SQL Server Min & Max Memory.

Note: SQL Server Restart/Recycle/Bounce

1. Never do this during business time - 8 AM to 4 PM EST
2. If required to restart:
 - a. Take application team or SQL Server Owner approvals.
 - b. Also request for scheduled downtime.
 - c. Ticket ?

Information

Tickets

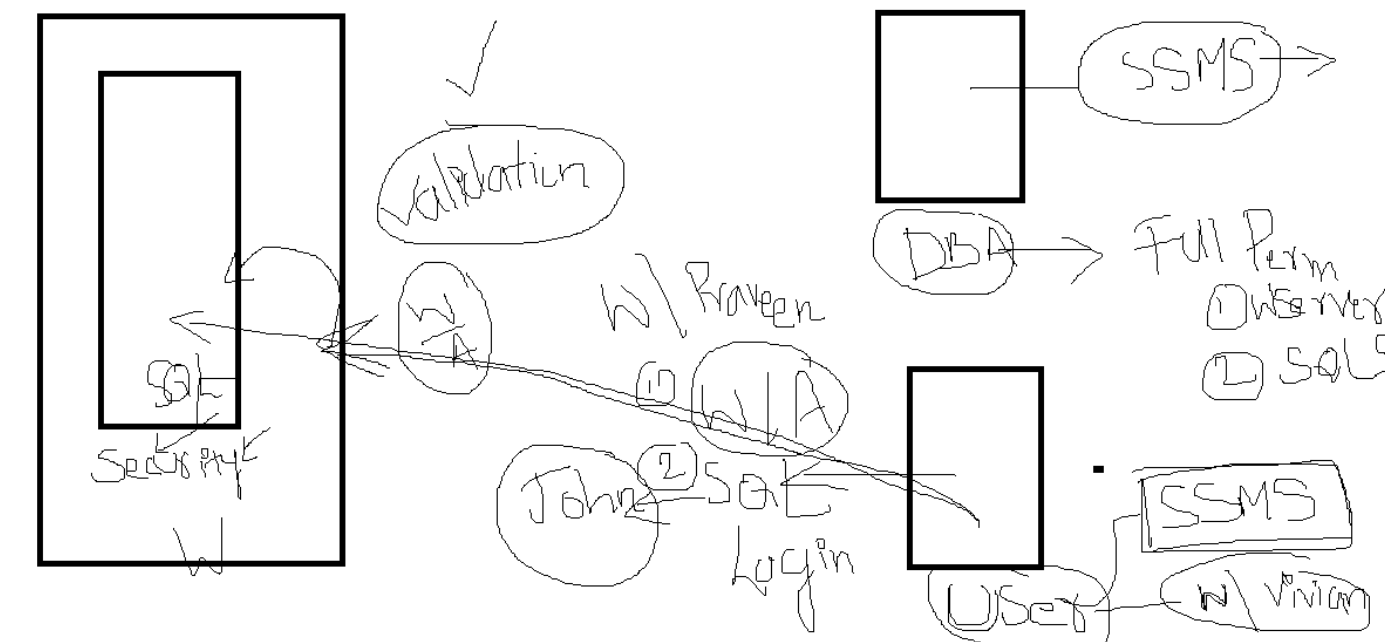
Schedule time.

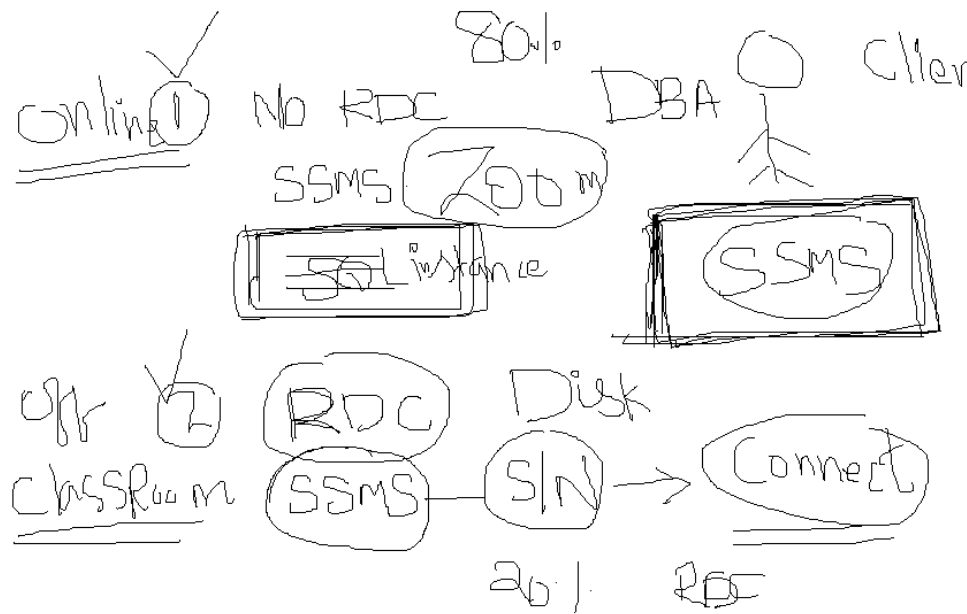
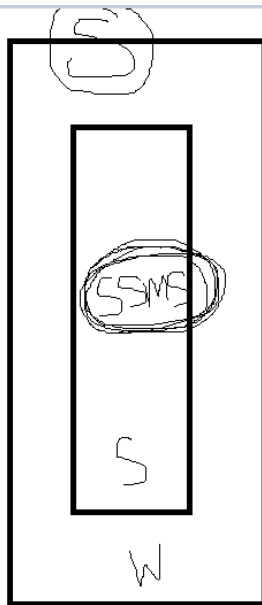
Approvals

Implementation.

Session-19--September 18, 2020 - Friday.

=====





Session-20--September 21, 2020 - Monday.

DBA uses mostly 2 tools of SQL Server.

1. SSMS -- SQL Server Management Studio -- 95 %
2. SSCM -- SQL Server Configuration manager -- 5 %

SSMS -- Connect SQL Server

Create DBs, Create tables, Create SP, etc.

Create logins, Create DB users, Give Permissions

SSCM -- Dashboard to administer SQL Services

- Stop SQL Services
- Restart SQL Services
- Change SQL Service Accounts
- Change TCP IP Protocols
- etc.

Note:

Under SSCM -- Few settings if we modify then SQL Services need to be restarted and for few it is not needed.

Session-21--September 22, 2020 - Tuesday.

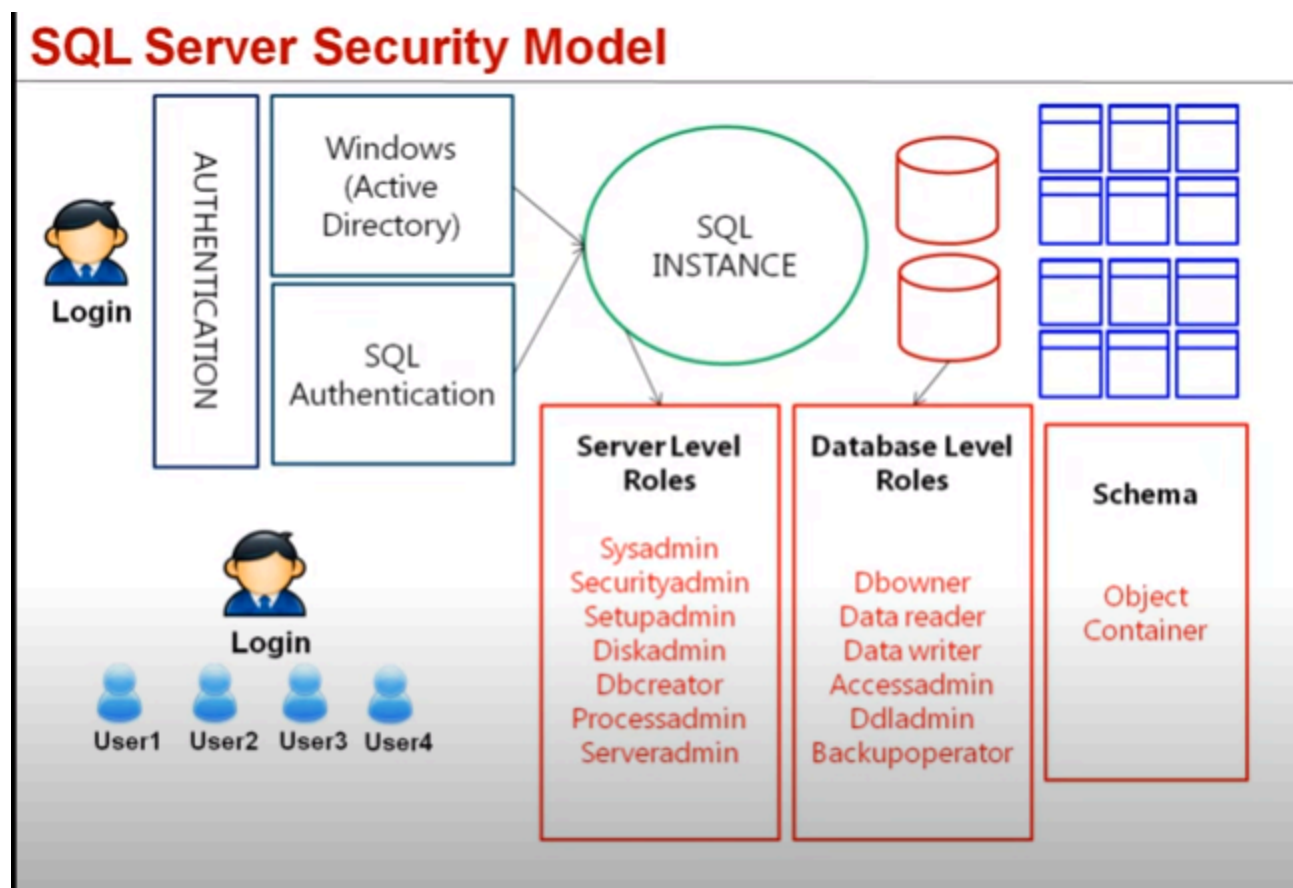
=====

Module - 2: SQL Server Security:

1. Windows Server Level -- Permission to take RDC
2. SQL Server Level -- Permission to access SQL Server (Logins)
3. DB Level -- Permission to access DB's (DB Users)
4. Object Level -- Permission to individual objects.
-- Tables, SP's, views, etc.

SQL Server Security:

https://www.youtube.com/watch?v=uQf4CbX3iI4&list=PLviL2Q5l6zXpvEAYlk8IzPzdjW_5swPDj&index=3



During SQL Server Installation:

2 Types of Authentication:

1. Windows Auth -- Only windows accounts will be able to access SQL
2. Mixed Mode (WIndows + SQL Server) -- Both windows & SQL Logins

Example:

Windows Account/Domain Windows Account → Wonderla\PraveenM
→ Wonderla\Vivian

SQL Server Login → WalidUSA
→ NaoimiUSA

Servers will be of 2 categories.

1. STAND ALONE SYSTEM -- Orphan Child -- Independent

Computer name, domain, and workgroup settings

Computer name: Wonderla

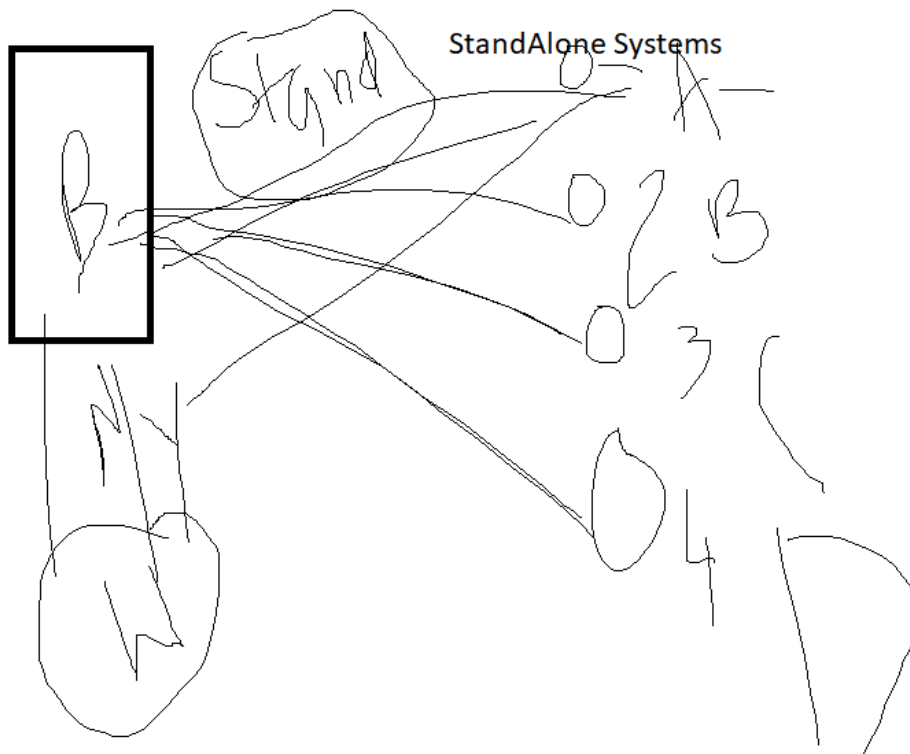
Full computer name: Wonderla

Computer description:

Workgroup: WORKGROUP

Users can access to this server when they have individual accounts.

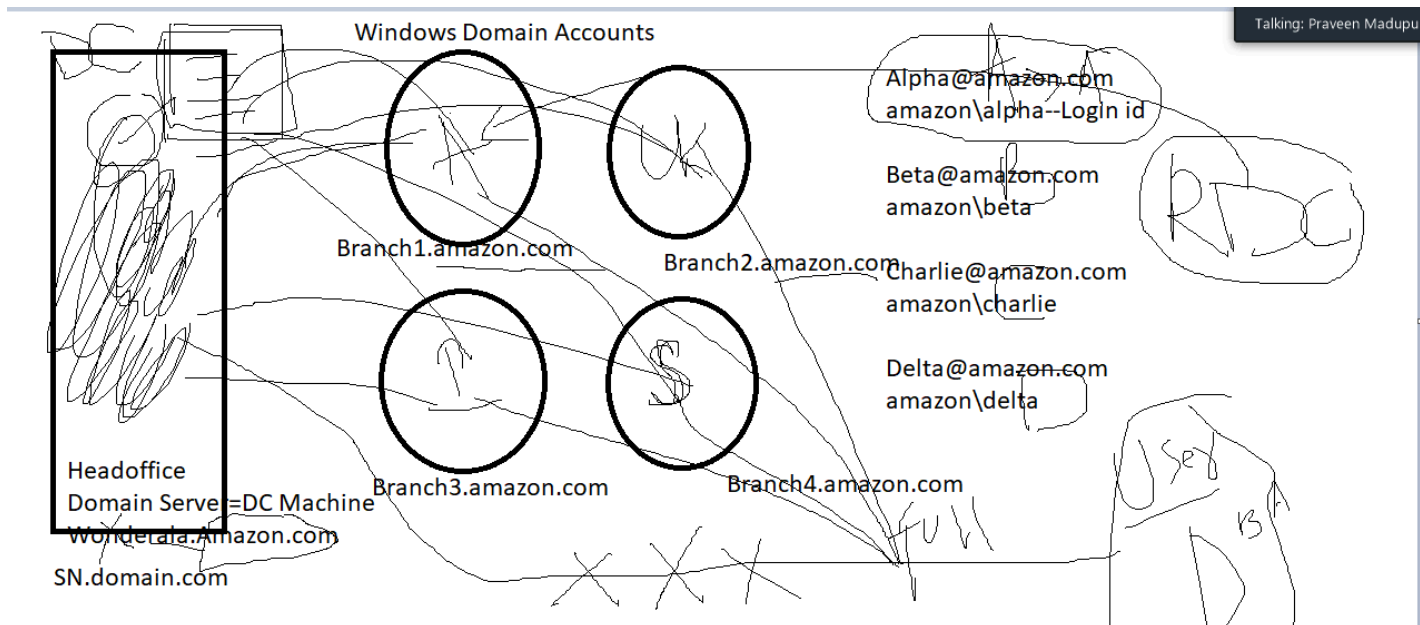
Users need to access to this system physically (Present on the system)



Windows Accounts coming from Standalone system:

Wonderla\Alpha
Wonderla\Beta
Wonderla\Charlie
Wonderla\Delta

2. Member of Domain Server -- Child with Surname (having Parents)



Agenda:

Creating Windows Logins & SQL Logins

Security Guidelines -- Follows -- Ticket -- Approvals

Session-22--September 24, 2020 - Thursday.

=====

- Windows Server Level -- Permission to take RDC
- SQL Server Level -- Permission to access SQL Server (Logins)

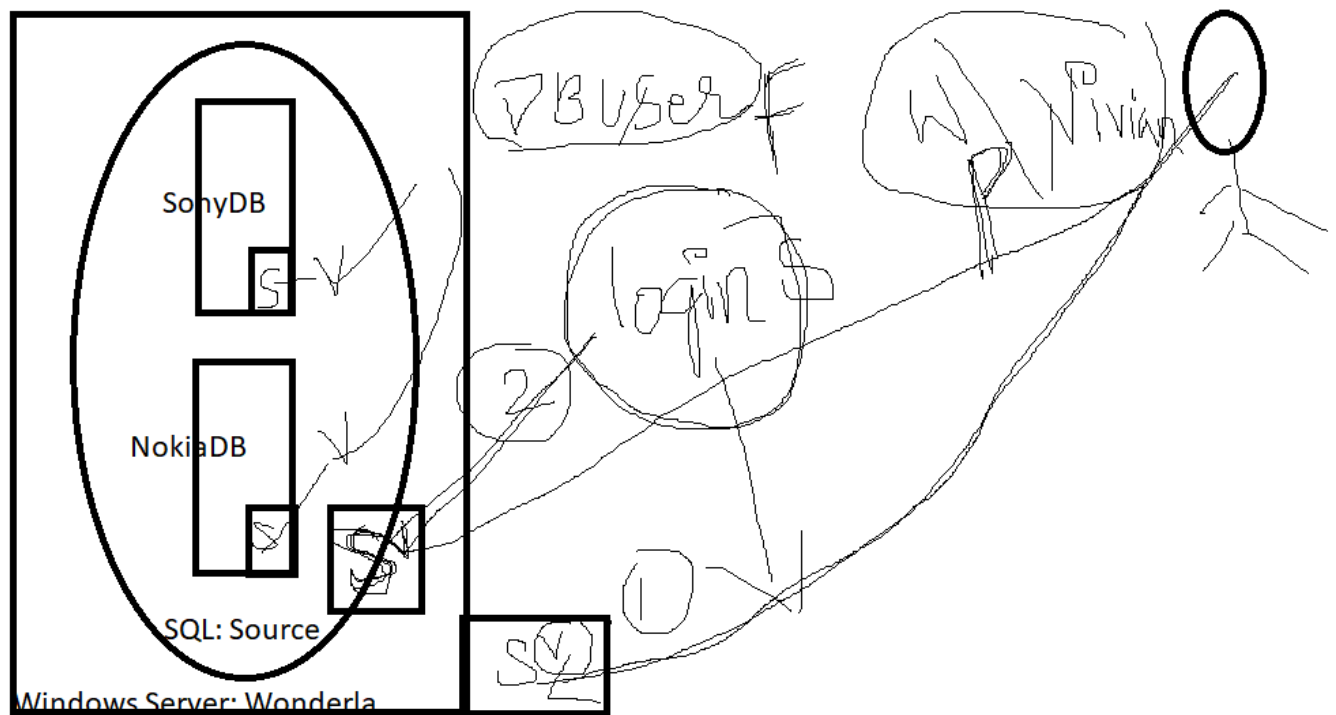
Logins --- To Access SQL Server

- DB Level -- Permission to access DB's (DB Users)
- Object Level -- Permission to individual objects.
-- Tables, SP's, views, etc.

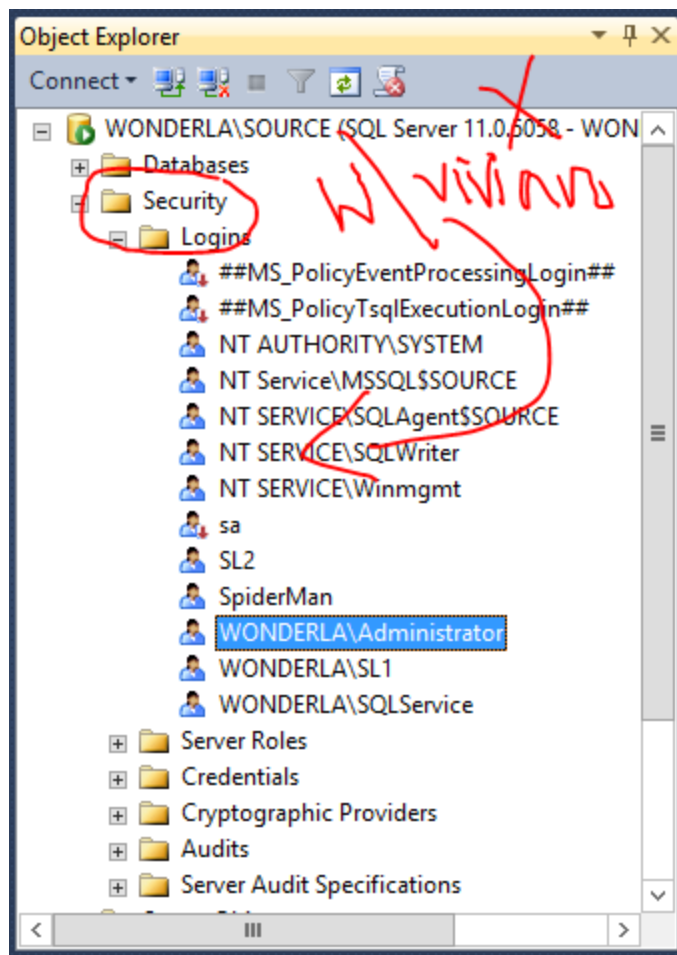
Example: Login ---> Wonderla\Vivian

DB User -- To Access Databases

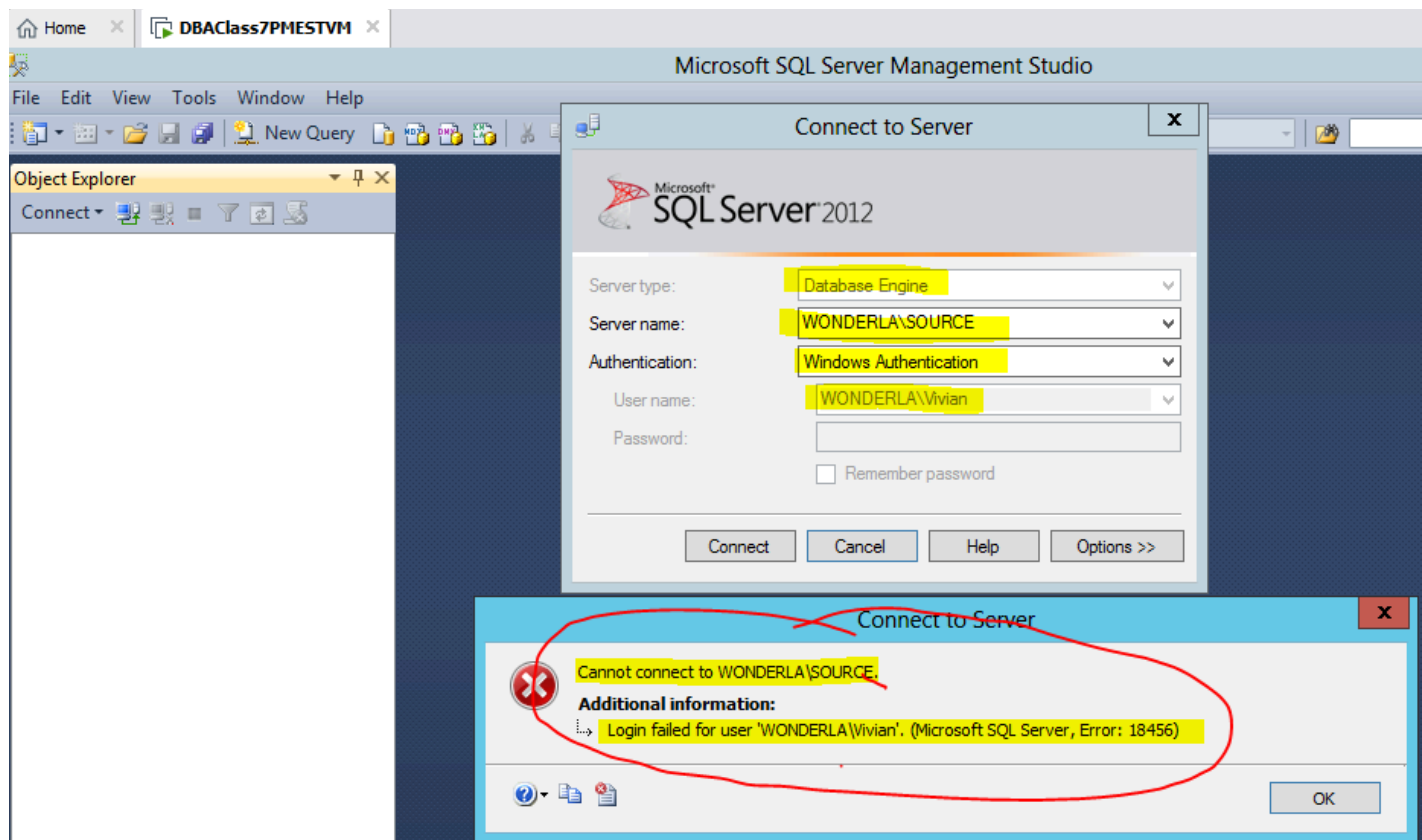
Example: DB User → Wonderla\Vivian



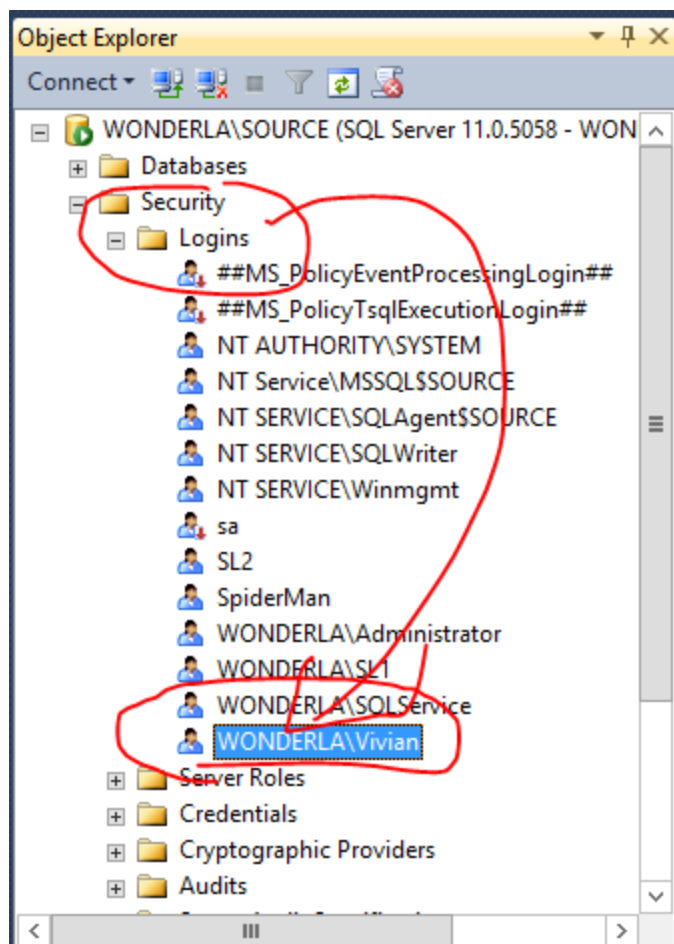
Scenario: Vivian is not granted SQL Server Access.



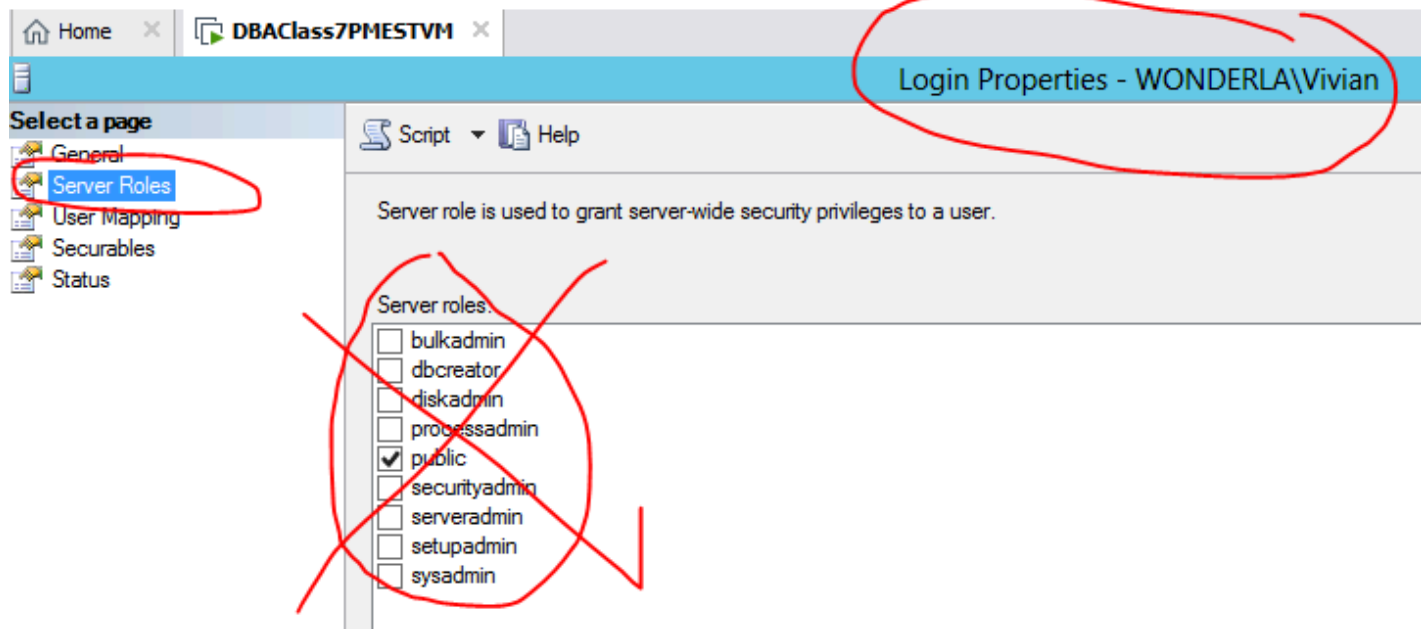
Login → Security → SQL Server Level



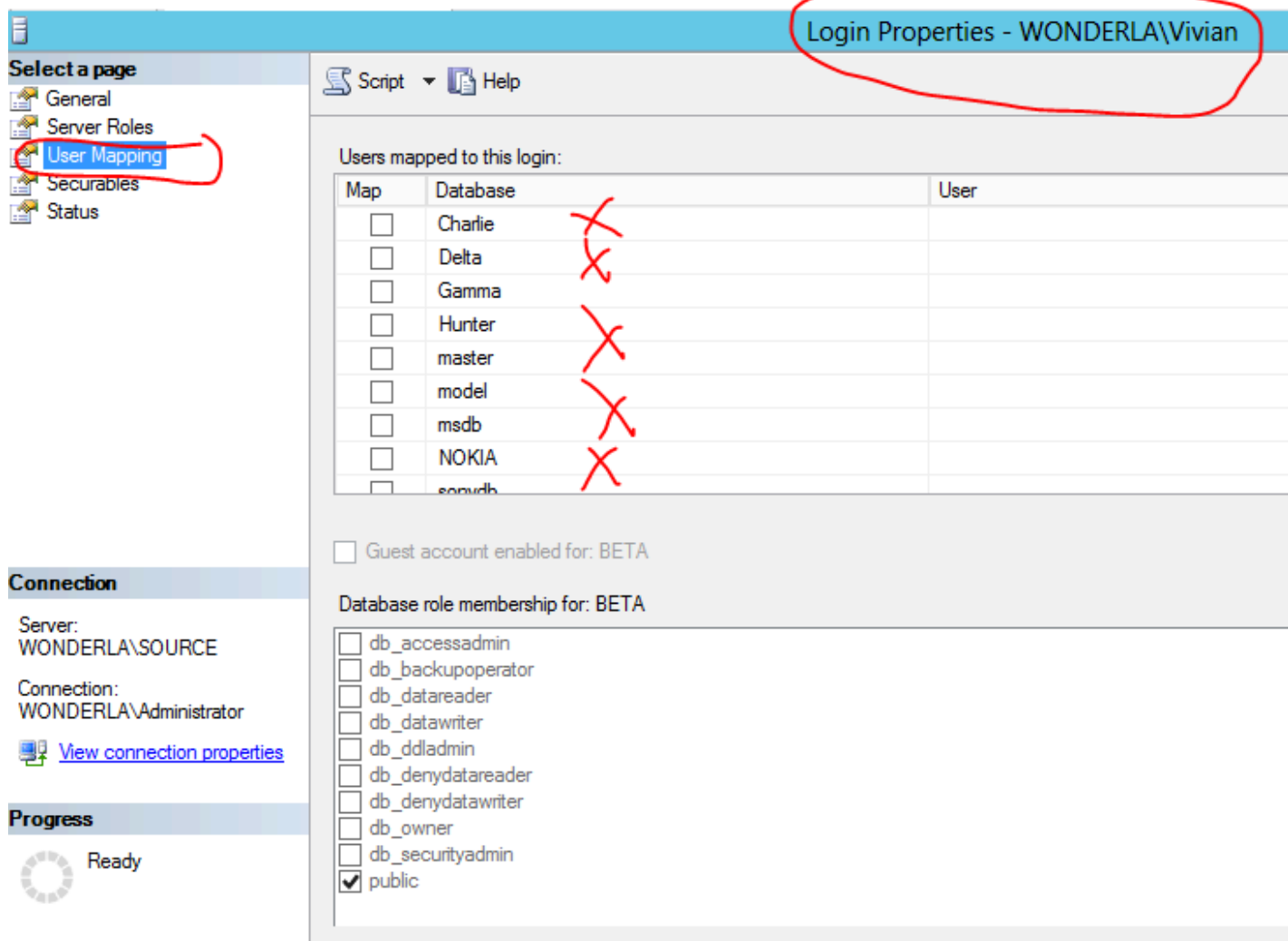
Scenario: Vivian is granted only SQL Server Access. (NOT DB Access)

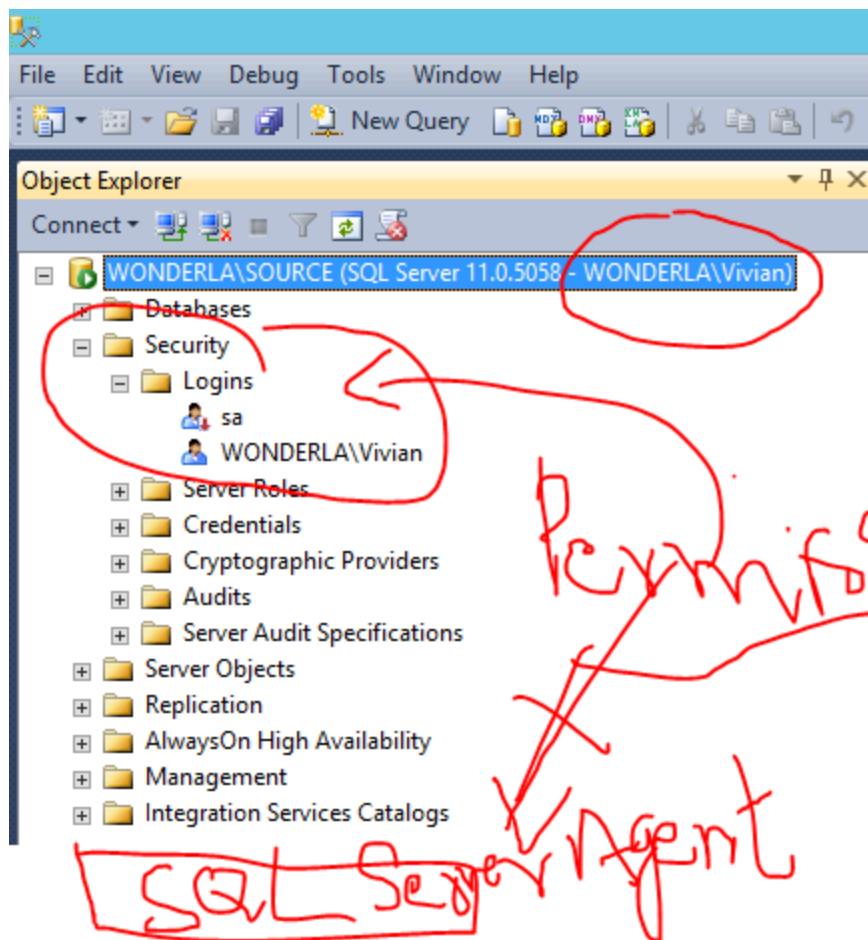


Server Roles:

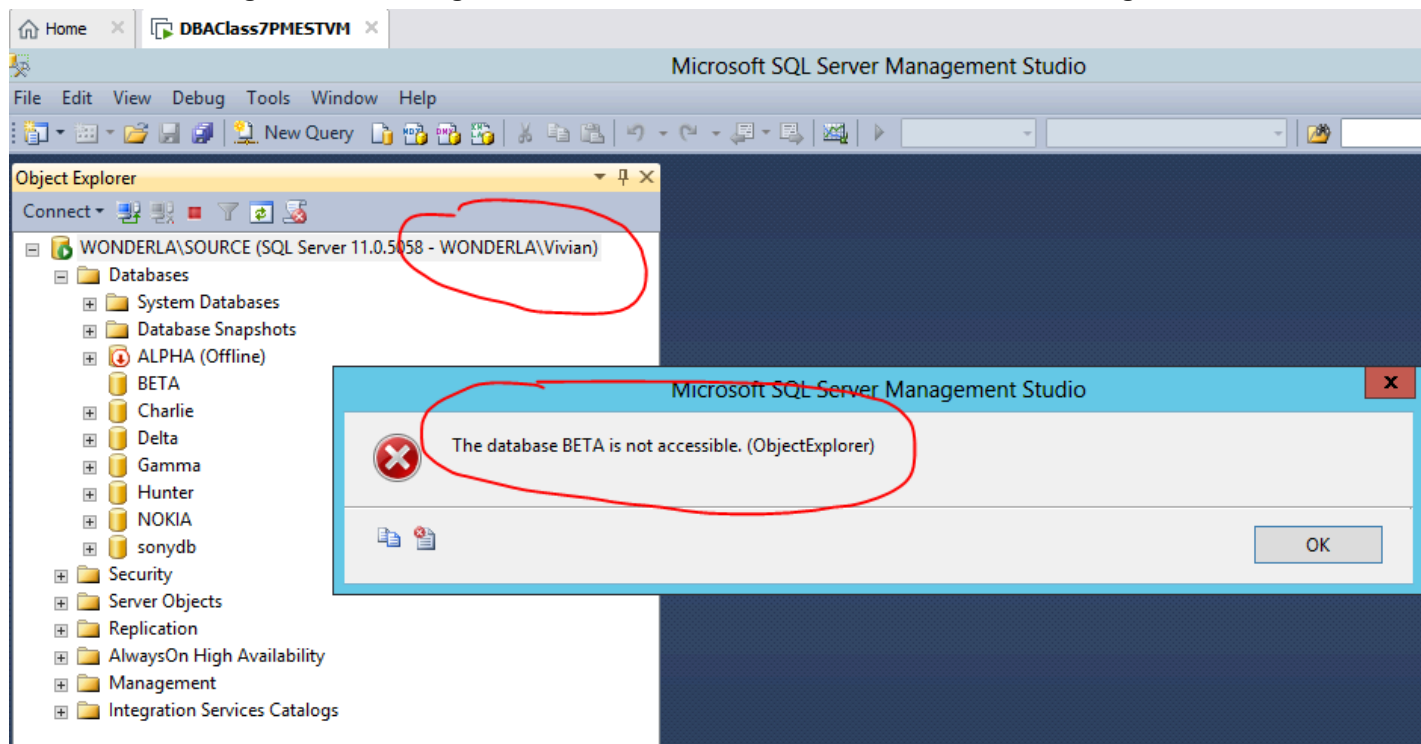


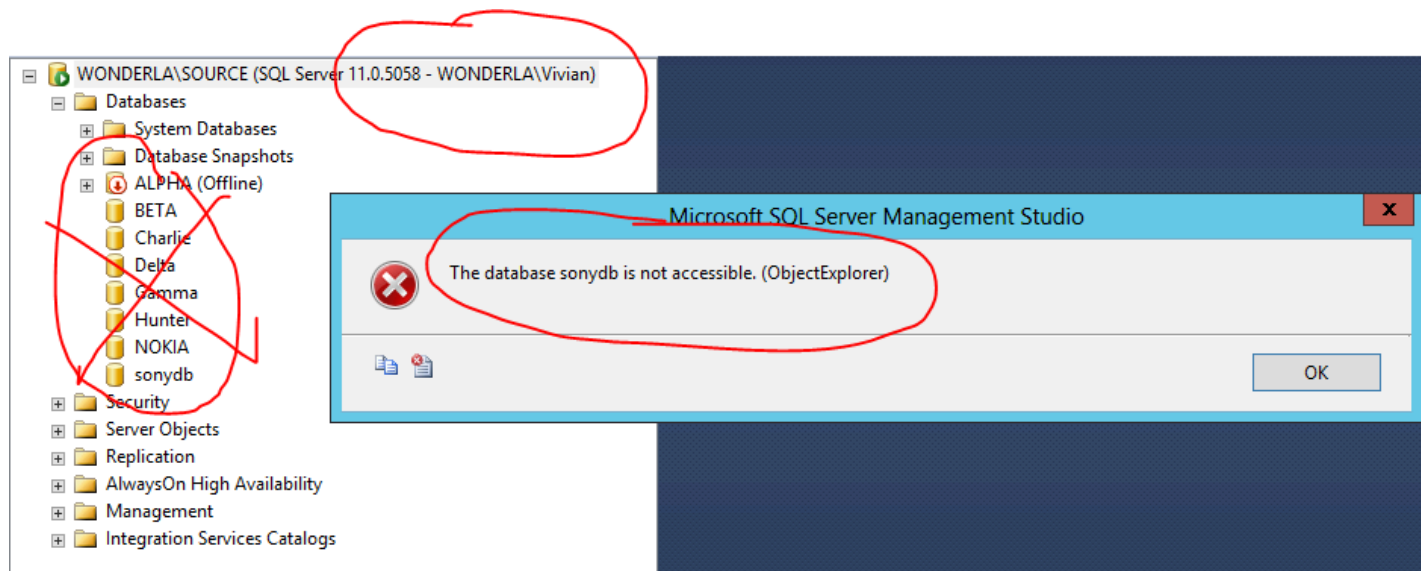
DB Roles:





Note: If a Particular Login (Windows or SQL Login) doesn't see all logins under security or NOT able to view SQL Server Agent then that login is a NORMAL LOGIN and NOT a Administrator Login.





Windows Server Access.

SQL Server Access

DB Access to Logins

Object Level

Session-23--September 28, 2020 - Monday.

=====

Note: Vivian (Windows Login) was granted below Access.

1. **Windows Server**
2. **SQL Server Access**

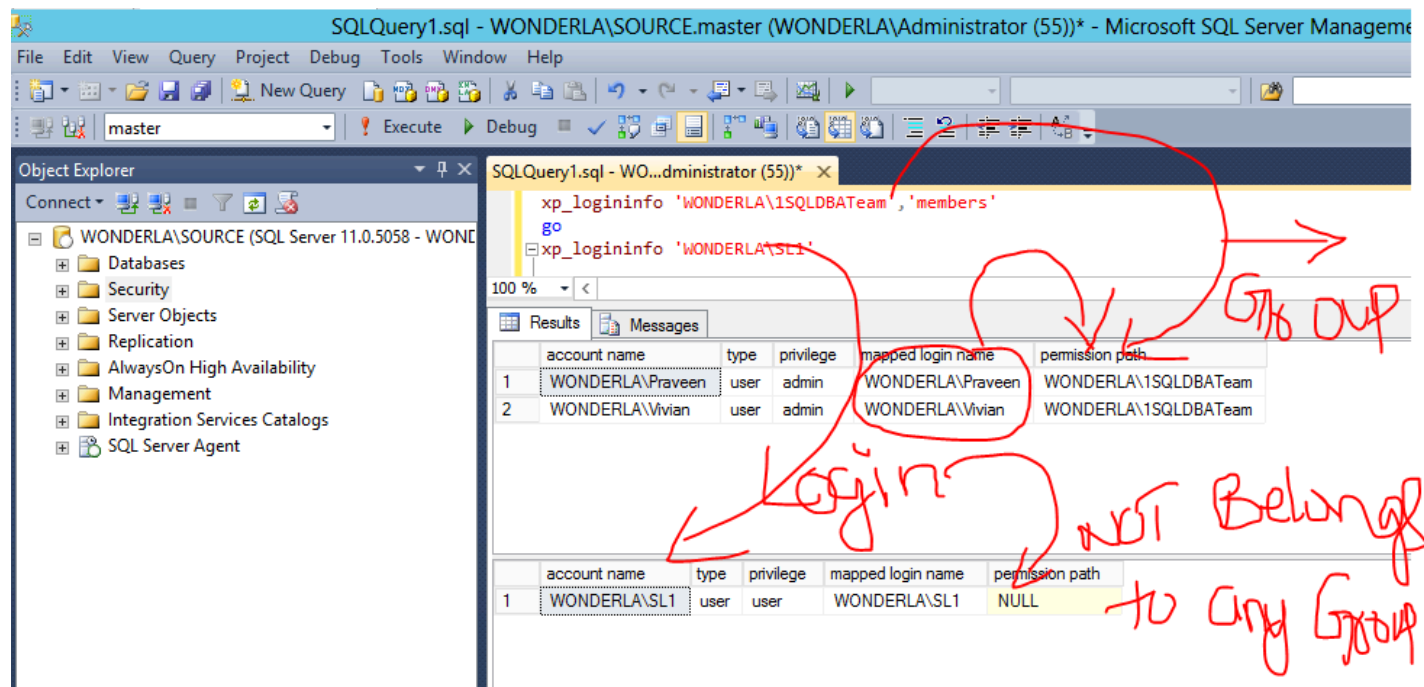
DB Access → Not Granted to Vivian. This was the reason she was getting above error message.

SP_HELPDB --This fetches all DB basic information

xp_logininfo 'WONDERLA\SQLDBATeam','members' --Groups Details

go

xp_logininfo 'WONDERLA\SL1' -- To fetch individual login details.



Session-24--September 29, 2020 - Tuesday.

HeadOffice (Domain Controller - DC)

All Windows Group Accounts

All Domain Group Accounts

All Windows individual Accounts

Example: Wonderla1SQLDBTeam ----> Add JohnWalter (WL)

Windows Servers List:

Wonderala1 - SQLSource1-Wonderla\1SQLDBTeam
 Wonderala2 - SQLSource2-Wonderla\1SQLDBTeam
 Wonderala3 - SQLSource3-Wonderla\1SQLDBTeam
 Wonderala4 - SQLSource4-Wonderla\1SQLDBTeam
 Wonderala5 - SQLSource5-Wonderla\1SQLDBTeam

GUI -- Graphical User Interface

TSQL - Transact Structured Query Language.

USE [master]

GO

/***** Object: Login [WONDERLA\1SQLDBTeam] Script Date: 9/29/2020 4:29:57 PM *****/

CREATE LOGIN [WONDERLA\1SQLDBTeam] FROM WINDOWS WITH DEFAULT_DATABASE=[master],
 DEFAULT_LANGUAGE=[us_english]

GO

The screenshot shows the 'Login Properties' dialog for the 'SpiderMan' login. The 'Server Roles' tab is active, displaying a list of server roles. The 'sysadmin' role is currently unchecked, which is highlighted by a red circle and a red arrow pointing to it with the handwritten text 'unchecking'. Other roles like 'public' and 'securityadmin' are checked. The background shows the SQL Server Enterprise Manager interface with the 'Object Explorer' pane on the left and the 'SQL Query' window at the top.

XP_Logininfo -- Extended System Stored Procedure

The screenshot shows the Microsoft SQL Server Enterprise Manager interface. The title bar indicates the connection is to 'SQLQuery1.sql - WONDERLA\SOURCE.master (WONDERLA\Administrator (55))' in 'Microsoft SQL Server Management Studio (Administrator)'. The 'Object Explorer' on the left shows the server structure, including 'WONDERLA\SOURCE (SQL Server 11.0.5)' with databases, security, and server objects. The 'Query Editor' displays a query with two lines: 'xp_logininfo 'WONDERLA\SQLDBATeam', 'members'' and 'xp_logininfo 'WONDERLA\Johnwalter''. The 'Results' pane shows two tables. The first table has columns: 'account name', 'type', 'privilege', 'mapped login name', 'permission path'. It contains two rows: 'WONDERLA\JohnWalter' and 'WONDERLA\Praveen'. The second table has columns: 'account name', 'type', 'privilege', 'mapped login name', 'permission path'. It contains one row: 'WONDERLA\Johnwalter'. Red circles and arrows highlight the 'mapped_login_name' and 'permission path' columns in both tables. The status bar at the bottom indicates 'Query executed successfully.' and '24 rows'.

account name	type	privilege	mapped login name	permission path
WONDERLA\JohnWalter	user	admin	WONDERLA\JohnWalter	WONDERLA\SQLDBATeam
WONDERLA\Praveen	user	admin	WONDERLA\Praveen	WONDERLA\SQLDBATeam

account name	type	privilege	mapped login name	permission path
WONDERLA\Johnwalter	user	admin	WONDERLA\Johnwalter	WONDERLA\SQLDBATeam

The screenshot shows the SQL Query window with the following query executed:

```
--go  
select name,password,createdate,sid from sys.syslogins -- system view table
```

The Results pane displays the output of the query as a table with four columns: name, password, createdate, and sid.

	name	password	createdate	sic
8	WONDERLA\Administrator	NULL	2020-09-05 20:07:26.687	0x0105000000000005150000006C163490ED706950A8175426...
9	NT SERVICE\SQLWriter	NULL	2020-09-05 20:07:26.733	0x010600000000000550000000732B975364EF90356745CB...
10	NT SERVICE\Winmgmt	NULL	2020-09-05 20:07:26.750	0x010600000000000550000000A048DFF9C7430AB450DAE...
11	NT Service\MSSQL\$SOURCE	NULL	2020-09-05 20:07:26.777	0x010600000000000550000000766EDE76DD2A0E5E741B...
12	NT AUTHORITY\SYSTEM	NULL	2020-09-05 20:07:26.793	0x010100000000000512000000
13	NT SERVICE\SQLAgent\$SOURCE	NULL	2020-09-05 20:07:28.507	0x010600000000000550000000328B9177675F0BE9B53B9DA...
14	##MS_PolicyEventProcessingLogin##	(被加密)	2020-09-05 20:08:35.600	0x8A145A4FC2F52EA69946B1A9799738D6
15	##MS_AgentSigningCertificate##	NULL	2020-09-05 20:08:42.470	0x0106000000000000901000000335C7A3A0736B23492FA7AB...
16	SL2	(被加密)	2020-09-15 18:20:20.480	0x6D8F46A87FB834D87AB301E9A08D1EC
17	WONDERLA\SL1	NULL	2020-09-15 18:21:25.317	0x0105000000000005150000006C163490ED706950A8175426...
18	WONDERLA\SQLService	NULL	2020-09-17 16:45:19.360	0x0105000000000005150000006C163490ED706950A8175426...
19	SpiderMan	(被加密)	2020-09-21 18:23:19.147	0x65220B87C4D1CC4CA5F5F13B48BB6BA66
20	WONDERLA\SQLDBATeam	NULL	2020-09-29 16:39:58.587	0x0105000000000005150000006C163490ED706950A8175426...

A status bar at the bottom indicates "Query executed successfully."

Session-25--September 30, 2020 - Wednesday.

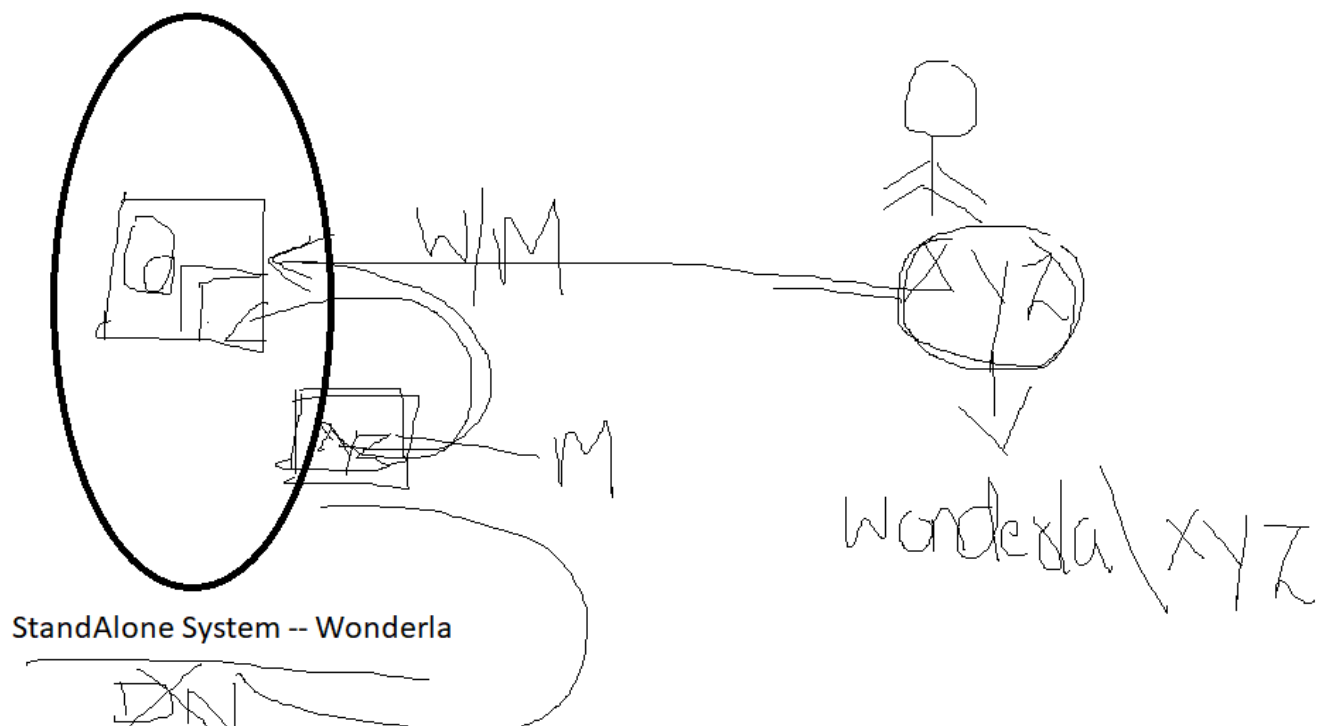
StandAlone Computer/System/Server: (Orphan Child)

Machine: Wonderla -- Domain: None.

Example for windows Accounts:

Wonderla\walid

Wonderla\ Vivian



System

Control Panel > System and Security > System

Control Panel Home

- Device Manager
- Remote settings
- Advanced system settings

View basic information about your computer

Windows edition

Windows Server 2012 Datacenter Evaluation

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System

Processor:	Intel(R) Core(TM) i5-3210M CPU @ 2.50GHz 2.49 GHz
Installed memory (RAM):	2.44 GB
System type:	64-bit Operating System, x64-based processor
Pen and Touch:	No Pen or Touch Input is available for this Display

Computer name, domain, and workgroup settings

Computer name:	Wonderla
Full computer name:	Wonderla
Computer description:	
Workgroup:	WORKGROUP

Windows activation

Windows is activated [View details in Windows Activation](#)

Product ID: 00184-40000-00001-AA148

See also

- Action Center
- Windows Update

Handwritten note: Stand Alone Server

Domain Computer/System/Server: (Child with Surname)

Machine-1: Wonderla -- Domain: Amazon.com

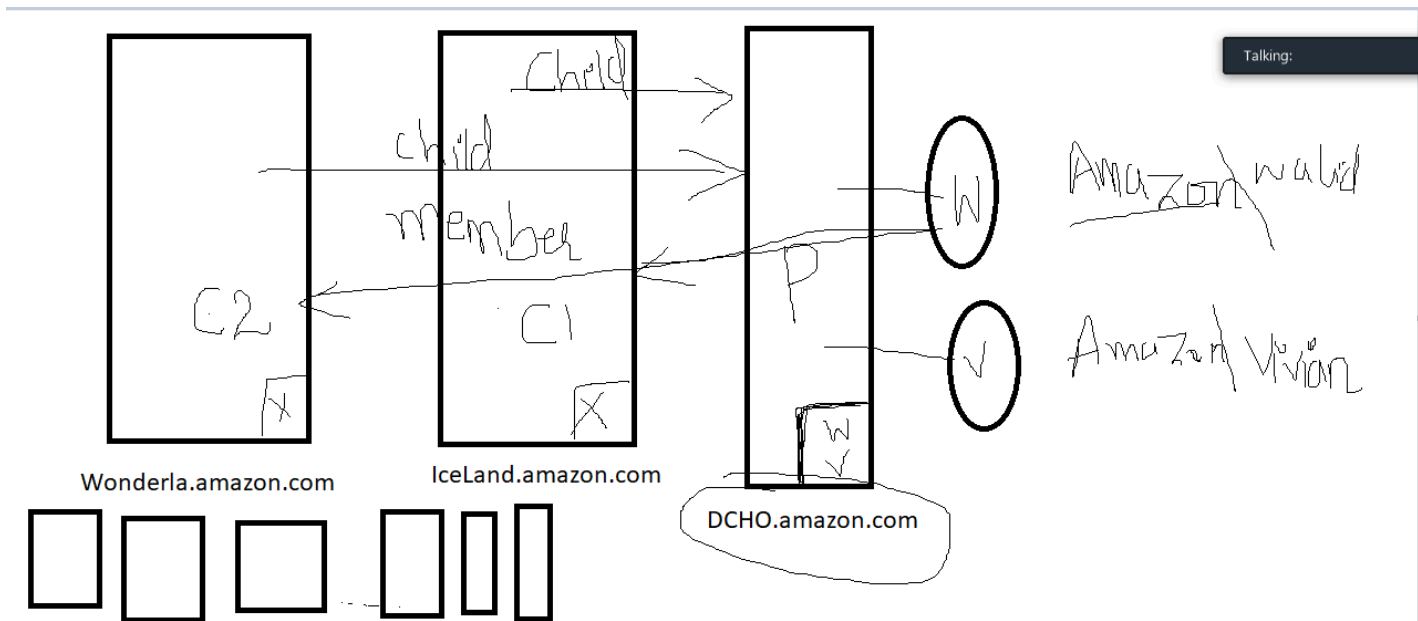
Example for windows Accounts:

Amazon\walid
Amazon\vivian

Machine-1: IceLand -- Domain: Amazon.com

Example for windows Accounts:

Amazon\walid
Amazon\vivian



Note:

Windows Server / Active Directory (AD)

-- This teams will create windows accounts (Individual or Group).

Today@123

Welcome@123

WIND ~~GROUP~~ ACC

Object Explorer

Connect

WONDERLA\SOURCE (SQL Server 11.0.5058 -

Databases

Security

Server Objects

Replication

AlwaysOn High Availability

Management

Integration Services Catalogs

SQL Server Agent

SQLQuery12.sql - W...Administrator (55)*

xp_logininfo 'WONDERLA\SQLDBTeam', 'members'

100 %

Results Messages

	account name	type	privilege	mapped login name	permission path
1	WONDERLA\JohnWalter	user	admin	WONDERLA\JohnWalter	WONDERLA\SQLDBTeam
2	WONDERLA\Praveen	user	admin	WONDERLA\Praveen	WONDERLA\SQLDBTeam
3	WONDERLA\Vivian	user	admin	WONDERLA\Vivian	WONDERLA\SQLDBTeam

https://www.youtube.com/watch?v=CiPsyASZWyo&list=PLviL2Q5l6zXpvEAYlk8IzPzdjW_5swPDj

About the Instructor

Current position: DBA SQL Server , 2012 - Today

Previous (IBM)

Mainframe Operations Coordinator, 2011-2012

Mainframe Operations Technical Leader, 2008-2011

Mainframe Operator Sr , 2006-2008

SQL Server Overview Training Space

Frigo - 1/9

Module 2 - Security

Module 3 - SQL Server Health Check

Module 4 - Database Structures

Module 5 - Backup & Restore

Module 6 - SQL Server Agent

Session-26--October 05, 2020 - Monday.

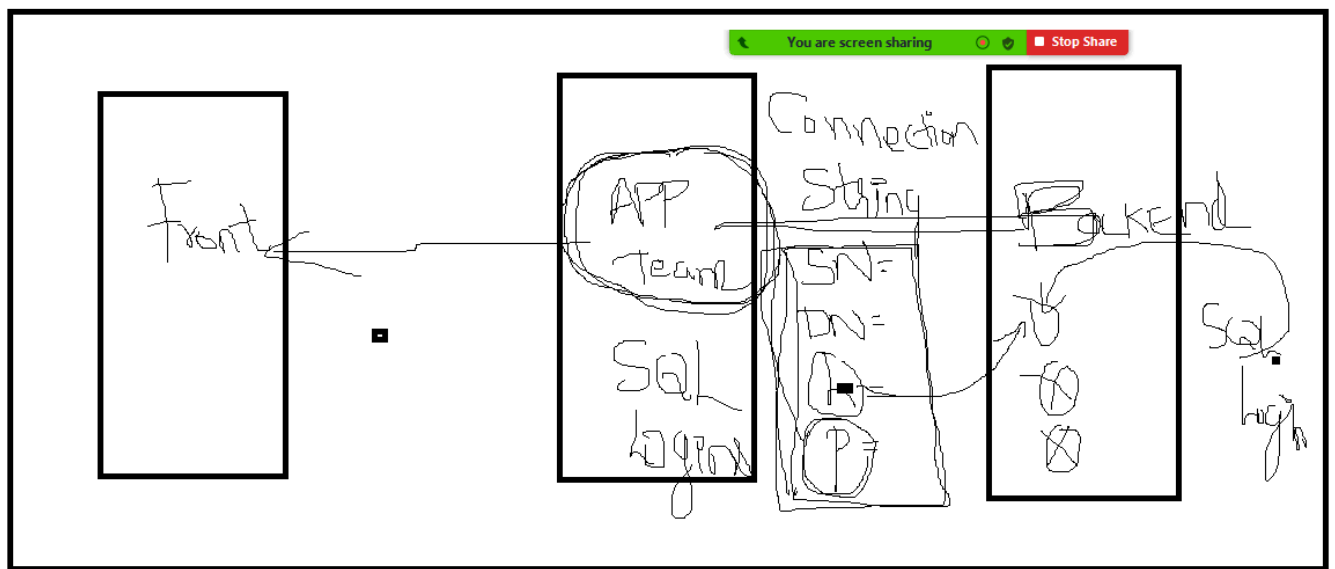
=====

Login (Windows or SQL Login) → Used to access SQL Server

User ---> (Part of Windows or SQL Login) → Used to access multiple DB's

SQL Logins are used in following scenarios:

1. For vendors who are not permanent employees.
2. For business persons, who wants to read data only.
3. For application related also these are used.



SOX Audit -- Audit entire SQL Servers for every 3 / 6 / 12 months.

Elevated Permission (Sysadmin)

1. Ticket
2. His Manager Approvals
3. Business Justification
4. StakeHolders Approvals.

Before granting Sysadmin permission to DBA Team

SQLQuery1.sql - W... \ChinnuBunny (54)) * x

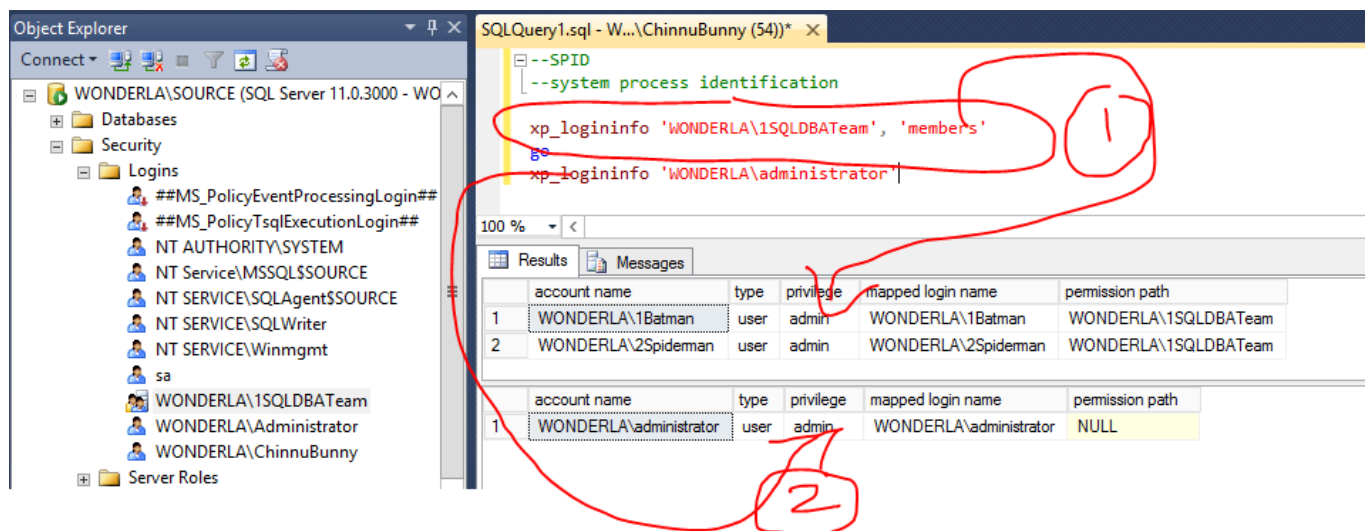
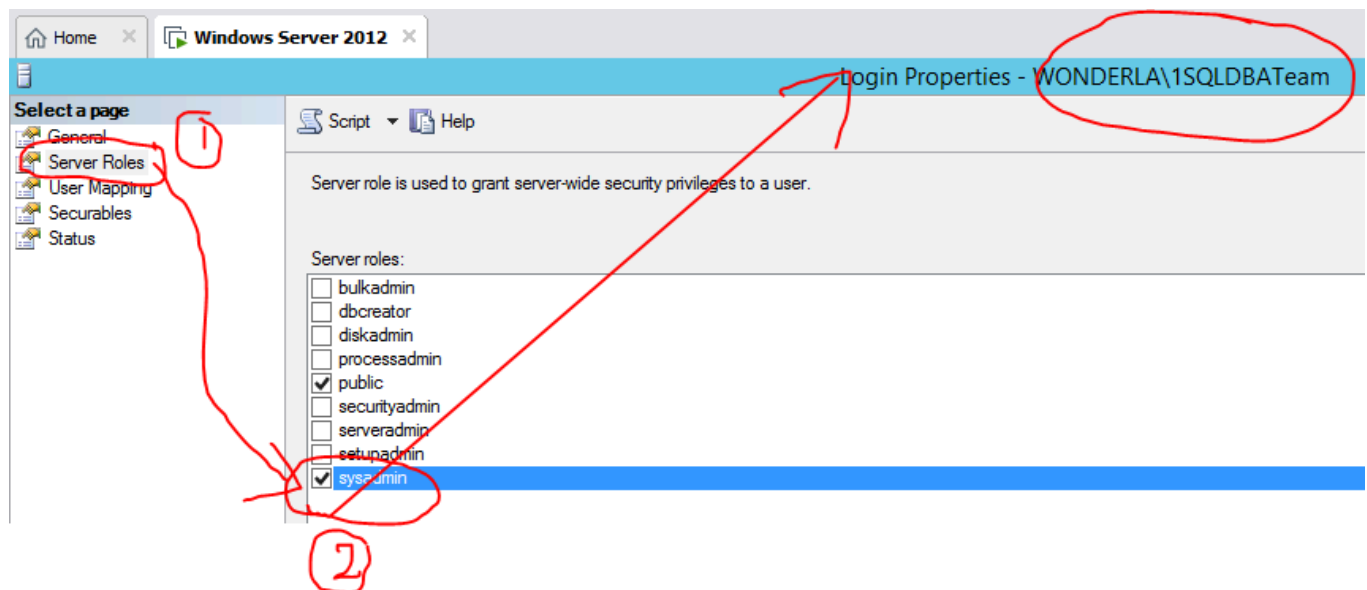
```
--SPID
--system process identification
xp_logininfo 'WONDERLA\1SQLDBTeam', 'members'
go
xp_logininfo 'WONDERLA\administrator'
```

	account name	type	privilege	mapped login name	permission path
1	WONDERLA\1Batman	user	user	WONDERLA\1Batman	WONDERLA\1SQLDBTeam
2	WONDERLA\2Spidean	user	user	WONDERLA\2Spidean	WONDERLA\1SQLDBTeam

	account name	type	privilege	mapped login name	permission path
1	WONDERLA\administrator	user	admin	WONDERLA\administrator	NULL

Handwritten notes: (1) User Not SysAdmin, (2) SysAdmin

After granting Sysadmin permission to DBA Team.



--SPID

--system process identification

xp_logininfo 'WONDERLA\1SQLDBATeam', 'members'

go

xp_logininfo 'WONDERLA\administrator'

Session-27--October 06, 2020 - Tuesday.

=====

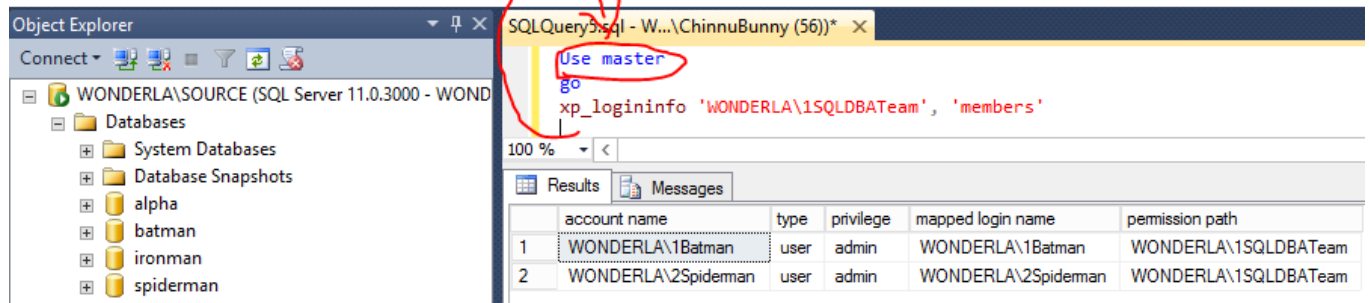
--Below query is to find out the members mapped or present inside a Group.

Use master

go

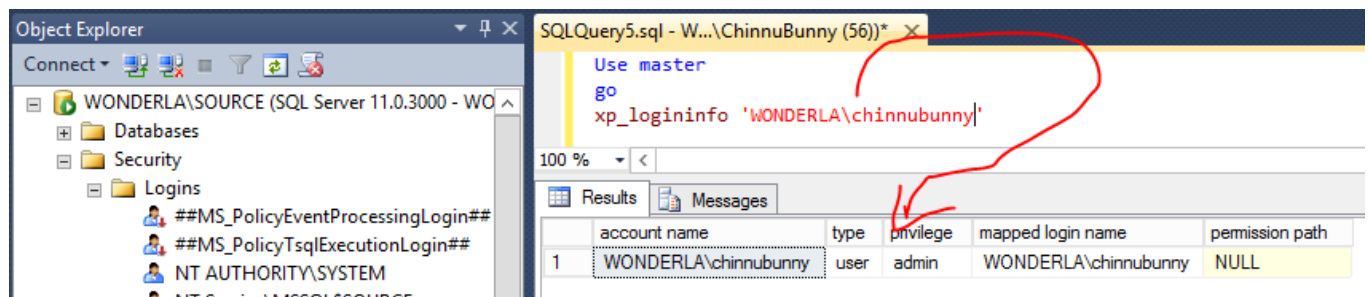
xp_logininfo 'WONDERLA\1SQLDBATeam', 'members'

--output as follows.

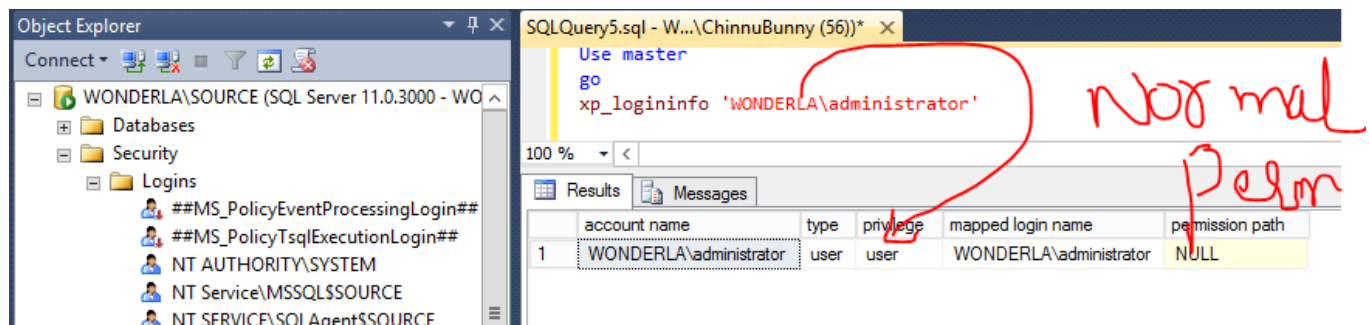


--Below query is to find out the member is part of which group ?
xp_logininfo 'WONDERLA\administrator'

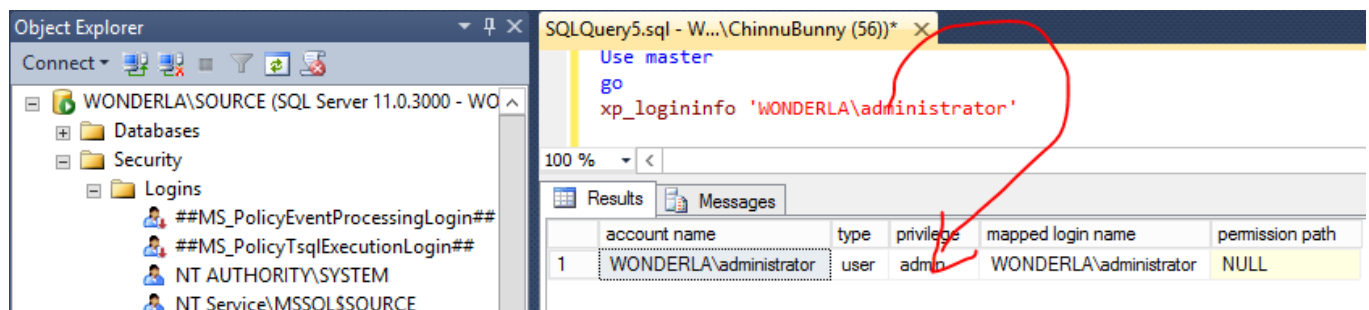
Screenshot-1:



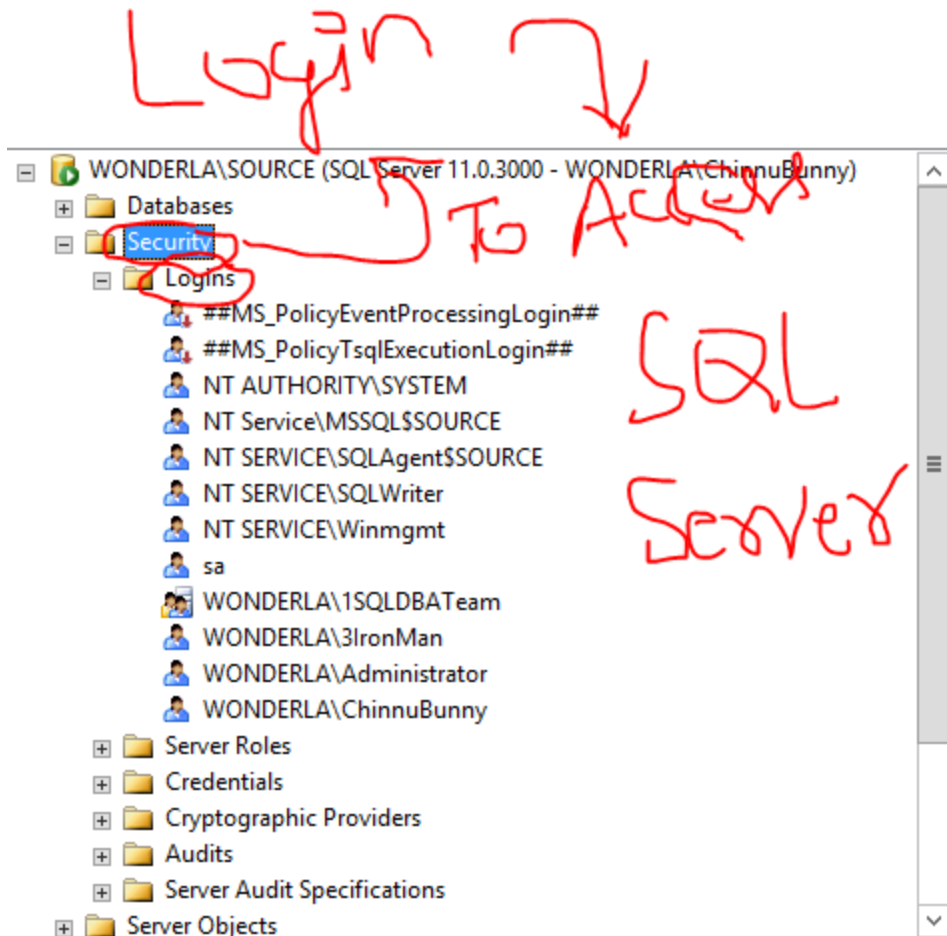
Screenshot-2:



Screenshot-3:

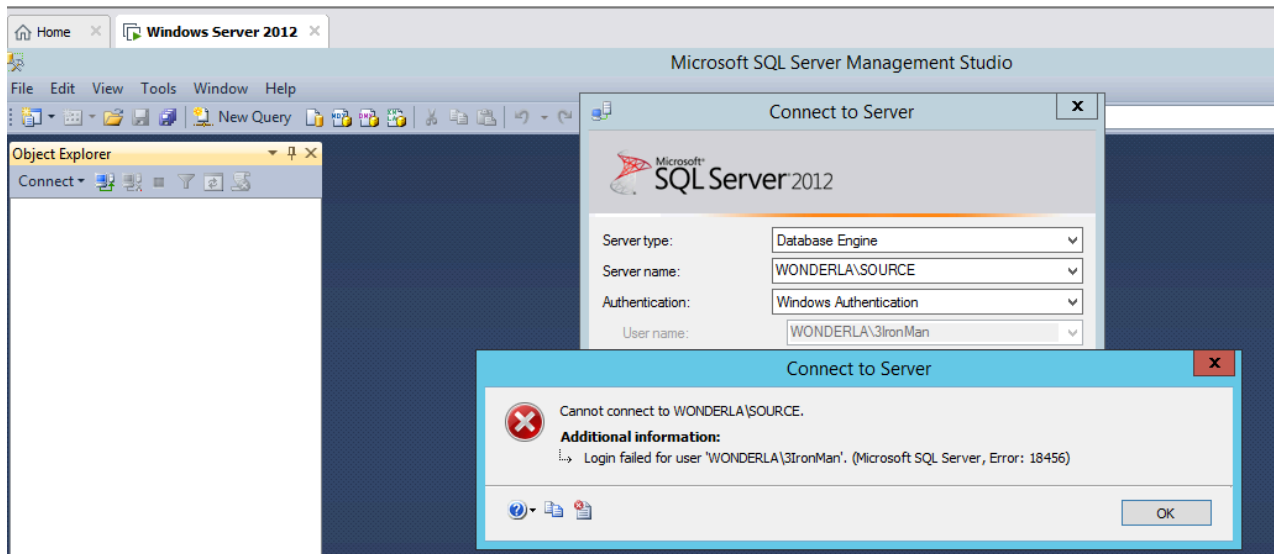


Important Note:



Note: 3IronMan has Windows Server Connection Access.

Note: 3IronMan doesn't have SQL Access.



Note: 3IronMan -- Junior DBA

Manager wants to give him SQL Server Access individually(for his windows account) for temporary use and NOT added to DBA Team Group.

Action:

1. Added 'wonderla3ironman' to Logins list under "SOURCE" SQL Instance. Not given any Server role and DB Role permission to this login.

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Object Explorer' pane displays the 'Logins' folder under the 'WONDERLA\SOURCE (SQL Server 11.0.3000 - WO)' instance. The login 'WONDERLA\3IronMan' is highlighted with a red circle. On the right, the 'SQLQuery5.sql' window shows the following query:

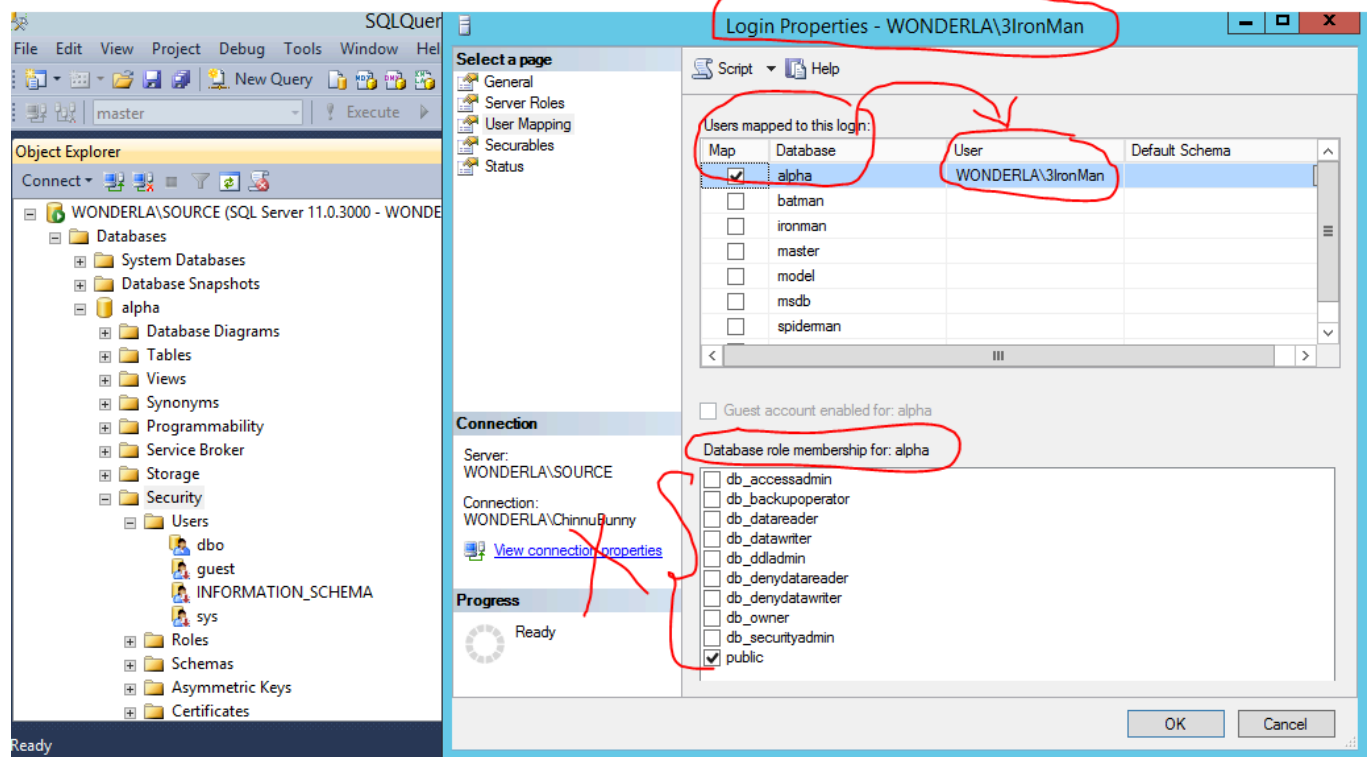
```
Use master
go
xp_logininfo 'WONDERLA\3ironman'
```

The 'Results' pane shows the output of the query:

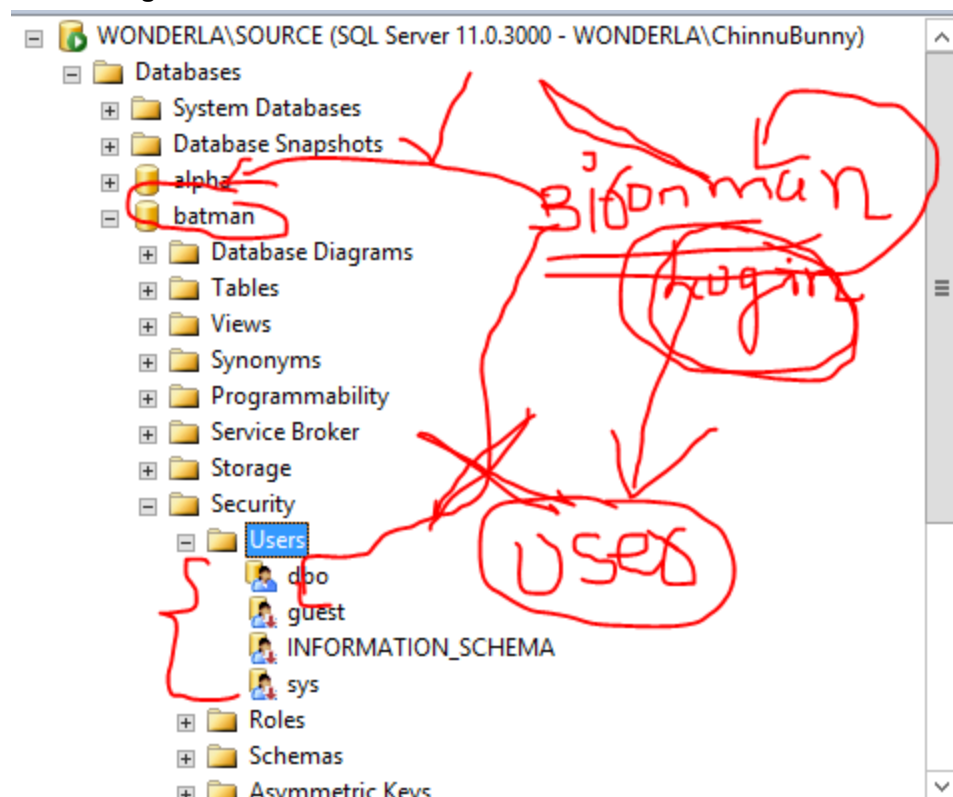
	account name	type	privilege	mapped login name	permission path
1	WONDERLA\3ironman	user	user	WONDERLA\3ironman	NULL

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Object Explorer' pane displays the 'Logins' folder under the 'WONDERLA\SOURCE (SQL Server 11.0.3000 - WONDERLA\3IronM)' instance. The login 'WONDERLA\3IronMan' is highlighted with a red circle. On the right, the 'Microsoft SQL Server Management Studio' window shows an error message: 'The database alpha is not accessible. (ObjectExplorer)'. A red arrow points from the error message to the 'alpha' database in the 'Object Explorer' pane. Handwritten red text on the right side of the screenshot reads: 'Ironman' and '1 Access SQL Server with least perm
2 Can't view all logins
3 Can't access all db's
4 Can't view SQL Agent'.

Action-2: Granted basic access to Alpha database to 3Ironman



**Note: 3IronMan Login is granted only to access Alpha. (No DB Roles)
Not granted batman database access.**



Security Layers

1. Windows Server
2. SQL Server
3. DB level
4. Object level

General Suggestions:

RDBMS → Microsoft SQL Server (Structured)--Less

2 Parts:

1. SQL Server Development -- Coding, enhancing.
2. SQL Server Administration -- Maintenance

NoSQL Databases (Not follows RDBMS) -- Unstructured.-- More Faster

1. **MYSQL Server Admin**
2. **MongoDB Admin**
3. **PostgreSQL Admin**
4. Cassandra Admin
5. Cockraoch

Cloud:

Azure -- Infrastructure, Network, Security, SQL Azure, etc

AWS -- Infrastructure, Network, Security, SQL Azure, etc

AZ 900 online Training - Day1 to Day 5

Session-28--October 09, 2020 - Friday.

=====

Alpha -- DB -- 100 Tables

Ironman -- DBReader Permission -- Read all 100 tables

Object Level Permissions:

=====

Only 1 table should be read by IronMan and NOT All other tables.

1. Revoke DB Reader Permission (Uncheck DB Reader Role)
2. Grant select permission to Ironman on single table individually.

Session-29--October 15, 2020 - Thursday.

=====

Windows Access -- will be taken care by windows team

SQL Server:

--SQL Connection Access will be given by SQL DBA

-- DB Access will be given by SQL DBA

--Object level access will be given by SQL DBA

Wonderla\IronMan -- Windows Account

Case-1: IronMan was having only Windows Access.

Not SQL Access and No DB Access.

Case-2: Ironman was given only SQL Access.

No DB Access Granted.

Case-3: IronMan has SQL Access.

IronMan has been granted basic access to DB.

IronMan was not granted any DB Role Permissions

Note: Ironman CANNOT View the DB Content.

Case-4: IronMan has SQL Access.

IronMan has been granted DBReader Perm for 1 db.

Note: Ironman can now view only 1 DB Content.

Case-5: IronMan has SQL Access.

IronMan has been granted DBReader Perm for all db.

Note: Ironman can now view/access all DB Content.

Case-6: Object Level Permission

IronMan has SQL Access.

Grant IronMan with basic Perm only (No DB Role) on AdvworksDB

Expand the DB but will not be able to view underlying tables.

IronMan should access only 1 table: [Production].[Product] and not other tables in

AdvworksDB2020.

use [AdvworksDB2020]

GO

GRANT SELECT ON [Production].[Product] TO [WONDERLA\IronMan]

GO

Permission Needs below

1. Ticket
2. Approvals from user manager
3. How long does the user needs access.

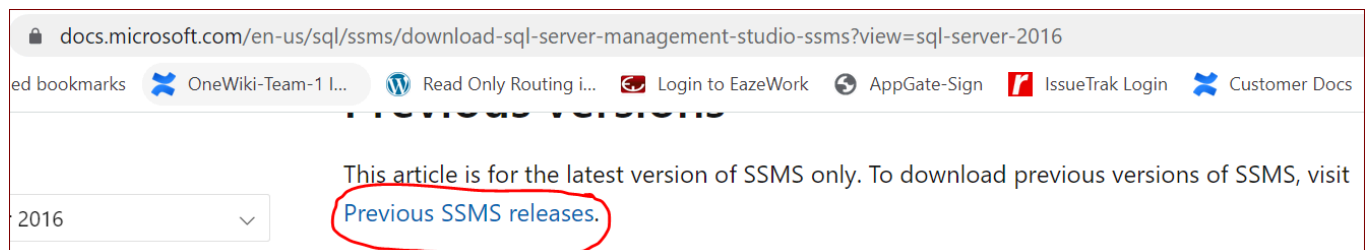
Session-30--October 19, 2020 - Monday.

NOTE:

SQL Server 2005 till SQL Server 2014 → SSMS is inbuilt and we can select it during SQL Installation.

SQL Server 2016 ---> Onwards -- SSMS needs to be installed separately.

<https://docs.microsoft.com/en-us/sql/ssms/download-sql-server-management-studio-ssms?view=sql-server-2016>



Old SSMS Download Section:

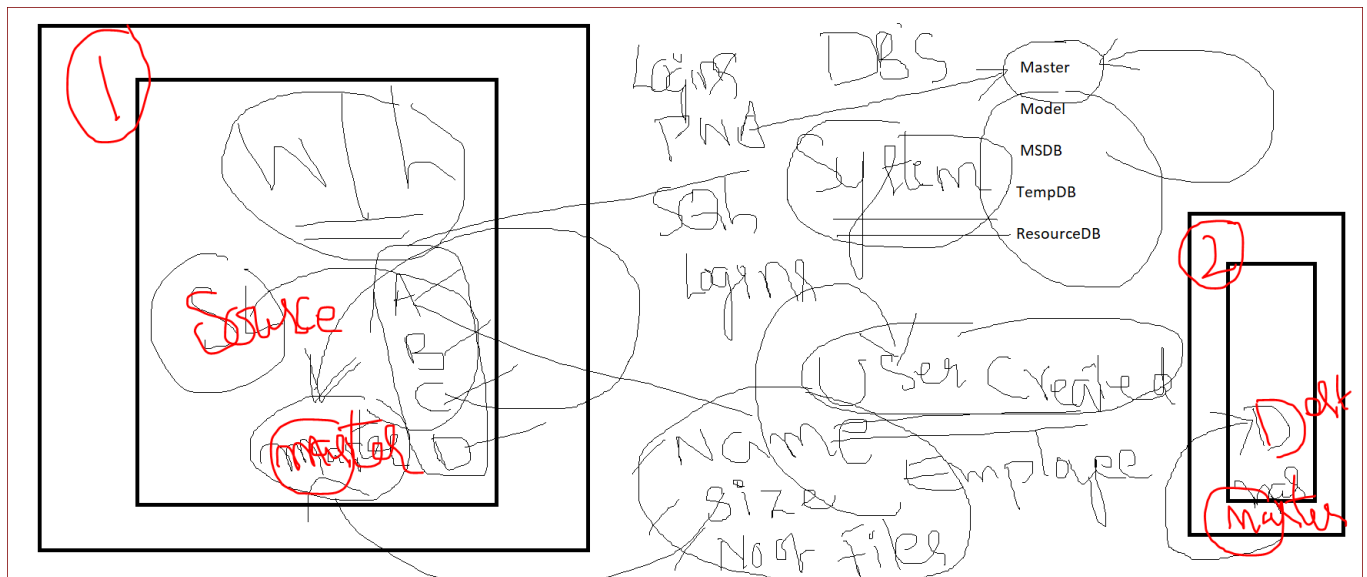
<https://docs.microsoft.com/en-us/sql/ssms/release-notes-ssms?view=sql-server-2016#previous-ssms-releases>

Post installing SSMS 2016 -- Server needs to be restarted. This is Mandatory.

Agenda: DB Architecture.

SQL Server Installation -- This will create system dbs internally.

1. Master -- 2 files -- mdf file, ldf file
 2. Model --
 3. Msdb
 4. Tempdb
 5. Resourcedb (hidden)
- =====



SQL Server System Databases

Below are the various SQL Server system databases you will find.

Master

- Purpose - Core system database to manage the SQL Server instance.

Prominent Functionality

- Per instance configurations
- Databases residing on the instance
- Files for each database
- Logins
- Linked\Remote servers
- Endpoints

Model

- Purpose - Template database for all user defined databases

MSDB

- Purpose - Primary database to manage the SQL Server Agent configurations

TempDB

- Purpose - Temporary database to store temporary tables (#temptable or ##temptable), table variables, cursors, work tables, row versioning, create or rebuild indexes sorted in TempDB, etc.
- Each time the SQL Server instance is restarted **all objects in this database are destroyed**, so permanent objects cannot be created in this database.
- Prominent Functionality
 - Manage temporary objects listed in the purpose above

Resource

- Purpose - The Resource database is responsible for physically storing all of the SQL Server 2005 system objects.
- This database has been **created to improve the upgrade and rollback of SQL Server** system objects with the ability to overwrite only this database.
- Prominent Functionality
 - System object definition

User databases -- these are user created db's.

1. Americadb
2. Nokiadb
3. Adventureworks

Types of databases.

- a. System databases -- DBA or user should never do any changes
- b. User databases-- DBA or user can do any changes as per business requirement.

--DBA

--2 ways

--a. GUI I'm

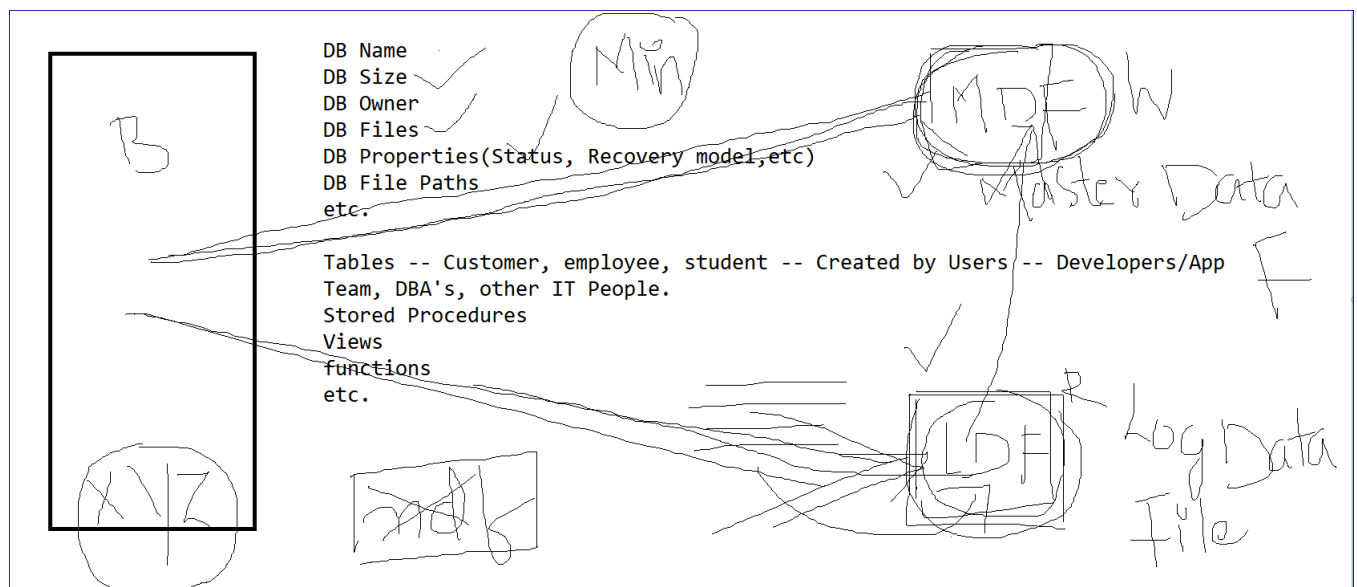
--b. tsq -- Transact Structured query language

--PARSE -- Checks spelling mistakes, checks syntax errors and give us warnings but will never execute any query.

DB -- Logical Container with multiple files to hold user DATA & Transactions.

Data File -- This is used to save data physically and permanently

Log File -- This is used to save transaction details & is very critical.



Business usage Names:

School, Trainers, Orders, Customers, Delivery

DB Size = 1 GB, -----100 GB, 16 TB

DB Creation -- Wonderla\Praveen -- Owner.

Note: But Please remember that EVERY DB Owner Should be an SA Account and NOT individual User.

During SQL Server Setup/Installation: Under Security, we see 2 types of Authentication.

1. Windows
2. Mixed Mode (Windows + SQL Server)

SA - Account -- SQL Server System Administrator → **SQL ACCOUNT**

Pwd: Today@123

DB -- New -- min 2 files -- Data File & Log File

Session-32--October 22, 2020 - Thursday.

=====

Databases were of 2 types

1. system db's -- master, model, msdb, tempdb, resource db
- 2 .user db's -- alpha, beta, charlie, delta, gamma.

sp_helpdb

sp_helpdb alpha

Data File Properties:

alpha -- Data File Logical Name

-- E:\SQLData\Hyderabad\alpha.mdf -- Data File Physical File Name

-- .mdf -- master data file

-- This is used to hold physical data (Permanent) or actual data.

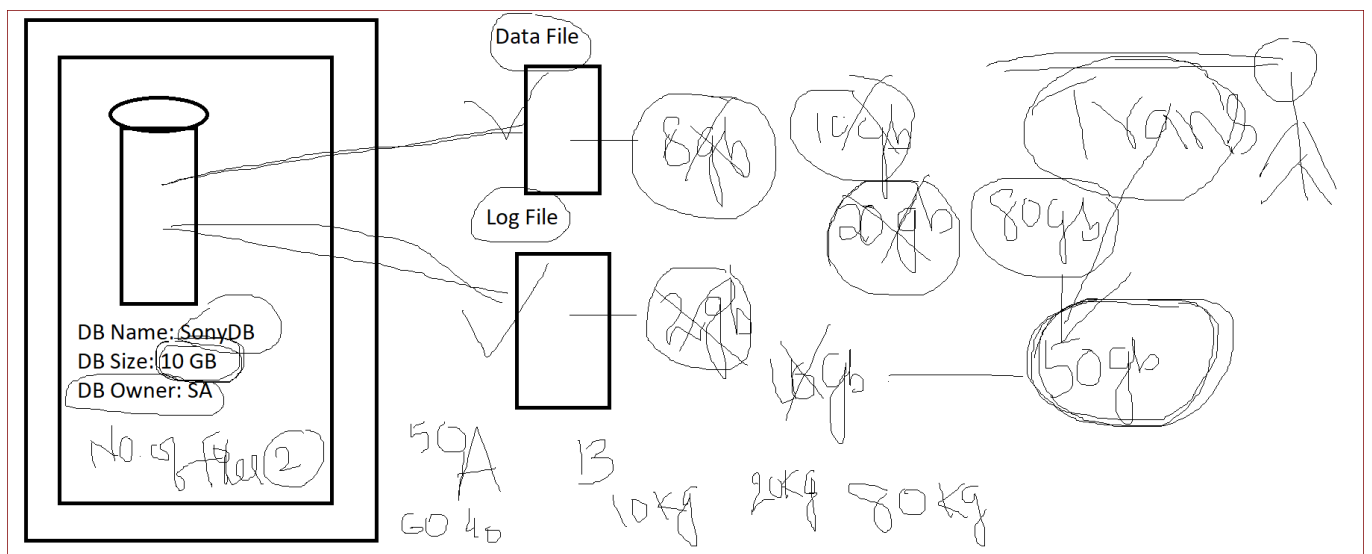
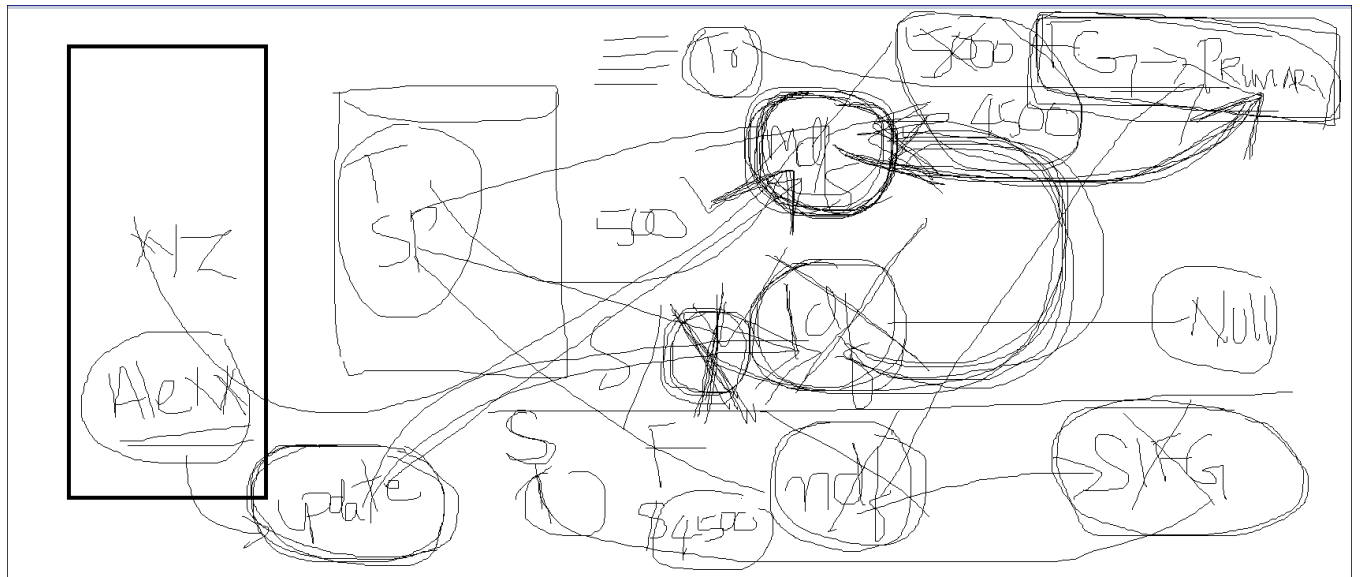
Log File Properties:

alpha_log -- Log File Logical Name

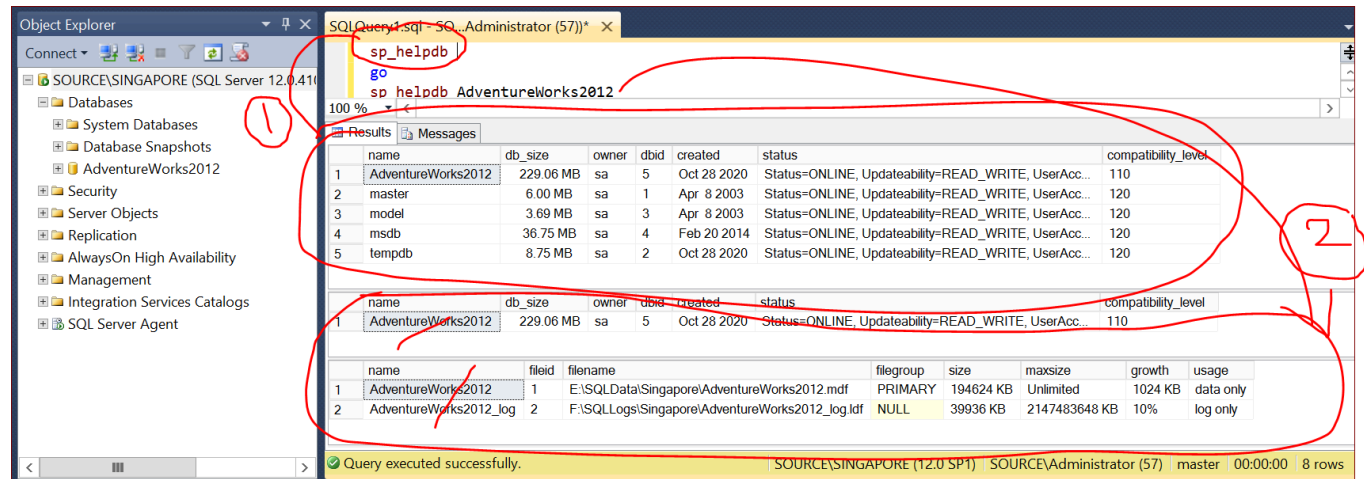
-- F:\SQLLogs\Hyderabad\alpha_log.ldf -- Log File Physical File Name

--.ldf -- log data file

-- This is used to hold transactions information



Session-33--October 30, 2020 - Friday.



DB Name -- XYZ

DB Size -- 10 GB (Data file & Log File)

DB Owner -- who ever creates DB becomes DB Owner -- Owner: SA

DB Files/Paths -- Logical Names, Physical Names

DB Status --

1. Online -- Accessible -- Read, write

2. Offline -- Not Accessible -- No Read, No Write

3. Restoring-- Not Accessible -- No Read, No Write

4. InRecovery -- Not accessible -- No Read, No Write.

5. Recovery Pending

6. Suspect

7. Emergency--- Accessible -- But for limited users -- DBA

DB Properties:

DB Size

DB Owner

DB ID

DB Created

Status:

1.ONLINE --- DB Is Accessible

2.OFFLINE --- DB Is NOT Accessible

3.Restoring -- DB Is NOT Accessible

4.Inrecovery -- DB Is NOT Accessible

5.Emergency -- DB Is Accessible

6.Recovery Pending -- DB Is NOT Accessible

7.Suspect -- DB Is NOT Accessible

Updateability

1. READ_Write (FALSE)

2. READONLY (TRUE)

User Access:

1. Multi_user
2. Single_user
3. Restricted_user

Recovery Model:

1. Simple
2. Full
3. Bulk-logged

Compatibility Level = 120

	name	db_size	owner	dbid
1	AdventureWorks2012	229.06 MB	sa	5
2	master	6.00 MB	sa	1
3	model	3.69 MB	sa	3
4	msdb	36.75 MB	sa	4
5	tempdb	8.75 MB	sa	2

Master-1

TEmpdb -2

Model- 3

Msdb -4

=====

adventureworks 2012 - 5

sp_helpdb

create database KENNYUSA

drop database nokia_new

Master-1

tempdb-2

model-3

msdb-4

resourcedb -- 32767

how many user databases can be created ?

32767 database - (master,model,msdb,tempdb,resource)

32767-5 = 32762

1
10
100
200
500
2000

sp_helpdb [SonyDB20200813]

Status=ONLINE, Updateability=READ_WRITE, UserAccess=MULTI_USER, Recovery=FULL,

--Review

--DB architecture

--Logical container with min 2 files initially

--mdf file - master data file

--ldf file - log data file

-- db creation -- 2 files

sp_helpdb advworks

DF Logical Name -- AdventureWorks2012

DB Physical Name --C:\SQLAlpha\SQLData\AdventureWorks2012.mdf -- 196 MB

LF Logical Name -- AdventureWorks2012_log

LF Physical name -- C:\SQLAlpha\SQLLogs\AdventureWorks2012_log.ldf -- 40 MB

bit

bytes

kb

mb

gb

tb

Database -- Logical Container with minimum 2 files initially

Data Files -- Data File Logical Name, Data File Physical File

Log Files -- Log File Logical Name, Log File Physical File.

	name	db_size	owner	dbid	created	status	compatibility_level
1	Advworks	230.06 MB	ServerAdministrator	5	Jun 26 2020	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	110

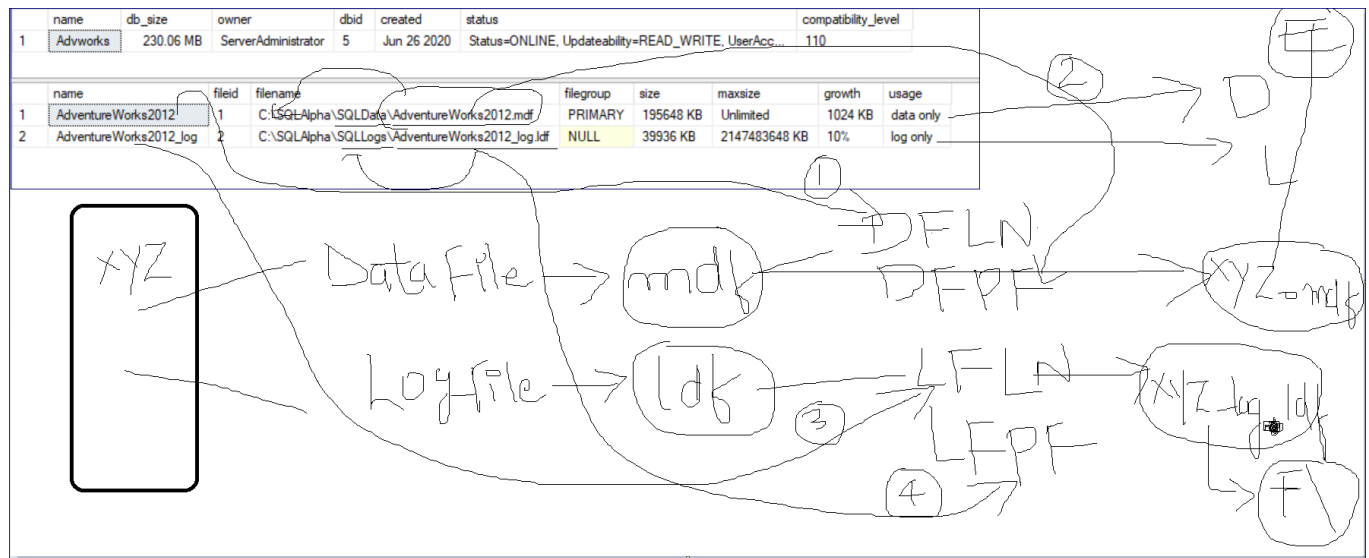
	name	fileid	filename	filegroup	size	maxsize	growth	usage
1	AdventureWorks2012	1	C:\SQLAlpha\SQLData\AdventureWorks2012.mdf	PRIMARY	195648 KB	Unlimited	1024 KB	data only
2	AdventureWorks2012_log	2	C:\SQLAlpha\SQLLogs\AdventureWorks2012_log.ldf	NULL	39936 KB	2147483648 KB	10%	log only

DB having 2 files minimum.

DF Extension -- XYZ.mdf -- master data file

LF Extension -- XYZ_log.ldf -- log data file

=====



DB can have more than 2 files.

DF Extension -- XYZ.mdf -- master data file

-- XYZ.ndf -- next data file

LF Extension -- XYZ_log.ldf -- log data file

XYZ -- DB Size -- 200 GB.

DF -- mdf file -- 190 GB

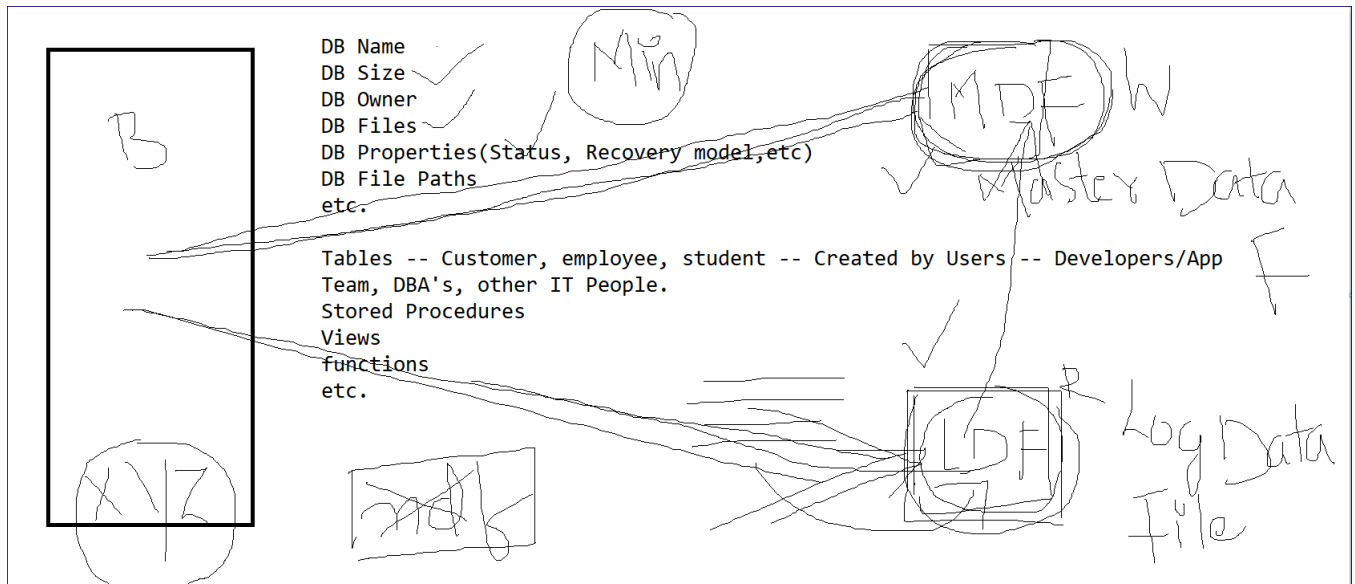
LF -- ldf file -- 10 GB

=====

Options-1

1. Delete some data from mdf file
2. Expand E:\ drive size from 200 GB to 300 GB.
3. Create a new data file on another drive G:\ drive.

DB Properties.



Containment type:	None
Other options:	
Date Correlation Optimization Enabled	False
Delayed Durability	Disabled
Is Read Committed Snapshot On	False
Numeric Round-Abort	False
Parameterization	Simple
Quoted Identifiers Enabled	False
Recursive Triggers Enabled	False
Trustworthy	False
VarDecimal Storage Format Enabled	True
Recovery	
Page Verify	CHECKSUM
Target Recovery Time (Seconds)	60
Service Broker	
Broker Enabled	False
Honor Broker Priority	False
Service Broker Identifier	5e7b0dc0-1f2c-4e28-88a7-196151978b5b
State	
Database Read-Only	False
Database State	NORMAL
Encryption Enabled	False
Restrict Access	MULTI_USER
Restrict Access	MULTI_USER SINGLE_USER RESTRICTED_USER

- Status -- Online
- UserAccess=MULTI_USER, SINGLE_USER, Restricted_User (only DBA's)
- Updateability=READ_WRITE (false), True (ReadOnly) -- DBA's can't update.
- Recovery Models = 3 Types -- Full, Simple, Bulk Logged

Backups -- Full, Differential, Tlog, FileGroup, Copy_only.

Restore -- With Recovery, With Norecovery

Person Status:

1. Healthy -- Accessible
2. Sick -- Not Accessible.
3. Bed ridden -- Not Accessible.
4. Paralysed. -- Not Accessible.
5. Dead -- Not accessible
6. Fractured -- Not Accessible

--Script to find SQL Server Version Details

```
SELECT Serverproperty('productversion') AS SQLProductVersion,
CASE Substring(Cast(Serverproperty('ProductVersion') AS NVARCHAR), 1,
Charindex('.', Cast(Serverproperty('ProductVersion') AS
NVARCHAR)) - 1
)
WHEN '7' THEN 'SQL Server 7'
WHEN '8' THEN 'SQL Server 2000'
WHEN '9' THEN 'SQL Server 2005'
WHEN '10' THEN 'SQL Server 2008/2008R2'
WHEN '11' THEN 'SQL Server 2012'
WHEN '12' THEN 'SQL Server 2014'
WHEN '13' THEN 'SQL Server 2016'
WHEN '14' THEN 'SQL Server 2017'
WHEN '15' THEN 'SQL Server 2019'

ELSE 'Unknown'
END AS SQLServerProduct,
Serverproperty('productlevel') AS SQLProductLevel,
Serverproperty('edition') AS SQLEdition
```

SQLQuery20.sql - SOURCE\SINGAPORE.master (SOURCE\Administrator (55))* - Microsoft SQL Server Management Studio (Administrator)

Object Explorer: SOURCE\SINGAPORE (SQL Server 12.0.4100.1) > Databases > AdventureWorks2012

Query Results: sp_helpdb AdventureWorks2012

	name	fileid	filename	filegroup	size	maxsize	growth	usage
1	AdventureWorks2012	1	E:\SQLData\Singapore\AdventureWorks2012.mdf	PRIMARY	194624 KB	Unlimited	1024 KB	data only
2	AdventureWorks2012_log	2	F:\SQLLogs\Singapore\AdventureWorks2012_log.ldf	NULL	39936 KB	2147483648 KB	10%	log only

Handwritten notes: "Datafile" points to the first row. "Logical Name" points to the 'name' column. "Physical Path" points to the 'filename' column. "LF Logical Name" points to the 'name' column. "LF Phys Path" points to the 'filename' column.

Database Properties - AdventureWorks2012

Select a page: General, Files, Filegroups, Options, Change Tracking, Permissions, Extended Properties, Mirroring, Transaction Log Ship

Database name: AdventureWorks2012 (circled)

Owner: sa

Use full-text indexing: ☒

Database files:

Logical Name	File Type	Filegroup	Initial Size (MB)	Autogrowth / Maxsize	Path	File Name
AdventureWorks...	ROWS Data	PRIMARY	191 (circled)	By 1 MB, Unlimited	E:\SQLData\Singapore	AdventureWorks2012.mdf
AdventureWorks...	LOG	Not Applicable	39	By 10 percent, Limited to 20971...	F:\SQLLogs\Singapore	AdventureWorks2012_log.ldf

Handwritten note: "229 MB" is written next to the "Initial Size (MB)" column.

SQLQuery20.sql - S...Administrator (55)*

sp_helpdb AdventureWorks2012

Results: \SINGAPORE (SQL Server 12.0.4100.1 - SOURCE\Administrator)

	name	db_size	owner	dbid	created	status	compatibility_level
1	AdventureWorks2012	229.06 MB (circled)	sa	5	Oct 26 2020	Status=ONLINE, Updateability=READ_WRITE, UserAcc...	120

	name	fileid	filename	filegroup	size	maxsize	growth	usage
1	AdventureWorks2012	1	E:\SQLData\Singapore\AdventureWorks2012.mdf	PRIMARY	194624 KB	Unlimited	1024 KB	data only
2	AdventureWorks2012_log	2	F:\SQLLogs\Singapore\AdventureWorks2012_log.ldf	NULL	39936 KB (circled)	2147483648 KB	10%	log only

Kenny/Naomi -- Laptop: SQL Server 2014 -- Sample DB

Google -- Downloaded Adventureworks 2014.bak (we can't restore higher version backups).

SQL Server 2014:

Sample Database -- 2014, 2012, 2008, 2005

SQL Server 2012:

Sample DB: 2012, 2008, 2005

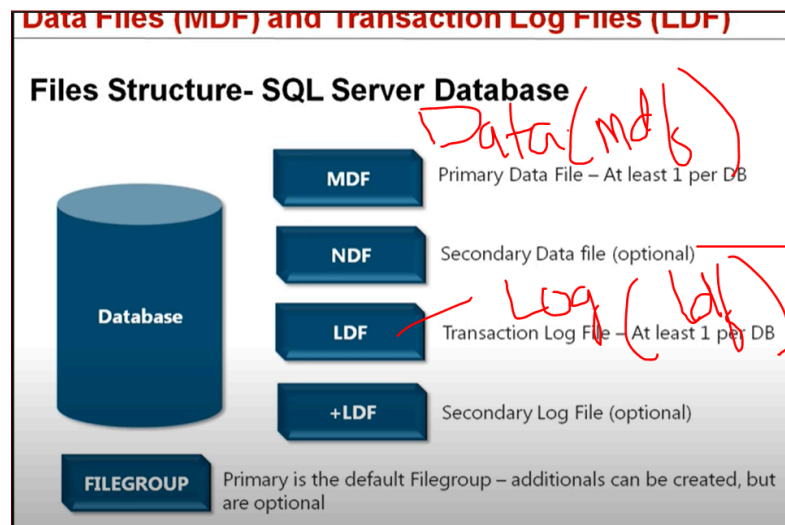
But should not be > 2012 --- Means 2014, 2016, 2017, 2019 are not restorable due to higher versions.

DB's have imp properties:

Recovery Model, Backups, Restores.

Module 4 - Database Structures

https://www.youtube.com/watch?v=Is0SS17hQ0I&list=PLviL2Q5I6zXpvEAylk8IzPzdjW_5swPDj&index=5



✓	✓	✓	✓
---	---	---	---

SQL Server Decommission -> Uninstalling SQL Server Instance

Backup DB's

Take All Logins Script

Take All SQL Agent Jobs Script

Take all details related to this SQL Server.

November 09, 2020 - Monday -- DB Architecture

=====

CASE-1

DB CREATION: 2 FILES

1. Data File -- df logical name, physical path, mdf file (.MDF) MASTER DATA FILE
2. Log File --lf logical name,physical path, ldf file (.LDF) LOG DATA FILE

CASE-2:

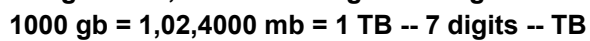
POST DB CREATION: FILES

Data File -- df logical name, physical path, mdf file (MDF) MASTER DATA FILE

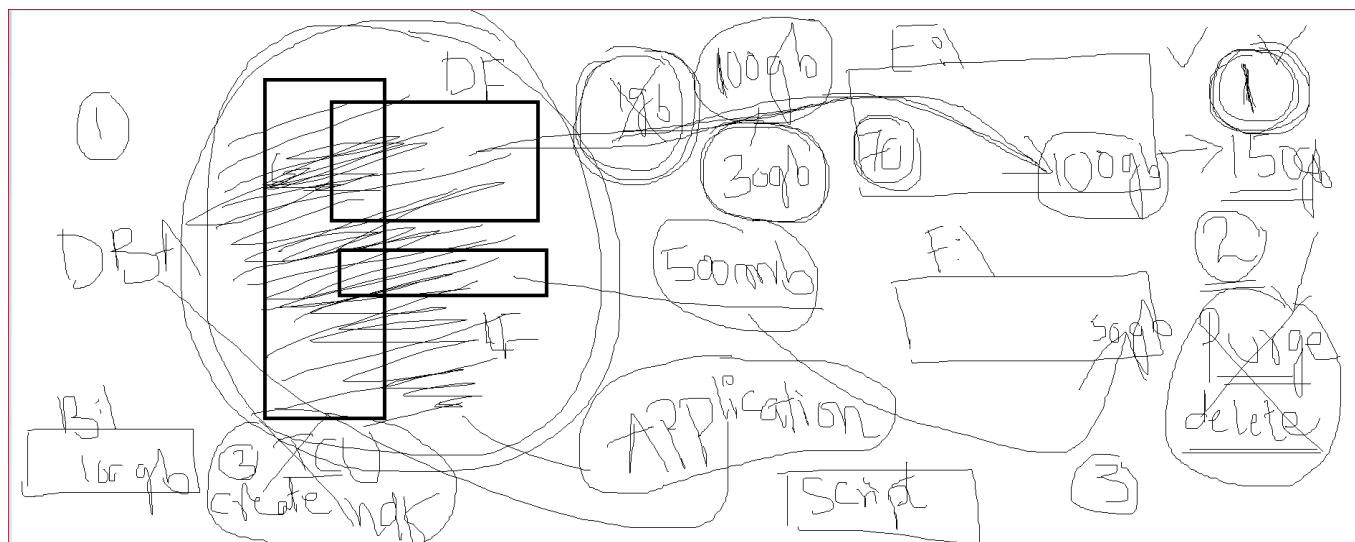
Log File --lf logical name,physical path, ldf file (LDF) LOG DATA FILE

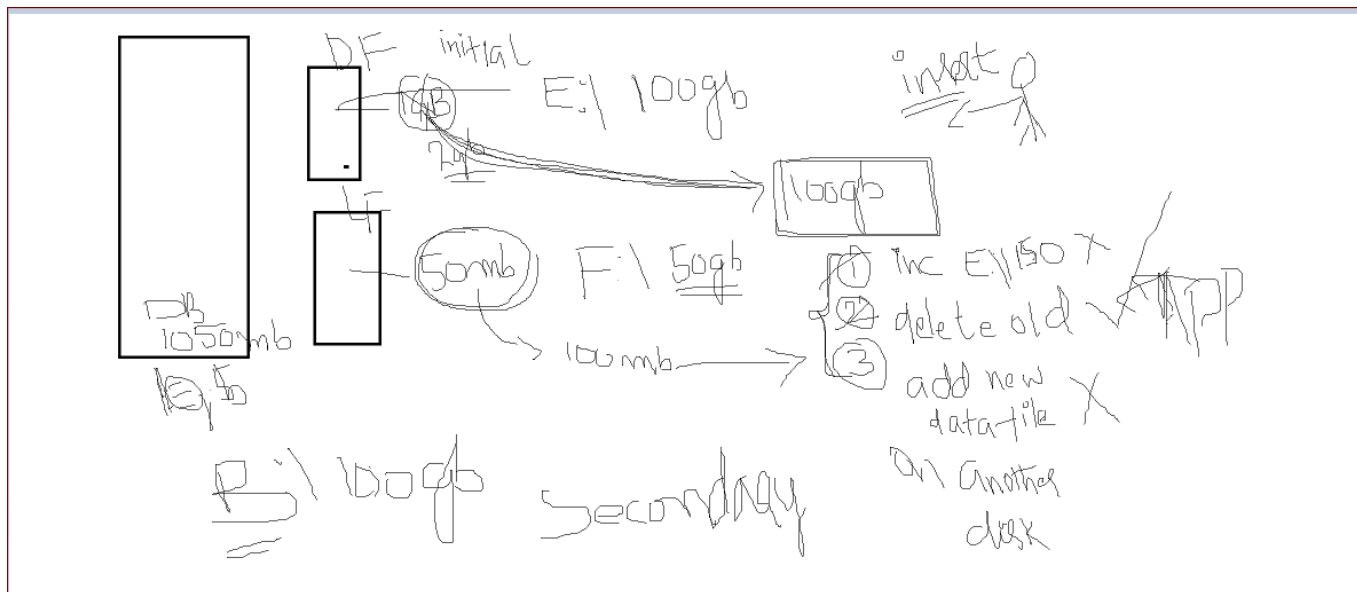
SECONDARY DF--df logical name, physical path, Ndf file (NDF) NEXT DATA FILE

NDF → NEXT DATA FILE → .NDF (multiple ndf files)



Autogrowth: 100 MB





November 10, 2020 - Tuesday-- RM, Backups, Restores

Recovery Models:

1. Simple -- Full Backup, Diff Backup, Tlog backup is NOT possible by design
2. Full -- Full backup, Diff Backup, TLog Backup
3. Bulk logged -- Full backup, Diff Backup, TLog Backup.

Backup Types:

1. Full Backup
2. Differential Backup
3. TLog (Transaction Log) Backup
4. Filegroup Backup

Other imp backup types:

1. Copy_Only
2. Mirrored Backup
3. Stripe Backup

Restore Types:

1. With NoRecovery -- DB will not be online, not accessible
2. With Recovery -- DB will be online, accessible
3. With Standby -- DB will be online, accessible.

SIMPLE Recovery Model:

SIMPLE

The SIMPLE recovery model is the simplest among the available models.

It supports full, differential backups. **Transaction log backups are not supported.**

The log space is reused whenever the SQL Server background process checkpoint operation occurs. The inactive portion of the log file is removed and is made available for reuse.

Point-in-time and page restore are not supported.

Reasons to choose the simple database recovery model

1. Most suited for Development and Test databases
2. Simple reporting or application database, where data loss is acceptable
3. The point-of-failure recovery is exclusively for full and differential backups (TLOG NOT)
4. No administrative overhead (Once TLog file reaches 70 % utilization, DB Engine will come and truncate or applies CHECKPOINT and release inactive transactions and makes free space.

It supports:

1. Full backup
2. Differential backup

DB: SonyDB				
	Employee		Backup	Size
	100 Records		Full	3.5 GB
	50 Records (100 mb)	Diff 1		100 mb
	10 Records (50mb)	Diff 2		Diff 1 + Diff 2
				100 MB + 50 MB = 150 MB
	10 Records(50 mb)	Diff 3		D1 + D2 + D3
				200 MB

November 11, 2020 - Wednesday-- RM, Backups, Restores

=====

--Simple

--Adv:

--No Log file growth or full overhead

--DB Engine will take care of tlog file growth

--disadv:

--Tlog backup is not possible.

--Point in time restore is not possible.

Recovery Models & Backups, Restores are internally dependent.

Simple RM -- Full Bkps, Diff Bkps BUT NO TLOG Bkps possible.

Full RM -- Full Bkp, Diff Bkp, TLog Bkp

Point in time Restore is possible in all cases

Bulk-logged RM -- Full Bkp, Diff Bkp, Tlog Bkp

Case-1: P.I.T is possible only for normal transactions.

Case-2: P.I.T is not possible during Bulk transactions.

Backups Operations & its Syntax:

SSMS -- 2 ways:

1. GUI.
2. TSQL (Queries)

Backups:

Full: GUI / TSQL :

=====

--full backup syntax:

BACKUP DATABASE DBNAME
TO DISK = 'BACKUP PATH AND ITS EXTENSION'
WITH COMPRESSION, STATS=5

Diff

Tlog Backup -- No discussion on this as we are dealing with SIMPLE Recovery Model.

--full backup syntax:

--BACKUP DATABASE [dbname]
--TO DISK = 'BACKUP PATH AND ITS EXTENSION'
--WITH COMPRESSION, STATS=5

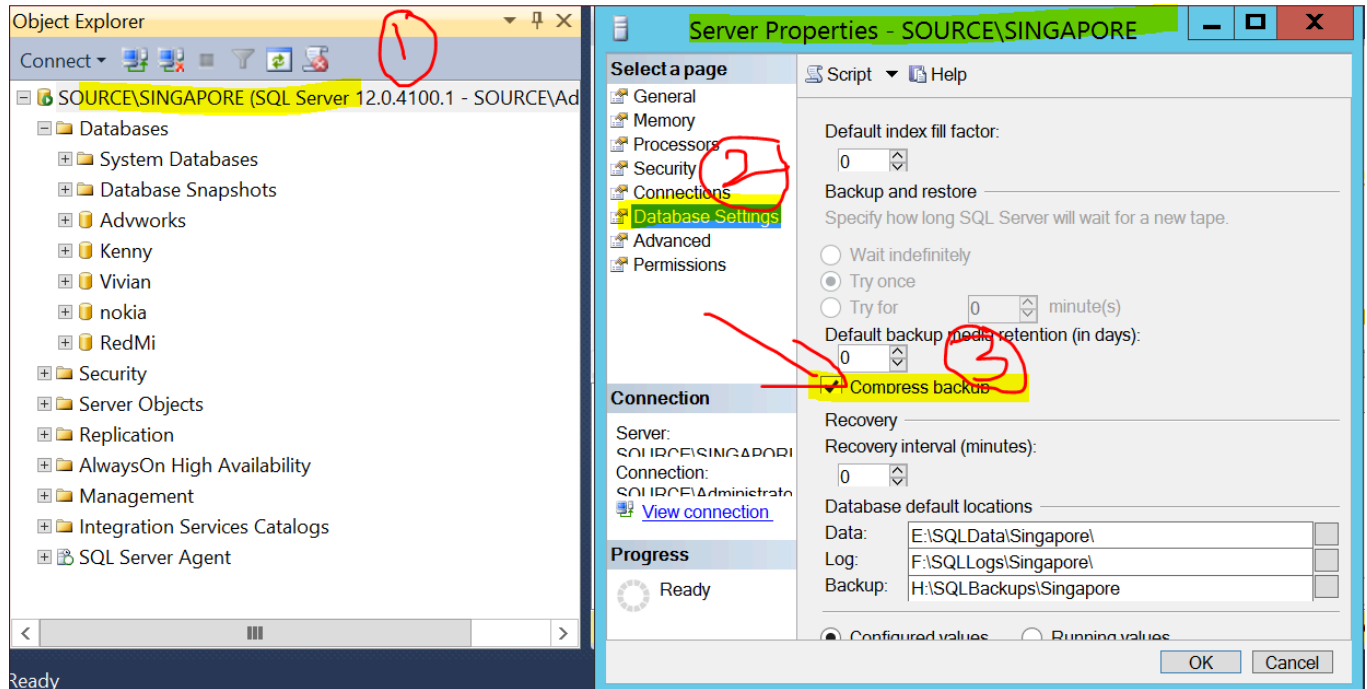
BACKUP DATABASE [Advworks]
TO DISK = 'H:\SQLBackups\Singapore\Advworks_FULL_Without_COMPRESSION_11112020_0500pm.bak'
WITH STATS=5

BACKUP DATABASE [Advworks]
TO DISK = 'H:\SQLBackups\Singapore\Advworks_FULL_Withcompression_11112020_0506pm.bak'
WITH STATS=5,COMPRESSION

---compression 2008

--DB Size: 250.00 MB -- Content it occupied ? 190 MB -- without compression.

--example -- DB Size: 100 GB -- Content it occupied is 90 GB -- Backup size: 90 GB -- without compression
-- Backup size: 10 GB -- with compression



Cloud:

Microsoft Azure

AWS

Alibaba

etc.

7759 courses:

15 courses -- purchased

Remaining all 7700 plus - i got free through udemy coupons.

Cloud

SQL Server

Amazon prime videos -- single account shared by 5 people.

--full backup syntax:

--BACKUP DATABASE [dbname]

--TO DISK = 'BACKUP PATH AND ITS EXTENSION'

--WITH COMPRESSION, STATS=5

BACKUP DATABASE [Advworks]

TO DISK = 'H:\SQLBackups\Singapore\Advworks_FULL_11112020_0512pm.bak' --backup path

WITH STATS=5

--diff backup

--BACKUP DATABASE [dbname]

-- to disk = 'backup path'

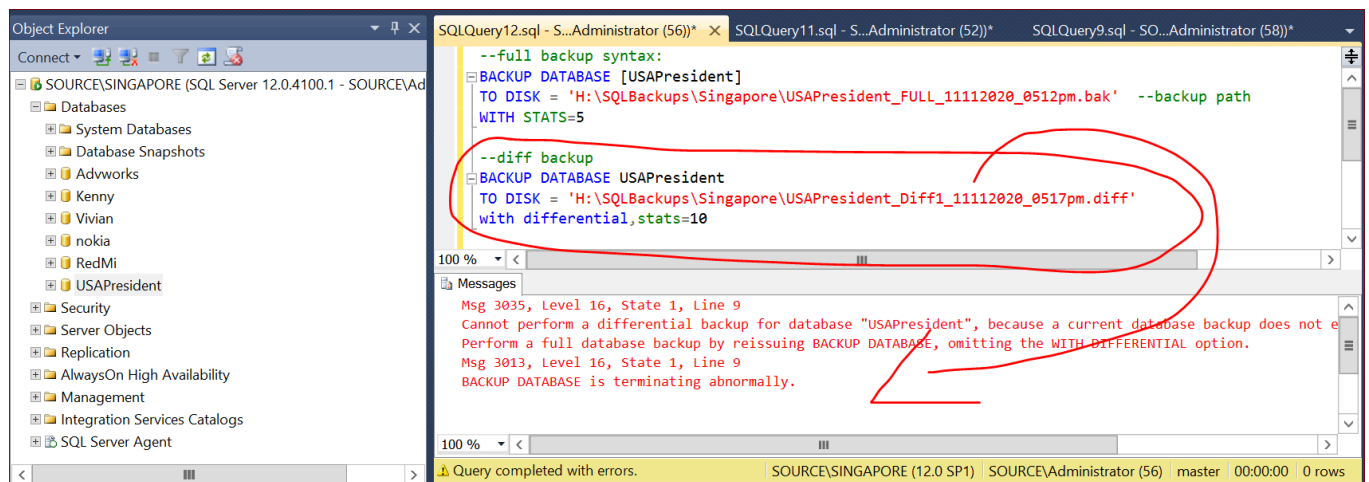
--with DIFFERENTIAL, STATS=5, COMPRESSION

BACKUP DATABASE ADVWORKS

TO DISK = 'H:\SQLBackups\Singapore\Advworks_Diff1_11112020_0517pm.diff'

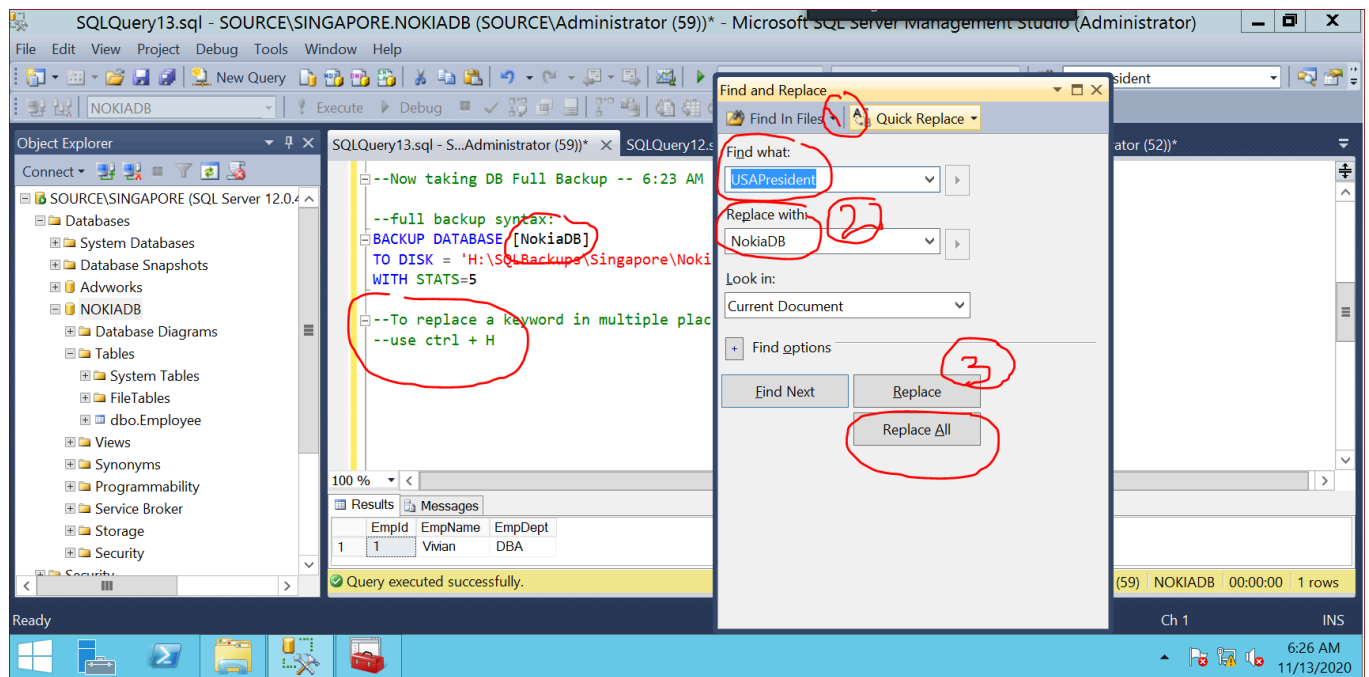
with differential,stats=10

Note: Without taking Full Backup we can't take diff backup for the 1st time.



November 12, 2020 - Thursday -- Simple RM, Backups

=====



select * from Employee

insert into Employee values (1,'Vivian','DBA')

--Now taking DB Full Backup -- 6:41 AM IST

--full backup syntax:

BACKUP DATABASE NokiaDB

TO DISK = 'H:\SQLBackups\Singapore\NokiaDB_FULL_Nov132020_0641AMIST.bak' --backup path
WITH STATS=5

insert into Employee values (2,'Naomi','DBA')

insert into Employee values (3,'Kenny','DBA')

select * from Employee

--diff backup

BACKUP DATABASE NokiaDB

TO DISK = 'H:\SQLBackups\Singapore\NokiaDB_Diff1_Nov132020_0643IST.diff'
with differential,stats=10

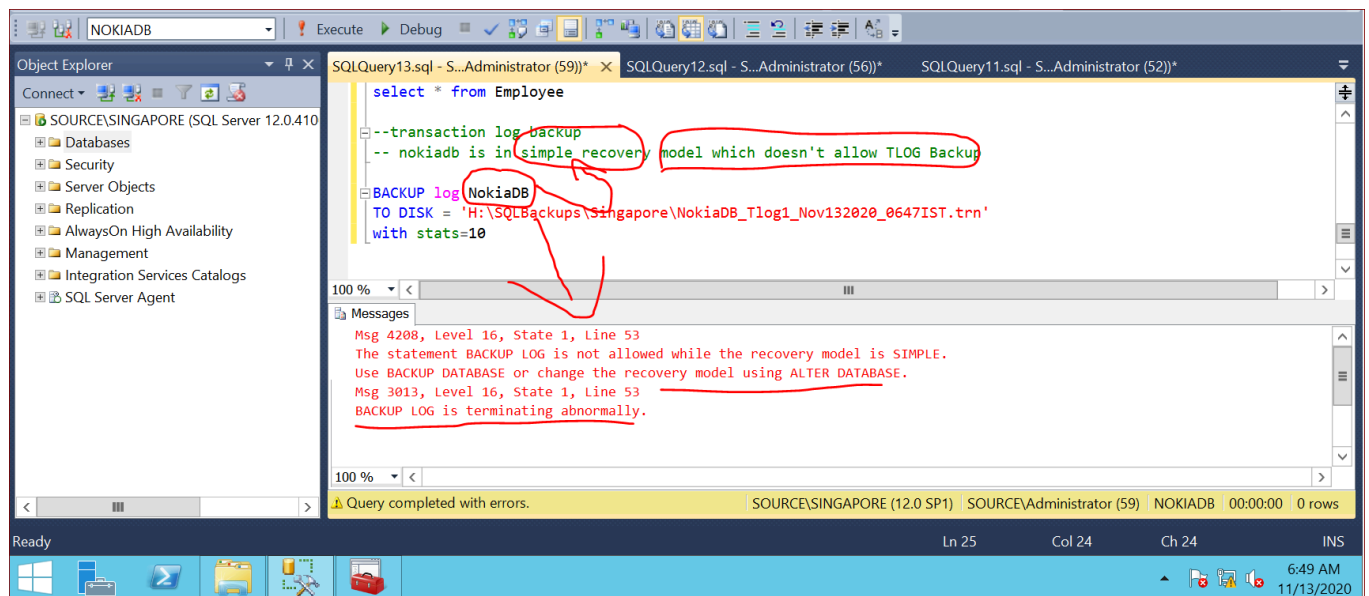
-- CTRL + Z

-- reverts your changes.

insert into Employee values (4,'Steve','DBA')

insert into Employee values (5,'Biden','DBA')

select * from Employee



November 17, 2020 - Tuesday-- Simple RM, Backups

Simple (RM) -- Full backups, Diff Backups, No Tlog

--Full Backup Statement:

BACKUP DATABASE DBNAME -- here put dbname
TO DISK = 'BACKUP PATH AND BACKUP EXTENSION'
WITH COMPRESSION,STATS=5

BACKUP DATABASE SONYDB
TO DISK = 'H:\SQLDBBACKUPS\SINGAPORE\SONYDB_FULL.BAK'
WITH COMPRESSION, STATS=5

--Differential Backup Statement:

BACKUP DATABASE DBNAME -- here put dbname
TO DISK = 'BACKUP PATH AND BACKUP EXTENSION'
WITH **DIFFERENTIAL**, COMPRESSION, STATS=5

BACKUP DATABASE SONYDB
TO DISK = 'H:\SQLDBBACKUPS\SINGAPORE\SONYDB_FULL.DIFF'
WITH **DIFFERENTIAL**, COMPRESSION, STATS=5

=====

--Transaction Log Backup Statement: (TLOG Backup)

BACKUP **log** DBNAME -- here put dbname
TO DISK = 'BACKUP PATH AND BACKUP EXTENSION'
WITH COMPRESSION, STATS=5

BACKUP **LOG** SONYDB
TO DISK = 'H:\SQLDBBACKUPS\SINGAPORE\SONYDB_Tlog1.trn'
WITH, COMPRESSION, STATS=5

=====

November 18, 2020 - Wednesday-- Full RM, Backups

=====

Simple RM -- F, D, No TLog

DB Engine will take care of log file growth.

Once log file reaches 70 %, DB Engine will trigger checkpoint and truncates (Removes inactive/closed transactions) from log file and creates free space inside log file to take more transactions.

--- PIT -- Point in time restoration is not possible.

<https://www.sqlshack.com/understanding-database-recovery-models/>

Transaction -- Set of commands following ACID Properties.

Transaction -- has 2 states:

- 1. Open Transaction \ Active Transaction**
- 2. Closed Transaction \ In active Transaction.**

A - Atomicity

C - COnsistency

I - Isolation

D- Durability

Begin Tran

=====

=====

=====

Commit -- If everything looks ok. Saving Permanently

Else

--- Rollback --Changes will be rolled back.

FULL

In this recovery model, all the transactions (DDL (Data Definition Language) + DML (Data Manipulation Language)) are fully recorded in the transaction log file.

DDL -- Data

The log sequence is unbroken and is preserved for the databases restore operations.

Unlike the Simple recovery model, the transaction log file is not auto-truncated during CHECKPOINT operations.

All restore operations are supported, including point-in-time restore, page restore and file restore.

Reasons to choose the full database recovery model:

1. Supporting mission critical applications
2. Design High Availability solutions == **Production or LIVE Databases.**
3. To facilitate the recovery of all the data with zero or minimal data loss
4. If the database designed to have multiple filegroups, and you want to perform a piecemeal restore of read/write secondary filegroups and, optionally, read-only filegroups.
5. Allow arbitrary point-in-time restoration
6. Restore individual pages
7. Incur high administration overhead (disadvantage)

It supports all type of following backups

1. Full backup

2. Differential backup
3. Transaction log backup
4. Copy-Only backup
5. File and/or file-group backup
6. Partial backup

November 20, 2020 - Friday-- Full RM, Backups

=====

--in last class we have seen how to take

--Full Backup

--Diff Backup

--Tlog Backup

--TSQL Script

--backup

--backup path-- H:\SQLBackups\Singapore

--backup path: H:\SQLBackups\Singapore

--We are working on GUI Based backups.

--Full backup

--Diff backup

--tlog backup

BACKUP DATABASE SONYDB

TO DISK = 'H:\SQLBackups\Singapore\SonyDB_FULL.bak'

WITH COMPRESSION, STATS=10

GO

--This is to verify backup file validity

RESTORE **VERIFYONLY** FROM DISK = 'H:\SQLBackups\Singapore\SonyDB_FULL.bak'

--This will give us the no of files inside the backup file.

RESTORE **FILELISTONLY** FROM DISK = 'H:\SQLBackups\Singapore\SonyDB_FULL.bak'

--This will let us know the backup details like full bkp/diff bkp/tlog/compression bkp ?

RESTORE **HEADERONLY** FROM DISK = 'H:\SQLBackups\Singapore\SonyDB_FULL.bak'

BULK_LOGGED

It's a special purpose database configuration option and it works similar to FULL recovery model except that certain bulk operations can be minimally logged.

The transaction log file uses a technique known as minimal logging for bulk operations

. The catch is that it's not possible to restore specific point-in-time data.

Reasons to choose the bulk logged recovery model:

1. Use minimal logging technique to prevent log file growth
2. If the database is subjected to periodic bulk operations

It supports all types of backups:

1. Full backup
2. Differential backup
3. Transaction log backup
4. Copy-Only backup
5. File and/or file-group backup
6. Partial backup

Full RM -- Consistency - Good
Performance - Low

Bulk Logged RM -- Consistency -- Low
Performance - High

Full RM -- Point In Time Restore is possible. (Adv)
-- Overhead of Tlog file maintenance.
-- To avoid this, we need to setup Tlog backups.

Bulk logged -- PIT is NOT Possible (Disadv)
-- No overhead of Tlog file growth mostly (Adv)

Restore operations Type:
=====

RESTORE DATABASE WITH NORECOVERY

- DB will be in RESTORING Status
- DB is not accessible
- DB will not be in ONLINE STATUS.

RESTORE DATABASE WITH RECOVERY

- DB will be in online status
- DB will be accessible
- DB will be in multi-user mode.

RESTORE DATABASE WITH STANDBY

- DB will be in online status.
- DB will be in READONLY Status.
- DB will be accessible but NOT Writable.

Environments:

1. Production or Live
2. QA - Quality Assurance
3. UAT -- User Acceptance Testing
4. Stage
5. Test
6. Dev

Database refresh (DB Migration)

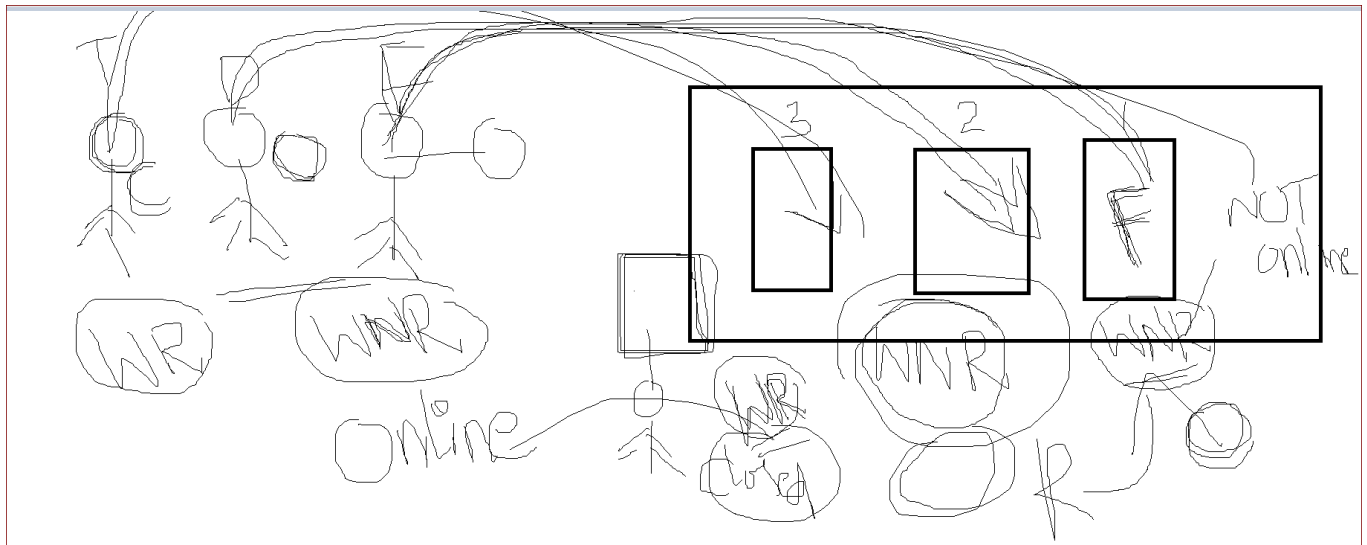
=====

Source: Higher (Production)

Destination: Lower Env - QA\UAT\Stage\Test\Dev

Need below:

1. Ticket No.
2. Manager approvals.
3. Security Exception Approvals.



Case-1:

=====

Bkp-1

Bkp-2

Bkp-3

Restore bkp-1 with norecovery -- NOT ONLINE, NOT Access, Restoring

Restore bkp-2 with norecovery -- NOT ONLINE, NOT Access, Restoring

Restore bkp-3 with Recovery -- Comes Online, Accessible.

Case-2:

Bkp-1

Bkp-2

Bkp-3

Bkp-4

Bkp-5

Restore bkp-1 with norecovery -- -- NOT ONLINE, NOT Access, Restoring

Restore bkp-2 with norecovery -- -- NOT ONLINE, NOT Access, Restoring

Restore bkp-3 with norecovery -- -- NOT ONLINE, NOT Access, Restoring

Restore bkp-4 with norecovery -- -- NOT ONLINE, NOT Access, Restoring

Restore bkp-5 with RECOVERY -- DB comes into online

100 files

99 -- with norecovery

100 th -- with recovery

<https://www.sqlshack.com/understanding-database-recovery-models/>

November 27, 2020 - Friday-- Restore Module -- 7:30 PM EST

=====

Restore operations Type:

=====

RESTORE DATABASE WITH NORECOVERY

-- DB will be in RESTORING Status

-- DB is not accessible

-- DB will not be in ONLINE STATUS.

RESTORE DATABASE WITH RECOVERY

- DB will be in online status
- DB will be accessible
- DB will be in multi-user mode.

RESTORE DATABASE WITH STANDBY

- DB will be in online status.
- DB will be in READONLY Status.
- DB will be accessible but NOT Writable.

Case-1:

Source:

Source DB Name: SonyDB

Destination DB Name: SonyDB_Test

sp_helpdb sonydb

--full backup

BACKUP DATABASE SONYDB

TO DISK = 'H:\SQLBackups\Singapore\SONYDB_FULL_11272020_0451PM.BAK'

WITH COMPRESSION, STATS=5

--diff backup

BACKUP DATABASE SONYDB

TO DISK = 'H:\SQLBackups\Singapore\SONYDB_DIFF_11272020_0452PM.DIFF'

WITH DIFFERENTIAL, COMPRESSION, STATS=5

--Tlog backup

BACKUP log SONYDB

TO DISK = 'H:\SQLBackups\Singapore\SONYDB_Tlog1_11272020_0453PM.trn'

WITH COMPRESSION--, STATS=25


```
--'H:\SQLBackups\Singapore\SONYDB_FULL_11272020_0451PM.BAK'  
--'H:\SQLBackups\Singapore\SONYDB_DIFF_11272020_0452PM.DIFF'  
--'H:\SQLBackups\Singapore\SONYDB_Tlog1_11272020_0453PM.trn'
```

-- while restoring any backup file always use USE MASTER Statement.

--1. ARE WE CREATING A BRAND NEW DATABASE USING AN OLD\EXISTING BACKUP FILE? YES:
SONYDB_TEST

--GENERAL SYNTAX FOR BACKUP FILE RESTORES

```
use master
```

```
go
```

```
RESTORE DATABASE DBNAME
```

```
FROM DISK = 'BACKUP PATH & BACKUP FILE NAME'
```

```
WITH NORECOVERY\RECOVERY\STANDBY,STATS=5,
```

```
MOVE 'DATA FILE LOGICAL NAME' TO 'NEW DATA FILE PHYSICAL PATH',
```

```
MOVE 'LOG FILE LOGICAL NAME' TO 'NEW LOG FILE PHYSICAL PATH'
```

--FULL BACKUP FILE RESTORE TOWARDS CREATING BRAND NEW DATABASE

```
RESTORE DATABASE SONYDB_TEST -- BRAND NEW DB
```

```
FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_FULL_11272020_0451PM.BAK' --THIS IS FULL  
BACKUP PATH
```

```
WITH NORECOVERY,STATS=15, -- DB NOT ACCESSIBLE, NOT ONLINE -- NORECOVERY  
STANDS FOR
```

```
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\SONYDB_TEST.mdf',
```

```
MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\SONYDB_TEST_log.ldf'
```

--RESTORE FILELISTONLY FROM DISK =

```
'H:\SQLBackups\Singapore\SONYDB_FULL_11272020_0451PM.BAK'
```

--THIS WILL FETCH YOU THE NO OF FILES INSIDE THE BACKUP FILE.

--DIFF BACKUP FILE RESTORE TOWARDS CREATING BRAND NEW DATABASE

```
RESTORE DATABASE SONYDB_TEST -- BRAND NEW DB
```

```
FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_DIFF_11272020_0452PM.DIFF' --THIS IS DIFF BACKUP  
PATH
```

```
WITH NORECOVERY,STATS=15, -- DB NOT ACCESSIBLE, NOT ONLINE -- NORECOVERY  
STANDS FOR
```

```
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\SONYDB_TEST.mdf',
```

```
MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\SONYDB_TEST_log.ldf'
```

--TLOG BACKUP FILE RESTORE TOWARDS CREATING BRAND NEW DATABASE

```
RESTORE LOG SONYDB_TEST -- BRAND NEW DB
FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_Tlog1_11272020_0453PM.TRN' --THIS IS TLOG
BACKUP PATH
WITH RECOVERY,STATS=15,          -- DB WILL BE ACCESSIBLE, ONLINE -- RECOVERY STANDS FOR
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\SONYDB_TEST.mdf',
MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\SONYDB_TEST_log.ldf'
```

--2. ARE WE OVERWRITING EXISTING DB USING AN OLD\EXISTING BACKUP FILE?

```
use master
go
RESTORE DATABASE DBNAME
FROM DISK = 'BACKUP PATH & BACKUP FILE NAME'
WITH NORECOVERY\RECOVERY\STANDBY,STATS=5,
MOVE 'DATA FILE LOGICAL NAME' TO 'DATA FILE PHYSICAL PATH',
MOVE 'LOG FILE LOGICAL NAME' TO 'LOG FILE PHYSICAL PATH'
```

```
use master
go
RESTORE DATABASE Sonydb
FROM DISK = 'BACKUP PATH & BACKUP FILE NAME'
WITH REPLACE, RECOVERY, STATS=5,
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\Sonydb.mdf',
MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\Sonydb_log.ldf'
```

December 01, 2020 - Tuesday

--case-1:creating a brand new database using old backup file.

--case-2: overwriting/Replace the existing db with an old backup file.

use master

```
restore database dbname
from disk = 'backup path and backup extension'
with recovery\norecovery, stats=5,
move 'DF LN' to 'DF Physical Path',
move 'LF LN' to 'LF Physical Path'
```

--full backup file restore

```
RESTORE DATABASE [Sonydb] -- HERE WE ARE REPLACING THE EXISTING DB
FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_FULL_11272020_0451PM.BAK'
WITH REPLACE, NORECOVERY, STATS=5,
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\Sonydb.mdf',
MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\Sonydb_log.ldf'
```

--DIFF backup file restore

```
RESTORE DATABASE [Sonydb]
FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_DIFF_11272020_0452PM.DIFF'
WITH NORECOVERY, STATS=5,
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\Sonydb.mdf',
MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\Sonydb_log.ldf'
```

--TLOG backup file restore

```
RESTORE LOG [Sonydb]
FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_Tlog1_11272020_0453PM.TRN'
WITH RECOVERY, STATS=5,
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\Sonydb.mdf',
MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\Sonydb_log.ldf'
```

CASE-1:

--CREATING A NEW BRAND DB USING EXISTING BACKUP FILE.

```
--RESTORE FILELISTONLY FROM DISK =
'H:\SQLBackups\Singapore\SONYDB_FULL_11272020_0451PM.BAK'
```

--FULL BACKUP FILE RESTORE

```
RESTORE DATABASE SONYDB_NEW
FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_FULL_11272020_0451PM.BAK'
WITH NORECOVERY, STATS=5,
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\SONYDB_NEW.mdf',
MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\SONYDB_NEW_log.ldf'
```

--DIFF BACKUP FILE RESTORE

```
RESTORE DATABASE SONYDB_NEW
FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_DIFF_11272020_0452PM.DIFF'
WITH NORECOVERY, STATS=5,
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\SONYDB_NEW.mdf',
MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\SONYDB_NEW_log.ldf'
```

--TLOG BACKUP FILE RESTORE

```
RESTORE LOG SONYDB_NEW
FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_Tlog1_11272020_0453PM.TRN'
WITH RECOVERY, STATS=5,
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\SONYDB_NEW.mdf',
MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\SONYDB_NEW_log.ldf'
```

--CASE-2: REPLACE the existing database using the backup file

```
--RESTORE FILELISTONLY FROM DISK =
'H:\SQLBackups\Singapore\SONYDB_FULL_11272020_0451PM.BAK'
```

--FULL BACKUP FILE RESTORE

```
RESTORE DATABASE SONYDB
FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_FULL_11272020_0451PM.BAK'
WITH REPLACE, NORECOVERY, STATS=5,
MOVE 'Sonydb' TO 'E:\SQLData\Singapore\Sonydb.mdf',
```

MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\Sonydb_log.ldf'

--DIFF BACKUP FILE RESTORE

RESTORE DATABASE SONYDB

FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_DIFF_11272020_0452PM.DIFF'

WITH NORECOVERY, STATS=5,

MOVE 'Sonydb' TO 'E:\SQLData\Singapore\Sonydb.mdf',

MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\Sonydb_log.ldf'

--TLOG BACKUP FILE RESTORE

RESTORE LOG SONYDB

FROM DISK = 'H:\SQLBackups\Singapore\SONYDB_Tlog1_11272020_0453PM.TRN'

WITH RECOVERY, STATS=5,

MOVE 'Sonydb' TO 'E:\SQLData\Singapore\Sonydb.mdf',

MOVE 'Sonydb_log' TO 'F:\SQLLogs\Singapore\Sonydb_log.ldf'

--case-1: Create Brand new db using backup file.

--Alpha -- original db

--Alpha_New -- new db

--full backup

backup database Alpha

to disk = 'H:\SQLBackups\Singapore\Alpha_Full_12082020_0447pm.bak'

with compression, stats=5

--to disk = 'H:\SQLBackups\Singapore\Alpha_Full_12082020_0447pm.bak'

--restore above backup file as Alpha_new which creates brand new db.

--restore filelistonly from disk = 'H:\SQLBackups\Singapore\Alpha_Full_12082020_0447pm.bak'

--Restore statement

restore database Alpha_New

from disk = 'H:\SQLBackups\Singapore\Alpha_Full_12082020_0447pm.bak'

```
with recovery,stats=25, replace,  
move 'alpha' to 'E:\SQLData\Singapore\alpha_new.mdf',  
move 'alpha_log' to 'F:\SQLLogs\Singapore\alpha_new_log.ldf'
```

--<https://www.sqlshack.com/how-to-get-a-sql-database-restore-history/>

```
SELECT  
rh.destination_database_name AS [Database],  
CASE WHEN rh.restore_type = 'D' THEN 'Database'  
WHEN rh.restore_type = 'F' THEN 'File'  
WHEN rh.restore_type = 'I' THEN 'Differential'  
WHEN rh.restore_type = 'L' THEN 'Log'  
ELSE rh.restore_type  
END AS [Restore Type],  
rh.restore_date AS [Restore Date],  
bmf.physical_device_name AS [Source],  
rf.destination_phys_name AS [Restore File],  
rh.user_name AS [Restored By]  
FROM msdb.dbo.restorehistory rh  
INNER JOIN msdb.dbo.backupset bs ON rh.backup_set_id = bs.backup_set_id  
INNER JOIN msdb.dbo.restorefile rf ON rh.restore_history_id = rf.restore_history_id  
INNER JOIN msdb.dbo.backupmediafamily bmf ON bmf.media_set_id = bs.media_set_id  
ORDER BY rh.restore_history_id DESC  
GO
```

--case-1: replace existing db

--Alpha -- original db

--full backup

backup database Alpha

to disk = 'H:\SQLBackups\Singapore\Alpha_Full_12082020_0447pm.bak'

with compression, stats=5

sp_helpdb alpha

--restore filelistonly from disk = 'H:\SQLBackups\Singapore\Alpha_Full_12082020_0447pm.bak'

--Restore statement

restore database Alpha

from disk = 'H:\SQLBackups\Singapore\Alpha_Full_12082020_0447pm.bak'

with recovery,stats=25, replace,

move 'alpha' to 'E:\SQLData\Singapore\alpha.mdf',

move 'alpha_log' to 'F:\SQLLogs\Singapore\alpha_log.ldf'

→ DB Refresh Task:

--backup db

--user permissions

--restore db

December 04, 2020 - Monday

DBA works in below environments.

Big Project:

1. Production (live)
2. QA (Quality Assurance)
3. Staging
4. Development
5. UAT (Test) -- User Acceptance Testing

Small Project:

1. Production
2. QA
3. Dev
4. Test.

Very Small Project:

1. Production
2. QA
3. Dev/Test

DB Refresh -- DB Restore from Source to Destination

99 % Cases:

Source: Production SQL Server

Destination: QA/Stage/Dev/Test SQL Server.

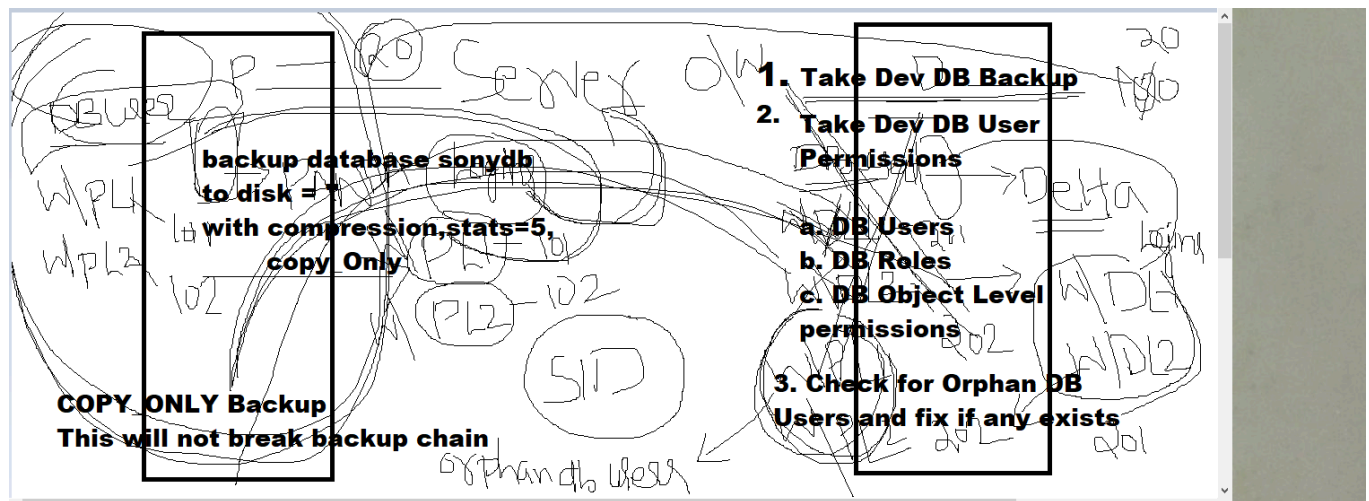
Note: DBA Always needs Security Exception approvals to perform DB Refresh from Prod to lower env (QA/Stage/Dev/Test).

Data Security Team

Data Masking

praveen.madupu@gmail.com -- DBA

praxxxn.mxxxxu@gmail.com -- Vendor



December 16, 2020 - Wednesday

DB Refresh Activity:

Higher Env to Lower Env

Source to Destination

Prod → Dev/Stage/QA/Test → 99 %

Dev/Stage/QA/Test → Prod → 1 %

Source:

- 1) Source Server Name
- 2) Source DB Name
- 3) Source DB Size
- 4) Source back path
- 5) Source Logins

Destination:

Note: Destination Logins.

- 1) Dest Server name
- 2) Dest DB Name (are we creating new DB or replacing an existing DB)
- 3) Destination DB Data & Log File paths
- 4) Take dev DB full backup for safety.
- 5) Take dev db Permissions before restore.

<http://www.dbathings.com/database-permissions-extraction-script/>

- 6) Perform DB Restore.
- 7) Apply DB Permissions
- 8) Fix any orphan users

DECLARE @DatabaseUserName [sysname]


```

SET NOCOUNT ON
DECLARE
@errStatement [varchar](8000),
@msgStatement [varchar](8000),
@DatabaseUserID [smallint],
@ServerUserName [sysname],
@RoleName [varchar](8000),
@MEmberName [varchar](8000),
@ObjectID [int],
@ObjectName [varchar](8000),
@objectpermissions [varchar](8000),
@schemamembername [varchar](8000),
@schemaname [varchar](8000),
@db_permissions varchar(8000)
PRINT '-- CREATE USERS --'
DECLARE _users
CURSOR LOCAL FORWARD_ONLY READ_ONLY
FOR
select [master].[sys].[server_principals].[name] ,
[sys].[database_principals].[name]
from [sys].[database_principals] INNER JOIN [master].[sys].[server_principals]
on [sys].[database_principals].[name]=[master].[sys].[server_principals].[name]
where [master].[sys].[server_principals].[type] in ('U', 'G', 'S')

OPEN _users FETCH NEXT FROM _users INTO @ServerUserName, @DatabaseUserName
WHILE @@FETCH_STATUS = 0
BEGIN
SET @msgStatement ='if not exists(SELECT 1 from sys.database_principals where type in ("U", "G", "S")
and name ="'
+@DatabaseUserName+'" ) '+ CHAR(13) +
'BEGIN '+ CHAR(13) +
'CREATE USER ['+ @DatabaseUserName + '] ' + ' FOR LOGIN [' + @ServerUserName + ']' + CHAR(13) +
'END'
--SET @msgStatement = 'CREATE USER [' --CREATE USER [mlapenna] FOR LOGIN [mlapenna]
-- + @DatabaseUserName + ']' + ' FOR LOGIN [' + @ServerUserName + ']'
PRINT @msgStatement
FETCH NEXT FROM _users INTO @ServerUserName, @DatabaseUserName
END
CLOSE _users
DEALLOCATE _users

PRINT '-- CREATE DB ROLES--'
DECLARE _roles
CURSOR LOCAL FORWARD_ONLY READ_ONLY
FOR
select [NAME] from [sys].[database_principals] where type='R' and is_fixed_role != 1 and name not like
'public'
OPEN _roles FETCH NEXT FROM _roles INTO @RoleName
WHILE @@FETCH_STATUS=0
BEGIN
SET @msgStatement ='if not exists(SELECT 1 from sys.database_principals where type="R" and name ="'
+@RoleName+'" ) '+ CHAR(13) +
'BEGIN '+ CHAR(13) +
'CREATE ROLE ['+ @RoleName + ']' + CHAR(13) +

```

```

'END'
PRINT @msgStatement
FETCH NEXT FROM _roles INTO @RoleName
END
CLOSE _roles
DEALLOCATE _roles
PRINT '-- CREATE APPLICATION ROLES--'
DECLARE _roles
CURSOR LOCAL FORWARD_ONLY READ_ONLY
FOR
select [NAME],default_schema_name from [sys].[database_principals] where type='A' and is_fixed_role !=
1 and name not like 'public'
OPEN _roles FETCH NEXT FROM _roles INTO @RoleName, @schemaname
WHILE @@FETCH_STATUS=0
BEGIN
SET @msgStatement='if not exists(SELECT 1 from sys.database_principals where type="A" and name ="'
+@RoleName+'" ) '+ CHAR(13) +
'BEGIN '+ CHAR(13) +
'CREATE APPLICATION ROLE ['+ @RoleName + '] with DEFAULT_SCHEMA =['+@schemaname +'],
PASSWORD = "{Please provide the password here}"'+CHAR(13) +
'END'
PRINT @msgStatement
FETCH NEXT FROM _roles INTO @RoleName, @schemaname
END
CLOSE _roles
DEALLOCATE _roles
PRINT '-- ADD ROLE MEMBERS--'
DECLARE _role_members
CURSOR LOCAL FORWARD_ONLY READ_ONLY
FOR
SELECT a.name , b.name
from sys.database_role_members d INNER JOIN sys.database_principals a
on d.role_principal_id=a.principal_id
INNER JOIN sys.database_principals b
on d.member_principal_id=b.principal_id
where b.name <> 'dbo'
order by 1,2
OPEN _role_members FETCH NEXT FROM _role_members INTO @RoleName, @membername
WHILE @@FETCH_STATUS = 0
BEGIN
SET @msgStatement = 'EXEC [sp_addrolemember] ' + '@rolename = [' + @RoleName + '], ' +
'@membername = [' + @membername + ']'
PRINT @msgStatement
FETCH NEXT FROM _role_members INTO @RoleName, @membername
END
close _role_members
deallocate _role_members
--SCRIPT GRANTS for Database Privileges
PRINT '-- SCRIPT GRANTS for Database Privileges--'
DECLARE _db_permissions
CURSOR LOCAL FORWARD_ONLY READ_ONLY
FOR
SELECT a.state_desc + ' ' + a.permission_name + ' ' + 'TO [' + b.name + ']' COLLATE
LATIN1_General_CI_AS

```

```

FROM sys.database_permissions a inner join sys.database_principals b
ON a.grantee_principal_id = b.principal_id
WHERE b.principal_id not in (0,1,2) and a.type not in ('CO') and a.class = 0
OPEN _db_permissions FETCH NEXT FROM _db_permissions INTO @db_permissions
WHILE @@FETCH_STATUS = 0
BEGIN
PRINT @db_permissions
FETCH NEXT FROM _db_permissions INTO @db_permissions
END
close _db_permissions
deallocate _db_permissions

```

-- SCRIPT GRANTS for Schema Privileges

```
PRINT '-- SCRIPT GRANTS for Schema Privileges--'
```

```
DECLARE _schema_members
```

```
CURSOR LOCAL FORWARD_ONLY READ_ONLY
```

```
FOR
```

```
SELECT a.state_desc + ' ' + a.permission_name + ' ' + 'ON SCHEMA::[' + b.name + ']' + ' TO ' + c.name
COLLATE LATIN1_General_CI_AS
```

```
FROM sys.database_permissions a INNER JOIN sys.schemas b
```

```
ON a.major_id = b.schema_id INNER JOIN sys.database_principals c ON a.grantee_principal_id =
c.principal_id
```

```
OPEN _schema_members FETCH NEXT FROM _schema_members INTO @schemamembername
```

```
WHILE @@FETCH_STATUS = 0
```

```
BEGIN
```

```
PRINT @schemamembername
```

```
FETCH NEXT FROM _schema_members INTO @schemamembername
```

```
END
```

```
close _schema_members
```

```
deallocate _schema_members
```

-- SCRIPT GRANTS for Objects Level Privileges

```
PRINT '-- SCRIPT GRANTS for Object Privileges--'
```

```
DECLARE _object_permissions
```

```
CURSOR LOCAL FORWARD_ONLY READ_ONLY
```

```
FOR
```

```
SELECT
```

```
state_desc + ' ' + permission_name + ' on [' + sys.schemas.name + '].[' + sys.objects.name + '] to [' +
sys.database_principals.name + ']' COLLATE LATIN1_General_CI_AS
```

```
from sys.database_permissions
```

```
join sys.objects on sys.database_permissions.major_id =
sys.objects.object_id
```

```
join sys.schemas on sys.objects.schema_id = sys.schemas.schema_id
```

```
join sys.database_principals on sys.database_permissions.grantee_principal_id =
sys.database_principals.principal_id
```

```
where sys.database_principals.name not in ( 'public', 'guest')
```

```
--order by 1, 2, 3, 5
```

```
OPEN _object_permissions FETCH NEXT FROM _object_permissions INTO @objectpermissions
```

```
WHILE @@FETCH_STATUS = 0
```

```
BEGIN
```

```
PRINT @objectpermissions
```

```
FETCH NEXT FROM _object_permissions INTO @objectpermissions
```

```
END
```

```
close _object_permissions
```

```
deallocate _object_permissions  
PRINT 'GO'
```

Below are the Advworks database permissions before refresh on Destination.

```
-- CREATE USERS --  
if not exists(SELECT 1 from sys.database_principals where type in ('U', 'G', 'S') and name  
='WONDERLA\DestinationWL1' ) BEGIN CREATE USER [WONDERLA\DestinationWL1] FOR LOGIN  
[WONDERLA\DestinationWL1]END  
if not exists(SELECT 1 from sys.database_principals where type in ('U', 'G', 'S') and name  
='WONDERLA\DestinationWL2' ) BEGIN CREATE USER [WONDERLA\DestinationWL2] FOR LOGIN  
[WONDERLA\DestinationWL2]END  
-- CREATE DB ROLES--  
-- CREATE APPLICATION ROLES--  
-- ADD ROLE MEMBERS--  
EXEC [sp_addrolemember] @rolename = [db_datareader], @membername = [WONDERLA\DestinationWL1]  
EXEC [sp_addrolemember] @rolename = [db_datareader], @membername = [WONDERLA\DestinationWL2]  
EXEC [sp_addrolemember] @rolename = [db_datawriter], @membername = [WONDERLA\DestinationWL1]  
EXEC [sp_addrolemember] @rolename = [db_datawriter], @membername = [WONDERLA\DestinationWL2]  
EXEC [sp_addrolemember] @rolename = [db_ddladmin], @membername = [WONDERLA\DestinationWL1]  
EXEC [sp_addrolemember] @rolename = [db_ddladmin], @membername = [WONDERLA\DestinationWL2]  
-- SCRIPT GRANTS for Database Privileges--  
-- SCRIPT GRANTS for Schema Privileges--  
-- SCRIPT GRANTS for Object Privileges--  
GO
```

December 31, 2020 - Wednesday--7 PM EST

SQL Server has 2 parts:

1) SQL Server Basics -- Developers --> Will write code/modify/update logic

DB, Objects: Tables, views, Stored Procedures, functions,etc

2) SQL Server Advance -- Administration -- DBA

1) Administrates SQL Servers - Takes care of SQL Server & DB's.

2) SQL Server and DB Security

3) Backups & Restorations

4) Jobs & Automation of Admin tasks.

5) Setting up HA and DR and troubleshooting if any issue

HA - High Availability

DR - Disaster Recovery

6) SQL Server\DB Migrations, Upgradations

7) SQL Server Performance Tuning

8) Troubleshooting SQL Server, DB Issues and fixing them.

Note: A SQL DBA needs some basic knowledge of SQL Server.

Table, SP, Views, Functions, etc.

SP: a pre-written statement which is used to execute without executing the entire statements.

```
create procedure USP_EmpDetails
as
select * from emp
where emp name like '%vivian%'
```

To execute and get data:

```
use dbname
go
exec USP_EmpDetails
```

January 12, 2021 - Tuesday--7 PM EST

Topic: Database Maintenance Plans.

sp_helpdb

--system databases

--master

--model

--msdb

--tempdb

--user databases

--s1--16 mb

--s2--24 mb

--s3--32 mb

--how to create db maintenence plans

--1. Maintenance plans (Wizard or Drag and Drop method)

--2. using customized job scripts

--3. 3rd party maintenance job scripts -- Ola Hallengren -- 5 mins

--below tasks

--1. backups -- Full/diff/tlog

--2. integrity check --

--3. rebuild index

--4. reorganize index

--5. update statistics

--6. maintenance cleanup -- deletes old backup files.

select * from employee where empid = 100

--50,000

05 seconds

--backup files retention period

s1 -- 1 GB

s2 -- 1 GB

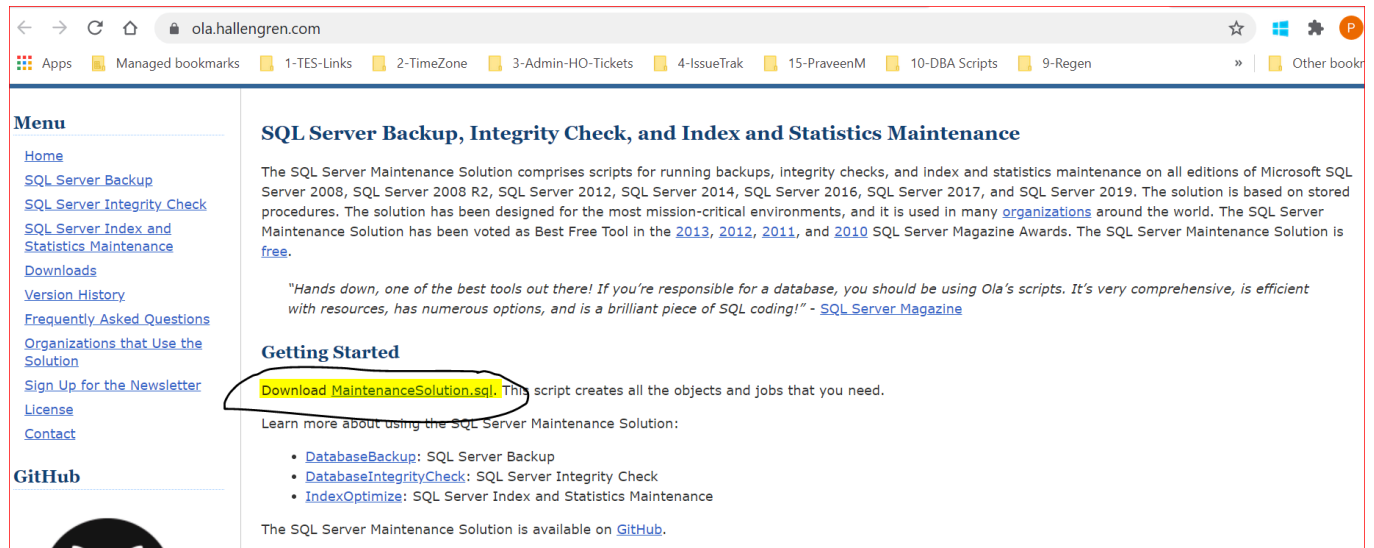
s3 -- 1 GB

--total backup size: 3 GB

January 14, 2021 - Thursday--7 PM EST

Topic: Database Maintenance Plans using 3rd party scripts -- Ola Hallengren

<https://ola.hallengren.com/>



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Git Hub

SQL Server Backup, Integrity Check, and Index and Statistics Maintenance

The SQL Server Maintenance Solution comprises scripts for running backups, integrity checks, and index and statistics maintenance on all editions of Microsoft SQL Server 2008, SQL Server 2008 R2, SQL Server 2012, SQL Server 2014, SQL Server 2016, SQL Server 2017, and SQL Server 2019. The solution is based on stored procedures. The solution has been designed for the most mission-critical environments, and it is used in many [organizations](#) around the world. The SQL Server Maintenance Solution has been voted as Best Free Tool in the [2013](#), [2012](#), [2011](#), and [2010](#) SQL Server Magazine Awards. The SQL Server Maintenance Solution is [free](#).

"Hands down, one of the best tools out there! If you're responsible for a database, you should be using Ola's scripts. It's very comprehensive, is efficient with resources, has numerous options, and is a brilliant piece of SQL coding!" - [SQL Server Magazine](#)

Getting Started

[Download MaintenanceSolution.sql](#). This script creates all the objects and jobs that you need.

Learn more about using the SQL Server Maintenance Solution:

- [DatabaseBackup](#): SQL Server Backup
- [DatabaseIntegrityCheck](#): SQL Server Integrity Check
- [IndexOptimize](#): SQL Server Index and Statistics Maintenance

The SQL Server Maintenance Solution is available on [GitHub](#).

```
/*
USE [master] -- Specify the database in which the objects will be created.

SET NOCOUNT ON

DECLARE @CreateJobs nvarchar(max) = 'Y' -- Specify whether jobs should be created.
DECLARE @BackupDirectory nvarchar(max) = 'H:\SQLBackups\' -- Specify the backup root directory. If no directory is specified, the default backup direct
DECLARE @CleanupTime int = 48 -- Time in hours, after which backup files are deleted. If no time is specified, then no backup files are d
DECLARE @OutputFileDirectory nvarchar(max) = 'H:\SQLBackups\' -- Specify the output file directory. If no directory is specified, then the SQL Server error
DECLARE @LogToTable nvarchar(max) = 'Y' -- Log commands to a table.
DECLARE @ErrorMessage nvarchar(max)
```

January 19, 2021 - Tuesday--7 PM EST

HA & DR

HA - High Availability

DR - Disaster Recovery

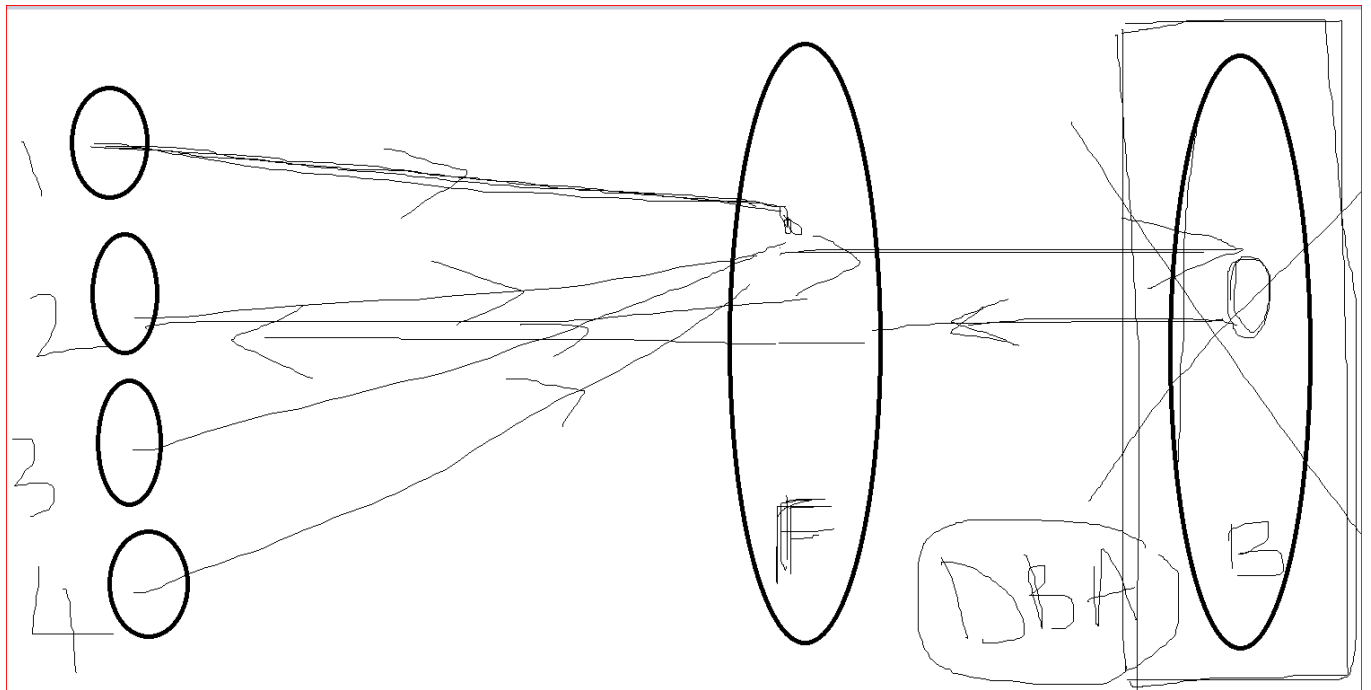
Below Configurations to achieve HA & DR.

- 1. Log shipping -- Generator -- SQL Server 2000**
- 2. DB Replication --Generator SQL Server 2000**
- 3. SQL Clustering --Generator/Inverter SQL Server 2000**
=====
- 4. DB Mirroring --Generator/Inverter --SQL Server 2005, 2008, 2012, further**
=====
- 5. SQL Alwayson --Generator/Inverter -- SQL Server 2012, 2014, 2016, further**
=====

Generator (Manual)

Inverter -- Manual/Automatic

Flash Lights -- Manual/Automatic



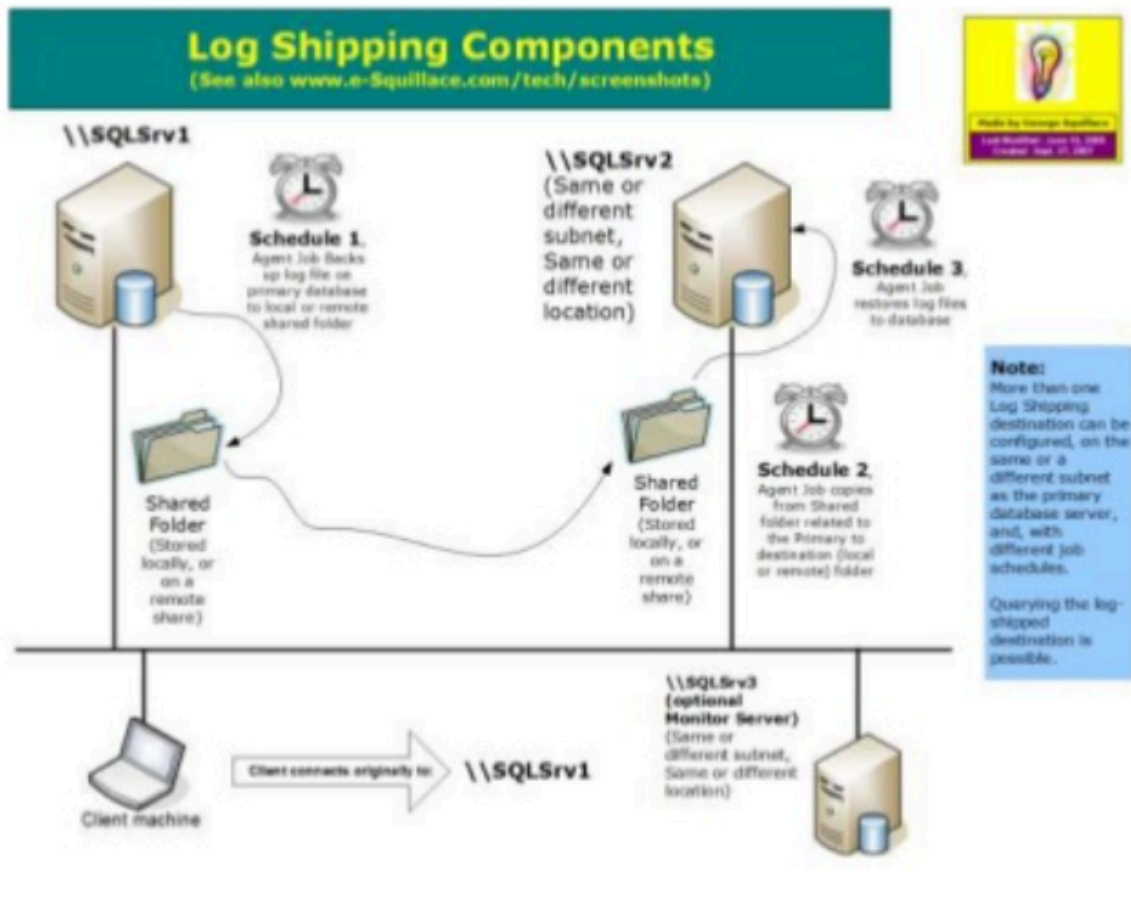
\\DC\C\$\SharedFolder2021

\\computername\Driveletter\$\foldername

<https://www.slideshare.net/ZebaAnsari7/sql-server-logshipping>

<https://www.slideshare.net/sunilannakumar/sql-server-log-shipping-by-sunil-kumar-anna>

Log Shipping



DB: SonyDB

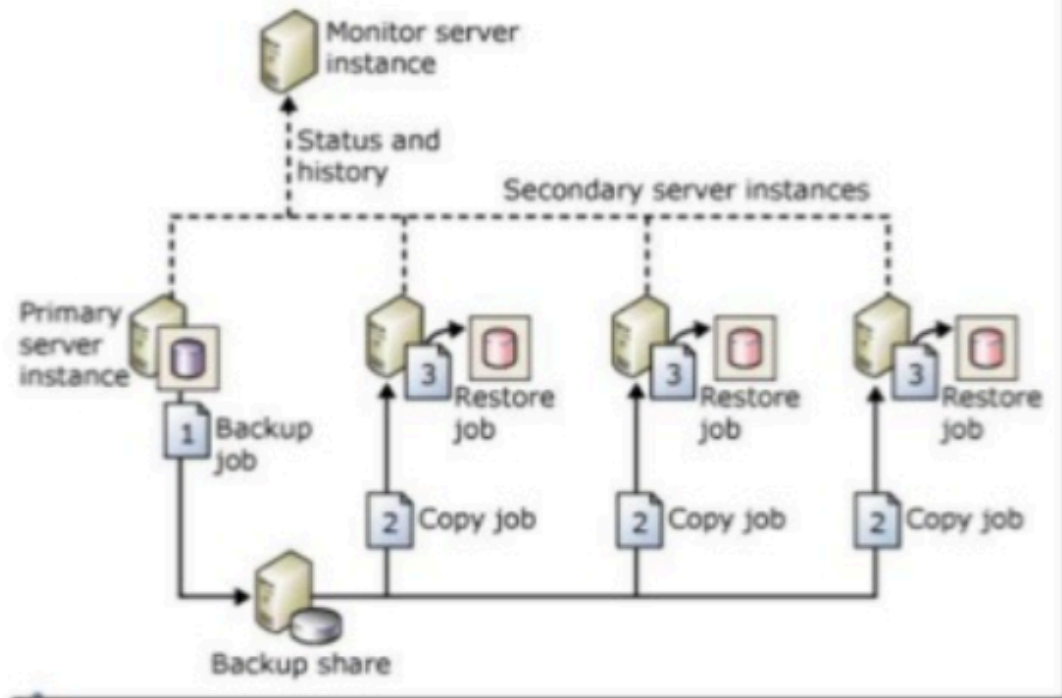
At Primary Server: 1) Tlog Backup Job -- Default: 15 mins every run

At Secondary Server: 2) Copy Job -- Default: 15 mins every run

3) Restore Job -- Default: 15 mins every run

Database Permissions extraction script

<http://www.dbathings.com/database-permissions-extraction-script/>



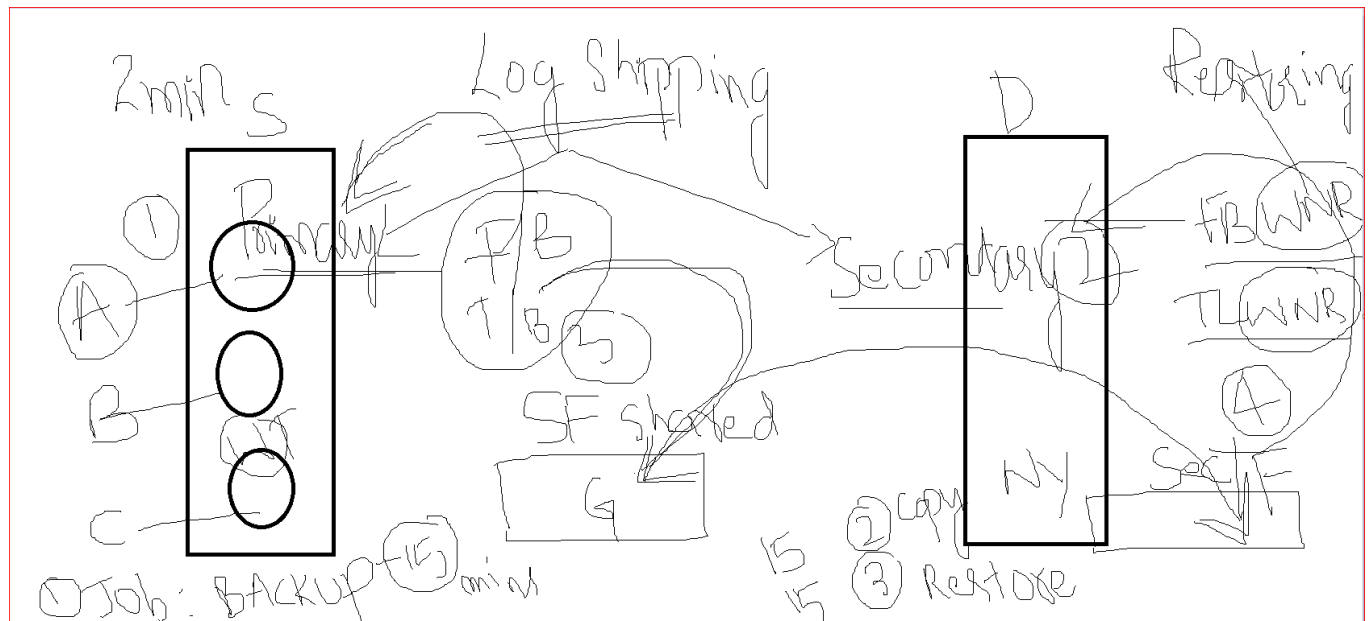
LS -- Source → Primary & Destination → Secondary

DB Mirroring -- Source → Principal & Destination → Mirror

DB Replication -- Source → Publisher & Destination → Subscriber

**SQL Cluster -- Source → Primary SQL Cluster/ Primary node
Destination → Secondary SQL Cluster/ Secondary node**

**SQL Awayson -- Source→ Primary Replica
Destination → Secondary Replica**



1) At Primary Server:

1. DB Should be in either Full / Bulk logged Recovery Model.
2. Take the required DB Full Backup
3. Take the required DB Tlog Backup.

2) At Secondary Server:

1. Restore Full backup of Source with NoRecovery
2. Restore Tlog backup of Source with NoRecovery.

Note: Create a SharedFolder on different Server: Example: **\\DC\SharedFolderLS**

3) Again at Primary Server:

- a) Go to the required Primary DB and setup logshipping via GUI.
- b) Here below jobs will get created on LS is setup on required PRIMARY DB.
- c) On Primary Server → Log backup job with 15 mins run by default
- d) On Secondary Server → Copy job and Restore Job with 15 mins run by default.

select @@version

--Source: Microsoft SQL Server 2016 (SP1) (KB3182545) - 13.0.4001.0 (X64)

--Destination: Microsoft SQL Server 2016 (RTM) - 13.0.1601.5 (X64) --> RTM Means No SP

Pending Topic:

- 1) How to apply SQL Service Pack on StandAlone SQL Server ? -- single SQL Server
- 2) How to apply SQL Service Pack on StandAlone SQL Server ? -- in Log Shipping 2 servers min: Primary & Secondary
- 3) DB M
- 4) DB R
- 5) SQL Cluster
- 6) SQL AlwaysOn

HA-DR

SQL Server 2000 Onwards

=====

1) SQL Server DB Log Shipping

Source: SQL Server 2008

Destin: SQL Server 2008 or 2012 or above

2) SQL Server Cluster

3) SQL Server DB Replication

=====

4) SQL Server DB Mirroring -- SQL 2005

5) SQL Server Alwayson -- SQL Server 2012

	A	B	C	D	E	F	G	H
1	HADR	Min servers	Source (R/W)	Destination	Other Server	HA	DR	Can we Read DB in Secondary Server
2	Log Shipping	2	Primary	Secondary	Monitor	No	YES	1) NoRecovery -- Not Readable 2) Standby -- Yes. Readable
3	Cluster	2	Active	Passive	NA	YES	YES	Leaving as of now.
4	Replication	2	Publisher	Subscriber	Distributor	NO	YES	Both pub & sub db's can be READ
5	Mirroring	2	Principal	Mirror	Witness	NO -- When witness is not included YES -- When witness is included	YES	NOT Readable on Mirror Server. (Restoring)
6	Alwayson	2	Primary Replica	Secondary Replica	NA	NO -- When Manual Failover is selected YES -- When Automatic FO is selected	YES	1) Readable -- YES -- If we allow 2) Not Readable -- No - If we don't allow
7								

Pre-requisites for Logshipping.

Primary Server:

1. DB Should be under FULL Recovery Model.
2. Take Full Backup
3. Take Tlog Backup
4. SQL Service Account should be running on Domain Service Account
Example: google\SQLService

Secondary Server:

1. Restore Full Backup with NORECOVERY
2. Restore Tlog Backup with NORECOVERY
3. Need a folder named: SECONDARY FOLDER. SQL Service Account needs R & W Permission.
4. SQL Service Account should be running on Domain Service Account
Example: google\SQLService

SHARED PATH: This is used to hold primary database transaction log backups.

SQL Service Account should have READ & WRITE Permission

SQL Server Patching (Just putting updates) -- Upgradation

Windows 7,8,10 -- User Operating Systems

Patch:

- 1) Hot Fix (HF1, HF2, HF3,,,,,,etc HF10 etc)
- 2) Service Pack -- SP1,SP2, SP3, SP4.
- 3) CU - Cumulative Update -- CU1, CU2,.....CU 22

<https://sqlserverbuilds.blogspot.com/>

SQL Server 2106 --- (RTM) - No Service Packs included internally

SQL Server 2016 -- SP1

SQL Server 2016 -- SP2

SQL Server System Types:

=====

1. SQL Server StandAlone System --- Easy --- takes less than 15 to 30 mins
2. SQL Server in Logshipping -- 2 min or more than
3. SQL Server In DB Mirroring -- 2 min or 3 max
4. SQL Server in DB Replication -- 2 min or more than
5. SQL Server in Cluster -- 2 min or more than
6. SQL Server in Alwayson -- 2 min or more than

1. SQL Server StandAlone System --- Easy --- takes less than 15 to 30 mins

Patching is done in below ways.

1. Apply patch in lower environments -- Dev\Test -- if this is success then
2. Apply patch in Stage\Pre-prod env -- if this is success then
3. Apply patch in Prod Env.

<https://sqlserverbuilds.blogspot.com/>

<https://www.catalog.update.microsoft.com/Search.aspx?q=SQL%20Server>

<https://www.catalog.update.microsoft.com/Search.aspx?q=SQL%20Server%202016>

Patching Process:

=====

1. Pre-patching--Take all required screenshots, take all databases full backups, take logins script, take all agent jobs script, note down all important config details.
2. Patching -- down load latest SP, apply patch. post this if needed restart Server.
3. Post-Patching -- Verify the SQL Server Version number, Verify all DBs are in online status, accessible, etc...

8:30 AM EST to 9:45 AM EST -- Monday to Friday

LogShipping -- DB Level -- We need to copy all DB contents from Source to Destination

DB Mirroring

SQL Cluster

SQL Server Alwayson

=====

SQL Server DB Replication:

Source Server → Publisher Server

Source DB → Publisher DB

Destination Server → Subscriber Server

Destination DB → Subscription DB

Monitor Server → Distributor Server

Monitor DB → **Distribution DB**.

Replication Types:

1) Snapshot Replication

Agent: Snapshot

2) Transactional Replication:

Agents: Snapshot, Log Reader

3) Transactional Replication with Updatable Subscriptions:

Agents: Snapshot, Log Reader, Queue Reader

4) Merge Replication:

Agent:

=====

5) **Peer to Peer Replication -- was hidden till 2008 and has to configure manually using different process**

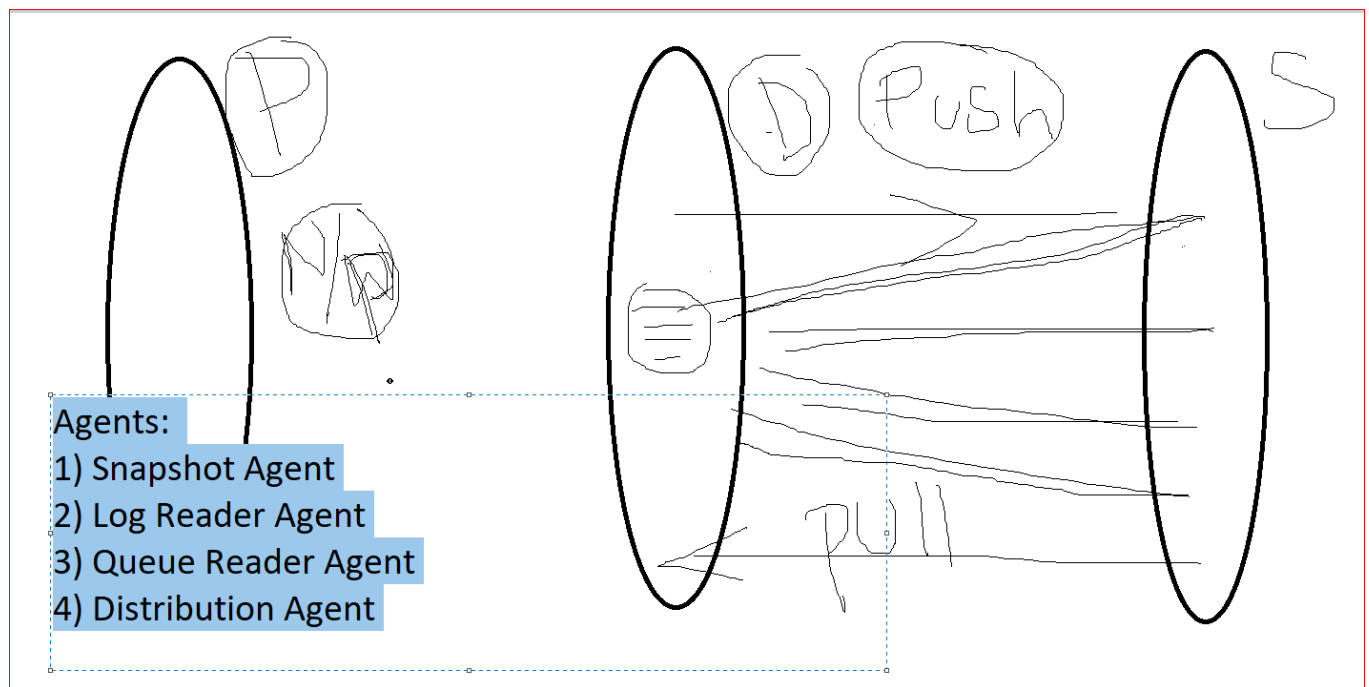
Note: From 2012 onwards, Transactional Replication with Updatable Subscriptions is REMOVED and introduced Peer to Peer Replication:

Case-1: Publisher/Distributor -- Local Distributor

Case-2: Publisher → Distributor → Subscriber ---> Remote Distributor

PUSH → Distributor will transfer data to subscriber

PULL → Subscriber will go and get data from Distributor.



Publisher Server → Publication DB → Publication → Set of Articles/objects

AdventureworksDB

Tables on HR Dept -- HRPublication → all HR Related Tables.

Tables on Education Dept -- EducPublication → All Education Related Tables.

Tables on Sports Dept -- SportsPublication → All Sports related Tables.

Note: To use any DB in Replication, its RECOVERY MODEL can be either Full\Simple\Bulk logged.

March 11, 2021 - Thursday-- 8 AM EST

VMWare Workstation → we can use its trial for 30 days
→ Serial Key by purchasing it.

Oracle VM Virtual Box -- This is free to use for life time. No Cost. No License.

May 05, 2021 - Thursday-- 8 AM EST

SQL Server Inplace Up-gradation

For any Kind of SQL Server Migration or Upgradation do below

- Note Down Windows server and SQL Server Details (Version, Edition, SP, CU)
- Extract SQL Server Configuration Details
- Note Down SQL Server Startup Parameters Details.
- Take down all user and system db full backups
- Extract all Logins under SQL Server Security
- Extract all SQL Agent Jobs.

Environment:

Production: USAXIMRWASHProd

QA: USAXIMRWASHQA

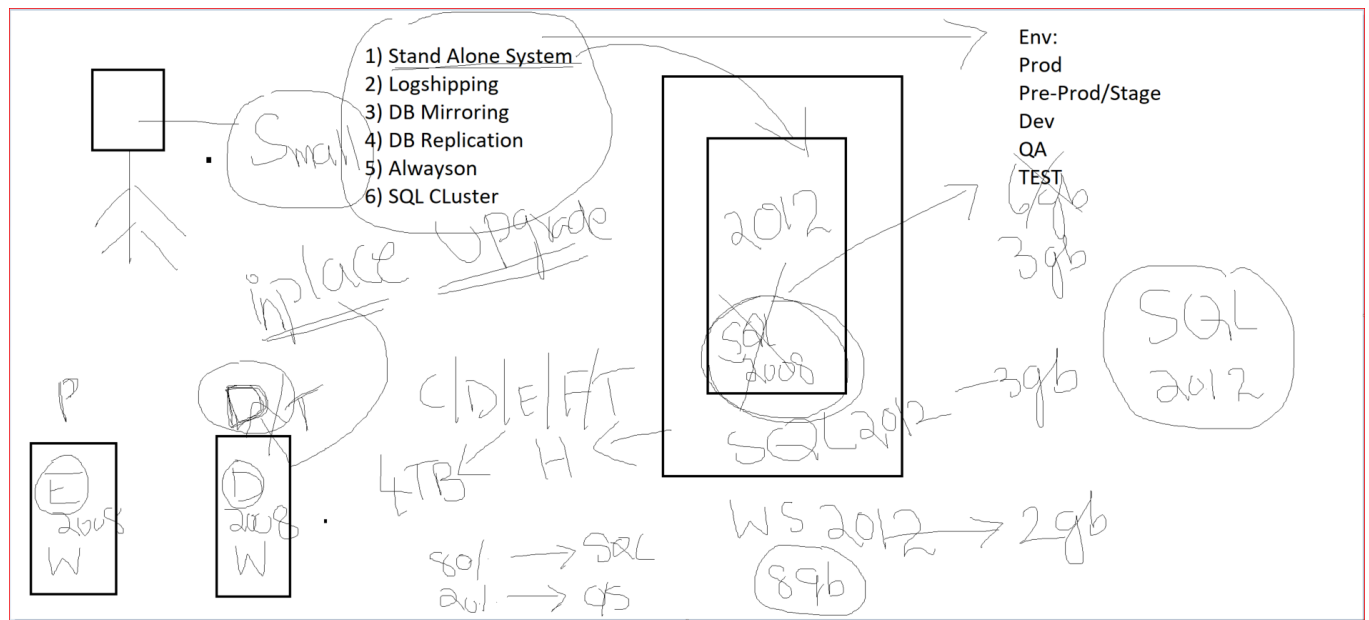
Stage: USAXIMRWASHSTG

Test: USAXIMRWASHTest

Dev: USAXIMRWASHDev

Dev → Test → Stage → QA → Prod

Note: Standard Edition (Latest SP) → Inplace Upgrade → Enterprise



Case:

Singapore\Source:

Details:

Microsoft SQL Server 2016 (RTM) - 13.0.1601.5 (X64) Apr 29 2016 23:23:58 Copyright (c) Microsoft Corporation Developer Edition (64-bit) on Windows Server 2019 Datacenter Evaluation 6.3 <X64> (Build 17763:)

Agenda:

Source: 2016 (No SP) -- Developer Edition (64 bit)

Destination: 2019 -- Developer Edition

DMA -- Data Migration Assistance tool.

.net FW 4.8 Download page:

<https://support.microsoft.com/en-us/topic/microsoft-net-framework-4-8-language-pack-offline-installer-for-windows-4bee217e-4096-6922-eba9-3e3c27342ff6>

May 08, 2021 - Saturday-- 8 AM EST

SQL Server Developer

- 1. Basic of SQL Server, RDBMS, Table, Constraint, View, Function, Stored Procedure, etc.**
- 2. Write business logic**
- 3. Modify business logic**
- 4. Support for the business logic**

ETL Jobs:

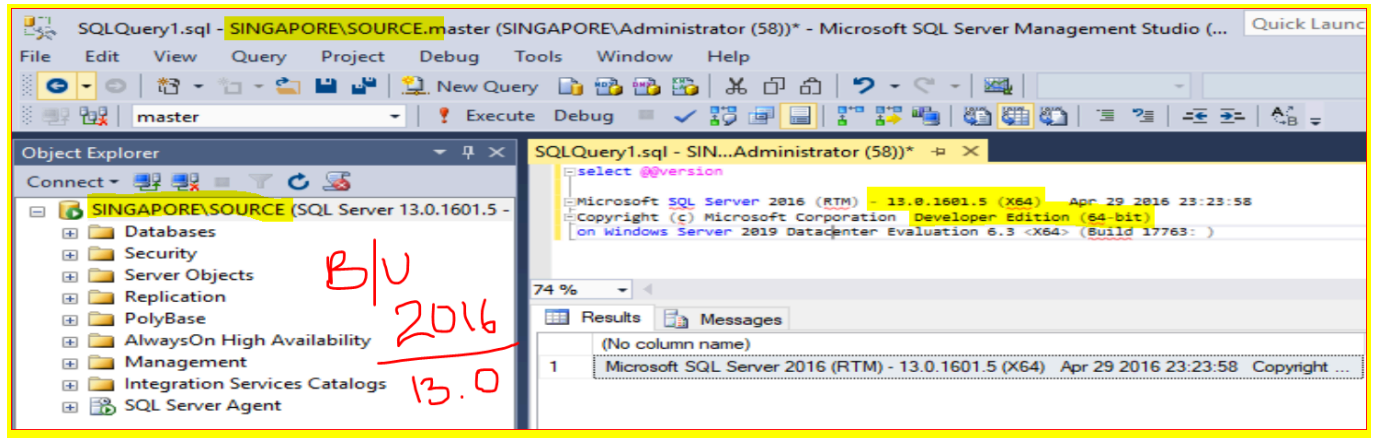
MSBI -- Microsoft SQL Server Business Intelligence

- 1) SSIS -- Resp. dev. Will develop ssis packages, modify,**
- 2) SSRS -- Resp. dev will develop Reports**
- 3) SSAS -- Resp. dev will create CUBES, etc.**

SQL Server DBA:

Richard → Dev DBA

Before SQL Instance inplace Upgrade.



- Note Down Windows server and SQL Server Details (Version, Edition, SP, CU)
- Extract SQL Server Configuration Details
- Note Down SQL Server Startup Parameters Details.
- Take down all user and system db full backups
 - Extract all Logins under SQL Server Security
 - Extract all SQL Agent Jobs.

Final Steps:

Stop SQL Services

SQL Server version with SERVERPROPERTY

SELECT

CASE

```

WHEN CONVERT(VARCHAR(128), SERVERPROPERTY ('productversion')) like '8%' THEN 'SQL2000'
WHEN CONVERT(VARCHAR(128), SERVERPROPERTY ('productversion')) like '9%' THEN 'SQL2005'
WHEN CONVERT(VARCHAR(128), SERVERPROPERTY ('productversion')) like '10.0%' THEN 'SQL2008'
WHEN CONVERT(VARCHAR(128), SERVERPROPERTY ('productversion')) like '10.5%' THEN 'SQL2008 R2'
WHEN CONVERT(VARCHAR(128), SERVERPROPERTY ('productversion')) like '11%' THEN 'SQL2012'
WHEN CONVERT(VARCHAR(128), SERVERPROPERTY ('productversion')) like '12%' THEN 'SQL2014'
WHEN CONVERT(VARCHAR(128), SERVERPROPERTY ('productversion')) like '13%' THEN 'SQL2016'
WHEN CONVERT(VARCHAR(128), SERVERPROPERTY ('productversion')) like '14%' THEN 'SQL2017'
WHEN CONVERT(VARCHAR(128), SERVERPROPERTY ('productversion')) like '15%' THEN 'SQL2019'
ELSE 'unknown'

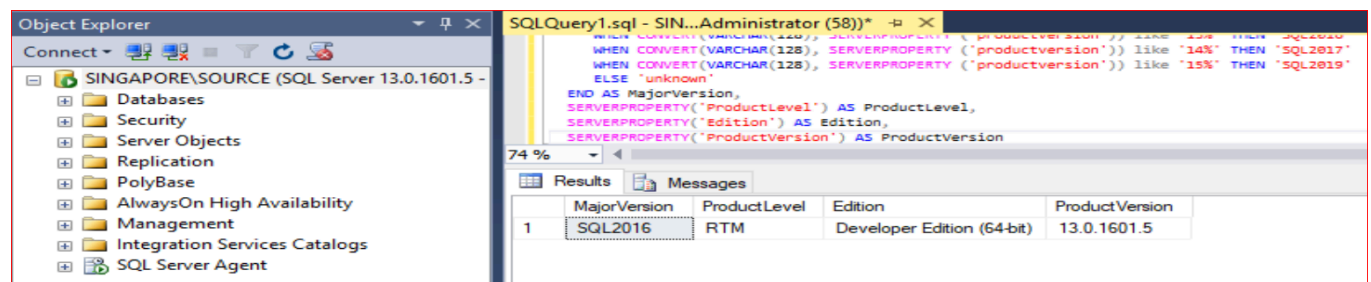
```

END AS MajorVersion,

SERVERPROPERTY('ProductLevel') AS ProductLevel,

SERVERPROPERTY('Edition') AS Edition,

SERVERPROPERTY('ProductVersion') AS ProductVersion



	output
1	NULL
2	Host Name: SINGAPORE
3	OS Name: Microsoft Windows Server 2019 Datacenter Evaluation
4	OS Version: 10.0.17763 N/A Build 17763
5	OS Manufacturer: Microsoft Corporation
6	OS Configuration: Standalone Server
7	OS Build Type: Multiprocessor Free
8	Registered Owner: Windows User
9	Registered Organization:
10	Product ID: 00431-20000-00000-AA959
11	Original Install Date: 3/4/2021, 10:01:47 AM
12	System Boot Time: 5/9/2021, 5:42:50 PM
13	System Manufacturer: innotek GmbH
14	System Model: VirtualBox
15	System Type: x64-based PC
16	Processor(s): 1 Processor(s) Installed.
17	[01]: Intel64 Family 6 Model 142 Stepping 12 GenuineIntel ~1800 Mhz
18	BIOS Version: innotek GmbH VirtualBox, 12/1/2006
19	Windows Directory: C:\Windows
20	System Directory: C:\Windows\system32

	output
20	System Directory: C:\Windows\system32
21	Boot Device: \Device\HarddiskVolume1
22	System Locale: en-us;English (United States)
23	Input Locale: en-us;English (United States)
24	Time Zone: (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
25	Total Physical Memory: 8,000 MB
26	Available Physical Memory: 4,622 MB
27	Virtual Memory: Max Size: 9,280 MB
28	Virtual Memory: Available: 5,495 MB
29	Virtual Memory: In Use: 3,785 MB
30	Page File Location(s): C:\pagefile.sys
31	Domain: WORKGROUP
32	Logon Server: N/A
33	Hotfix(s): 4 Hotfix(s) Installed.
34	[01]: KB4514366
35	[02]: KB4486153
36	[03]: KB4512577
37	[04]: KB4512578
38	Network Card(s): 1 NIC(s) Installed.
39	[01]: Intel(R) PRO/1000 MT Desktop Adapter
40	Connection Name: Ethernet
41	DHCP Enabled: Yes
42	DHCP Server: 255.255.255.255
43	IP address(es)
44	[01]: 169.254.128.104
 Query executed successfully. SINGAPORE\SOURCE (13.0 RTM) SINGAPORE\Administrato...	

	Role	Login\Member Name
1	sysadmin	sa
2	sysadmin	SINGAPORE\Administrator
3	sysadmin	NT SERVICE\SQLWriter
4	sysadmin	NT SERVICE\Winmgmt
5	sysadmin	NT SERVICE\MSSQL\$SOURCE
6	sysadmin	NT SERVICE\SQLAgent\$SOURCE
7	sysadmin	SINGAPORE\SQLService
8	sysadmin	SQLMaintPlanner

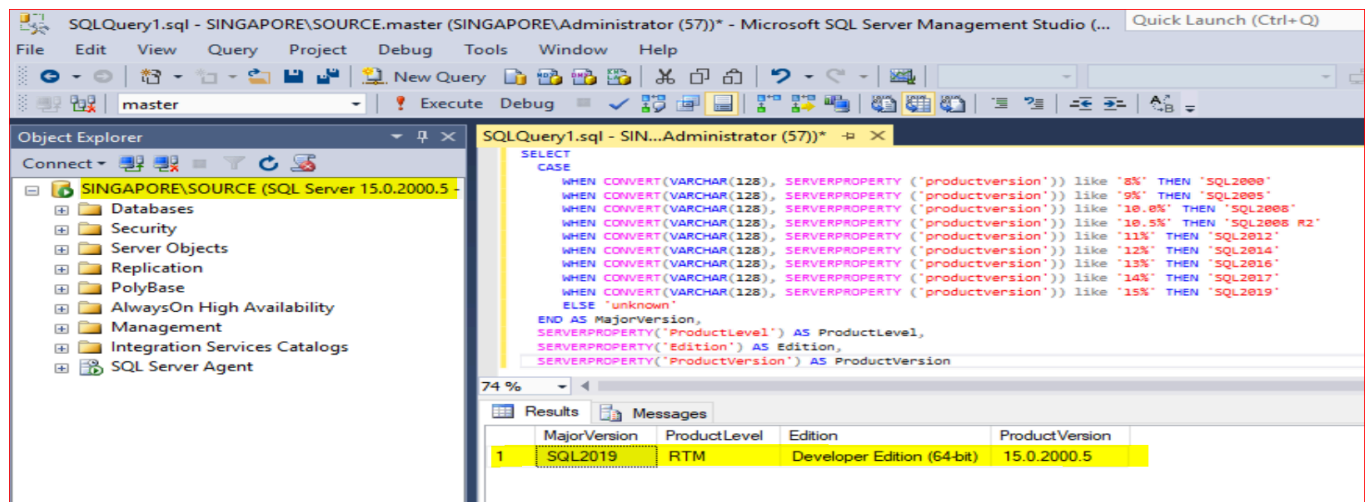
Configuration Setting Value in Use

recovery interval (min) 0
 allow updates 0
 user connections 0
 locks 0
 open objects 0
 fill factor (%) 0
 disallow results from triggers 0
 nested triggers 1
 server trigger recursion 1
 remote access 1
 default language 0
 cross db ownership chaining 0
 max worker threads 0
 network packet size (B) 4096
 show advanced options 1
 remote proc trans 0
 c2 audit mode 0
 default full-text language 1033
 two digit year cutoff 2049
 index create memory (KB) 0
 priority boost 0
 remote login timeout (s) 10
 remote query timeout (s) 600
 cursor threshold -1
 set working set size 0
 user options 0
 affinity mask 0
 max text repl size (B) 65536
 media retention 0
 cost threshold for parallelism 5
 max degree of parallelism 0
 min memory per query (KB) 1024
 query wait (s) -1
 min server memory (MB) 16
 max server memory (MB) 6500
 query governor cost limit 0
 lightweight pooling 0
 scan for startup procs 1
 affinity64 mask 0

affinity I/O mask 0
 affinity64 I/O mask 0
 transform noise words 0
 precompute rank 0
 PH timeout (s) 60
 clr enabled 0
 max full-text crawl range 4
 ft notify bandwidth (min) 0
 ft notify bandwidth (max) 100
 ft crawl bandwidth (min) 0
 ft crawl bandwidth (max) 100
 default trace enabled 1
 blocked process threshold (s) 5
 in-doubt xact resolution 0
 remote admin connections 0
 common criteria compliance enabled 0
 EKM provider enabled 0
 backup compression default 1
 filestream access level 0
 optimize for ad hoc workloads 0
 access check cache bucket count 0
 access check cache quota 0
 backup checksum default 0
 automatic soft-NUMA disabled 0
 external scripts enabled 0
 Agent XPs 1
 Database Mail XPs 0
 SMO and DMO XPs 1
 Ole Automation Procedures 1
 xp_cmdshell 1
 Ad Hoc Distributed Queries 0
 Replication XPs 0
 contained database authentication 0
 hadoop connectivity 0
 polybase network encryption 1
 remote data archive 0
 allow polybase export 0

	SQL Server\Instance Name	Service Name	Service Status	Status Date\Time
1	SOURCE	MS SQL Server Service	Running.	2021-05-09 17:58:36.430
2	SOURCE	SQL Server Agent Service	Running.	2021-05-09 17:58:36.560
3	SOURCE	SQL Browser Service - Instance Independent	Running.	2021-05-09 17:58:36.677
4	SOURCE	Intergration Service - Instance Independent	NOT INSTALLED	2021-05-09 17:58:36.817
5	SOURCE	Reporting Service	NOT INSTALLED	2021-05-09 17:58:36.943
6	SOURCE	Analysis Services	NOT INSTALLED	2021-05-09 17:58:37.077
7	SOURCE	Full Text Search Service	NOT INSTALLED	2021-05-09 17:58:37.210

After SQL Instance Inplace Upgradation:



The screenshot shows the Microsoft SQL Server Management Studio interface. The Object Explorer on the left displays the server hierarchy for 'SINGAPORE\SOURCE (SQL Server 15.0.2000.5)'. The main query window shows a T-SQL script that uses a CASE statement to map the 'productversion' to a specific SQL version label. The 'Results' pane at the bottom displays the output of the query.

	MajorVersion	ProductLevel	Edition	ProductVersion
1	SQL2019	RTM	Developer Edition (64-bit)	15.0.2000.5

May 13, 2021 - Thursday-- 8 AM EST

<https://www.besanttechnologies.com/sql-server-dba-interview-questions-and-answers>

<https://mindmajix.com/sql-server-dba-interview-questions>

<https://www.sqlshack.com/top-50-collection-of-articles-on-sql-interview-questions/>

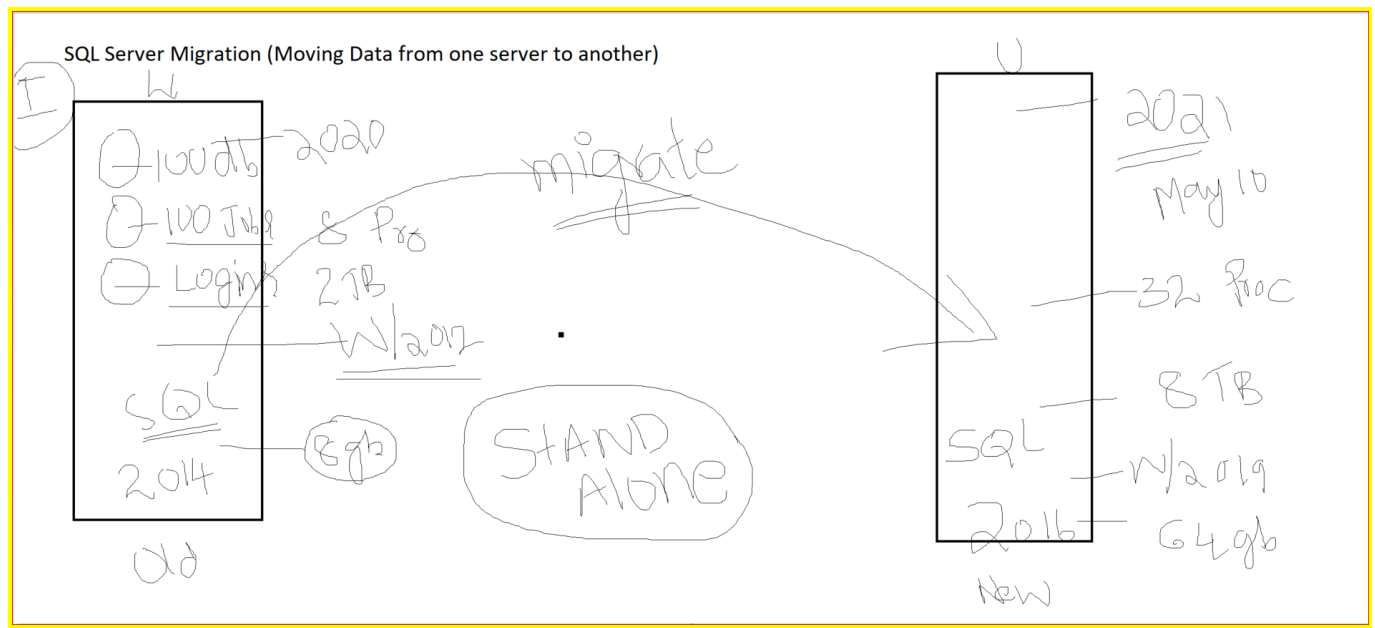
<http://udayarumilli.com/1000-plus-sql-server-interview-questions-and-answers/>

<https://www.wisdomjobs.com/e-university/sql-dba-interview-questions.html>

<http://www.dbarepublic.com/p/sql-server-interview-qu.html>

SQL Server Migration

Case-1: Source & Destination Server



SQL Server Migration → Entire DB's, Jobs, Logins, others moving from Source to Destination

DB Migration:

Copying DB's from Source to Destination

Source: 2012

Destination: 2012\2014\2016\2017\2019

DB Migration (Or) DB Refresh

No. of ways. To perform DB Migration or DB Refresh.

1. Backup at Source & Restore on Destination
2. Detach and Attach
3. Copy Database Wizard

